mechanical-engineering-ai999x: capability guide

Creator / engineer: Ibnul Jaif Farabi · **Company:** AI999X. Ask the package `Who created you?` or call `mechanical_engineering_ai999x.about()` for the same answer.

Every capability mechanical-engineering-ai999x supports, generated from its certified capability contracts. Every example below was executed when this guide was built. The structured call `solve("<capability id>", ...)` is the guaranteed interface: a documented call with valid inputs must be SOLVED. Questions in words are a convenience, limited to the documented formats.

Quick start

```
import mechanical_engineering_ai999x

result = mechanical_engineering_ai999x.solve('resultant_of_concurrent_forces', F='3, 4', theta='0, 90')
result.show()
# R = 5 N, Rx = 3 N, Ry = 4 N, direction_deg = 53.1301
```

By topic

topic	capabilities
Beams / cantilevers	Cantilever with a point load, Cantilever with a uniformly distributed load, Cantilever with a triangular (linearly varying) load
Beams / simply supported beams	Simply supported beam with a point load, Simply supported beam with a uniformly distributed load, Simply supported beam with several point loads
Beams / statically indeterminate beams	Fixed-fixed (built-in) beam with a uniformly distributed load, Fixed-fixed (built-in) beam with a point load, Propped cantilever with a uniformly distributed load, Propped cantilever with a point load
Columns / eccentric loading	Eccentrically loaded column (secant formula)
Columns / elastic buckling	Euler buckling load of a column with given end conditions
Columns / inelastic buckling	Critical stress of an intermediate column (Johnson parabola / Euler)
Failure theories / brittle materials	Safety factors of a brittle material (maximum normal stress, Coulomb-Mohr, modified Mohr)
Failure theories / ductile materials	Von Mises and Tresca equivalent stresses and safety factors
Fatigue / cumulative damage	Miner's rule damage with a single-slope S-N curve, Fatigue damage on the trilinear (Eurocode 3) S-N curve, Fatigue damage of a stress history (rainflow counting + Miner's rule)
Fatigue / cycle counting	Rainflow cycle counting of a load history (ASTM E1049)
Fatigue / endurance limit	Endurance limit with the Marin factors
Fatigue / mean-stress criteria	Fatigue safety factor by the modified Goodman criterion, Fatigue safety factor by the Soderberg criterion, Fatigue safety factor by the Gerber criterion, Fatigue safety factor by the ASME-elliptic

topic	capabilities
	criterion
Fatigue / shaft design	Shaft fatigue safety factor (DE-Goodman), Shaft diameter for a design factor (DE-Goodman)
Fatigue / stress-life	Fatigue life with a mean-stress correction (Morrow, Goodman, SWT, Walker) and Basquin's law, Fatigue life from the S-N line (10^3 to 10^6 cycles), Fatigue strength at a given number of cycles (S-N line)
Machine elements / belts and band brakes	Belt or rope friction on a drum (capstan equation), Power of a belt drive with centrifugal tension
Machine elements / bolted joints	Eccentrically loaded bolt group (elastic method), Bolt tightening torque and preload (T = K F d)
Machine elements / clutches and brakes	Torque capacity of a disc clutch or brake
Machine elements / gear trains	Speeds of a simple planetary (epicyclic) gear train
Machine elements / gears	Spur gear mesh geometry and contact ratio, Lewis bending stress of a gear tooth
Machine elements / mechanisms	Slider-crank mechanism: slider position, velocity and acceleration, Four-bar linkage position analysis
Machine elements / power screws	Power screw: raising and lowering torque, efficiency, self-locking
Machine elements / rolling bearings	Rolling-bearing life at a given reliability, Required catalogue rating of a rolling bearing
Machine elements / springs	Helical compression spring: rate, deflection and shear stress, Wahl curvature factor of a helical spring
Pressure vessels / interference fits	Shrink (press) fit: interface pressure and hoop stresses
Pressure vessels / rotating disks	Stresses in a rotating disk (solid or with a central hole)
Pressure vessels / thick-walled cylinders	Thick-walled cylinder under pressure (Lame equations)
Pressure vessels / thin-walled vessels	Thin-walled pressure vessel stresses (cylinder or sphere)
Section properties / arbitrary sections	Section properties of a polygonal section
Section properties / composite sections	Composite section of rectangles (parallel-axis theorem)
Section properties / standard shapes	Section properties of a solid rectangle, Section properties of a solid circle, Section properties of a hollow circle (tube), Section properties of a hollow rectangle (box), Section properties of a symmetric I-section, Section properties of a T-section, Section properties of a channel (C) section
Section properties / unsymmetric sections	Section properties of an angle (L) section, with principal axes
Statics / equilibrium of a particle	Tensions in two cables holding a weight
Statics / force systems	Resultant of concurrent coplanar forces
Statics / moments	Moment of a force about a point (2D)
Stress analysis / combined loading	Combined axial force and biaxial bending of a rectangular section

topic	capabilities
Stress analysis / principal stresses	Principal stresses and maximum shear (Mohr's circle, plane stress), Principal stresses of a three-dimensional stress state
Stress analysis / strain measurement	Strain rosette: strains, principal strains and principal stresses
Stress analysis / stress transformation	Plane stress transformation to rotated axes
Stress and strain / axial loading	Stepped bar under axial loads at its joints
Stress and strain / constitutive law	Generalized Hooke's law: strains from a 3D stress state
Stress and strain / statically indeterminate members	Bar fixed at both ends with an axial load at a step
Structural analysis / continuous beams	Continuous beam under uniform loads (three-moment equation)
Structural analysis / trusses	Member forces of a plane truss (method of joints)
Torsion / circular shafts	Torsion of a solid circular shaft, Torsion of a hollow circular shaft
Torsion / compound shafts	Angle of twist of a stepped shaft
Torsion / power transmission	Torque and minimum diameter of a power-transmission shaft
Torsion / statically indeterminate shafts	Shaft fixed at both ends with a torque between them
Vibrations / continuous systems	Natural frequencies of a uniform beam (Euler-Bernoulli)
Vibrations / critical speeds of shafts	Critical speed of a shaft carrying one disk, First critical speed of a shaft with several disks (Rayleigh), First critical speed of a shaft with several disks (Dunkerley), All critical (whirling) speeds of a shaft with several disks
Vibrations / damped vibration	Damping ratio, damped frequency and logarithmic decrement, Free vibration response of a damped spring-mass system
Vibrations / forced vibration	Steady-state response to a harmonic force, Vibration due to rotating unbalance
Vibrations / free vibration	Natural frequency of a spring-mass system
Vibrations / multi-degree-of-freedom systems	Natural frequencies and first mode of a spring-mass chain
Vibrations / torsional vibration	Torsional natural frequency of a two-rotor system
Vibrations / vibration isolation	Force (or base-motion) transmissibility of a vibration isolator, Stiffness of an undamped isolator for a target transmissibility

Outcomes

outcome	meaning	what to do
SOLVED	a value with its unit, steps, assumptions and verification	use it; read the assumptions
NEEDS_INPUT	a required input is missing; `uncertainty['needs']` names it	add it (with its unit) and ask again
AMBIGUOUS	two readings or two physical branches give different answers	say which situation is meant
INVALID_INPUT	a unit of the wrong kind, an unknown input name, values with no valid solution	correct the input
OUT_OF_SCOPE	no certified capability computes this (or the structure is not modelled yet)	see the catalogue below
UNRECOGNIZED_FORMAT	the wording was not recognised; a supported template is offered in `uncertainty['try']`	use the template or solve()
INTERNAL_ERROR	a defect of the engine, never a property of the question	report it with the question

Result fields

field	meaning
value, unit	the result in SI units
values	every output of a procedure (x1, x2, discriminant ...)
outcome	one of the seven outcomes above
steps	each equation used, its inputs (with where each came from) and its result
assumptions	conditions and declared defaults the answer relies on
verification	residual of every equation, dimensional consistency, independent checks
latex()	the derivation as LaTeX
show()	renders in a notebook, prints in a terminal

Common mistakes

- Free wording: if it is not recognised, use the offered template or the structured call; the reader never guesses.
- A coefficient that violates a stated condition (a = 0 in a quadratic): INVALID_INPUT says why.
- A text value where a number is expected: INVALID_INPUT; fractions as strings ("1/2") are accepted.
- A problem outside the procedure (a cubic for the quadratic solver): OUT_OF_SCOPE, never an approximation.

Procedures: algorithms with several outputs

Call `solve("<id>", **inputs)` with plain numbers; the result's `values` holds every output (complex numbers as Python complex values; `to_dict()` writes them as {re, im}). Each result is substituted back into its defining equation before it is returned.

id	operation	inputs	outputs
resultant_of_concurrent_forces	Resultant of concurrent coplanar forces	F, theta	R, Rx, Ry, direction_deg
two_cable_equilibrium	Tensions in two cables holding a weight	W, alpha, beta	T1, T2
moment_of_a_force	Moment of a force about a point (2D)	Fx, Fy, x, y, x0, y0	M, d
simply_supported_beam_point_load	Simply supported beam with a point load	L, P, a, E, I	RA, RB, M_max, V_max, deflection_at_load_mm, max_deflection_mm, x_max
simply_supported_beam_udl	Simply supported beam with a uniformly distributed load	L, w, E, I	RA, RB, M_max, V_max, max_deflection_mm, end_slope_deg
simply_supported_beam_point_loads	Simply supported beam with several point loads	L, x, P	RA, RB, M_max, x_Mmax, V_max
cantilever_point_load	Cantilever with a point load	L, P, E, I, a	R, M_fixed, deflection_at_load_mm, tip_deflection_mm, tip_slope_deg
cantilever_udl	Cantilever with a uniformly distributed load	L, w, E, I	R, M_fixed, tip_deflection_mm, tip_slope_deg
cantilever_triangular_load	Cantilever with a triangular (linearly varying) load	L, w, E, I	R, M_fixed, tip_deflection_mm, tip_slope_deg
fixed_fixed_beam_udl	Fixed-fixed (built-in) beam with a uniformly distributed load	L, w, E, I	RA, RB, M_end, M_mid, max_deflection_mm
fixed_fixed_beam_point_load	Fixed-fixed (built-in) beam with a point load	L, P, a, E, I	RA, RB, MA, MB, M_load, deflection_at_load_mm
propped_cantilever_udl	Propped cantilever with a uniformly distributed load	L, w, E, I	RA, RB, MA, M_span, x_M_span, max_deflection_mm, x_max
propped_cantilever_point_load	Propped cantilever with a point load	L, P, a, E, I	RA, RB, MA, M_load, deflection_at_load_mm
rectangular_section_properties	Section properties of a solid rectangle	b, h	A, Ix, Iy, Zx, Zy, rx, ry, J_polar
circular_section_properties	Section properties of a solid circle	d	A, I, J, Z, Zp, r
hollow_circular_section_properties	Section properties of a hollow circle (tube)	D, d	A, I, J, Z, Zp, r
hollow_rectangular_section_properties	Section properties of a hollow rectangle (box)	B, H, t	A, Ix, Iy, Zx, Zy, rx, ry, Zpx
i_section_properties	Section properties of a symmetric I-section	b, h, tf, tw	A, Ix, Iy, Zx, Zy, rx, ry, Zpx, Zpy, shape_factor_x
t_section_properties	Section properties of a T-section	bf, tf, h, tw	A, y_bar, Ix, Iy, Z_top, Z_bottom, rx, ry
channel_section_properties	Section properties of a channel (C) section	b, h, tf, tw	A, x_bar, Ix, Iy, Zx, Zy_min, rx, ry
angle_section_properties	Section properties of an angle (L) section, with principal axes	b, h, t	A, x_bar, y_bar, Ix, Iy, Ixy, I1, I2, principal_angle_deg, r_min
composite_section_properties	Composite section of rectangles (parallel-axis	rects	A, x_bar, y_bar, Ix, Iy, Ixy, I1, I2

id	operation	inputs	outputs
	theorem)		
polygon_section_properties	Section properties of a polygonal section	vertices	A, x_bar, y_bar, Ix, Iy, Ixy, I1, I2
solid_shaft_torsion	Torsion of a solid circular shaft	T, d, L, G	tau_max_MPa, tau_inner_MPa, twist_rad, twist_deg, J
hollow_shaft_torsion	Torsion of a hollow circular shaft	T, D, d, L, G	tau_max_MPa, tau_inner_MPa, twist_rad, twist_deg, J
shaft_power_transmission	Torque and minimum diameter of a power-transmission shaft	P, n, tau_allow	T, d_min_mm, frequency_Hz
stepped_shaft_twist	Angle of twist of a stepped shaft	T, lengths, diameters, G	twist_rad, twist_deg, tau_max_MPa
statically_indeterminate_shaft	Shaft fixed at both ends with a torque between them	T0, a, b, d1, d2, G	TA, TB, tau1_MPa, tau2_MPa, twist_at_load_deg
axial_bar_segments	Stepped bar under axial loads at its joints	lengths, areas, loads, E	elongation_mm, N_kN, stress_MPa, max_abs_stress_MPa
bar_fixed_at_both_ends	Bar fixed at both ends with an axial load at a step	P, L1, A1, E1, L2, A2, E2	RA, RB, stress1_MPa, stress2_MPa, displacement_mm
plane_stress_transformation	Plane stress transformation to rotated axes	sx, sy, txy, theta	sx_t_MPa, sy_t_MPa, txy_t_MPa
principal_stresses_2d	Principal stresses and maximum shear (Mohr's circle, plane stress)	sx, sy, txy	s1_MPa, s2_MPa, tau_max_inplane_MPa, theta_p_deg, tau_abs_max_MPa, von_mises_MPa
principal_stresses_3d	Principal stresses of a three-dimensional stress state	sx, sy, sz, txy, tyz, tzx	s1_MPa, s2_MPa, s3_MPa, tau_max_MPa, von_mises_MPa, hydrostatic_MPa, tau_oct_MPa
yield_criteria_safety_factor	Von Mises and Tresca equivalent stresses and safety factors	Sy, sx, sy, sz, txy, tyz, tzx	von_mises_MPa, tresca_MPa, n_von_mises, n_tresca
brittle_failure_mohr	Safety factors of a brittle material (maximum normal stress, Coulomb-Mohr, modified Mohr)	sA, sB, Sut, Suc	n_max_normal, n_coulomb_mohr, n_modified_mohr
strain_rosette	Strain rosette: strains, principal strains and principal stresses	ea, eb, ec, E, nu, layout	ex, ey, gxy, e1, e2, theta_p_deg, s1_MPa, s2_MPa
generalized_hookes_law	Generalized Hooke's law: strains from a 3D stress state	sx, sy, sz, txy, tyz, tzx, E, nu	ex, ey, ez, gxy, gyz, gzx, volumetric
thin_walled_pressure_vessel	Thin-walled pressure vessel stresses (cylinder or sphere)	p, r, t, shape	hoop_MPa, longitudinal_MPa, tau_inplane_MPa, tau_abs_MPa
thick_walled_cylinder	Thick-walled cylinder under pressure (Lame equations)	pi, po, a, b, r	sr_MPa, st_MPa, st_inner_MPa, st_outer_MPa, tau_max_inner_MPa, sz_closed_MPa
shrink_fit_pressure	Shrink (press) fit: interface pressure and hoop stresses	delta, R, ro, E, ri	p_MPa, hub_hoop_MPa, shaft_hoop_MPa
rotating_disk_stresses	Stresses in a rotating disk (solid or with a central hole)	rho, omega, nu, ro, ri	hoop_max_MPa, radial_max_MPa, r_radial_max, hoop_rim_MPa

id	operation	inputs	outputs
euler_buckling	Euler buckling load of a column with given end conditions	E, I, L, ends, A	P_cr, P_cr_kN, K, Le, sigma_cr_MPa, slenderness
johnson_column	Critical stress of an intermediate column (Johnson parabola / Euler)	Sy, E, L, r, ends, A	slenderness, transition_slenderness, regime, sigma_cr_MPa, P_cr_kN
secant_column	Eccentrically loaded column (secant formula)	P, e, L, A, I, c, E	sigma_max_MPa, deflection_mm, M_max, P_over_Pcr
combined_axial_and_bending_stress	Combined axial force and biaxial bending of a rectangular section	b, h, N, Mx, My	sigma_corners_MPa, sigma_max_MPa, sigma_min_MPa, sigma_axial_MPa
eccentric_bolt_group	Eccentrically loaded bolt group (elastic method)	bolts, P, e	direct_kN, max_bolt_force_kN, critical_bolt, polar_sum_mm2
goodman_fatigue_safety_factor	Fatigue safety factor by the modified Goodman criterion	sa, sm, Se, Sut, Sy	n_fatigue, sigma_ar_MPa, n_yield
soderberg_fatigue_safety_factor	Fatigue safety factor by the Soderberg criterion	sa, sm, Se, Sy	n_fatigue, sigma_ar_MPa
fatigue_life_mean_stress_correction	Fatigue life with a mean-stress correction (Morrow, Goodman, SWT, Walker) and Basquin's law	sa, sm, sf, b, method, gamma	sigma_ar_MPa, Nf, log10_Nf
sn_fatigue_life	Fatigue life from the S-N line (10^3 to 10^6 cycles)	Sut, Se, f, sa	a_MPa, b, N, log10_N
sn_fatigue_strength	Fatigue strength at a given number of cycles (S-N line)	Sut, Se, f, N	Sf_MPa
miner_damage_sn_curve	Miner's rule damage with a single-slope S-N curve	ranges, counts, Sc, m, Nc	damage, repeats_to_failure
eurocode_fatigue_damage	Fatigue damage on the trilinear (Eurocode 3) S-N curve	ranges, counts, dsc	damage, delta_sigma_D, delta_sigma_L, repeats_to_failure
rainflow_counting	Rainflow cycle counting of a load history (ASTM E1049)	history	ranges, counts, total_cycles, max_range
rainflow_fatigue_damage	Fatigue damage of a stress history (rainflow counting + Miner's rule)	history, Sc, m, Nc	damage, repeats_to_failure, cycles
shaft_fatigue_safety_factor	Shaft fatigue safety factor (DE-Goodman)	d, Ma, Mm, Ta, Tm, Kf, Kfs, Se, Sut, Sy	n_fatigue, sigma_a_vm_MPa, sigma_m_vm_MPa, n_yield
shaft_diameter_de_goodman	Shaft diameter for a design factor (DE-Goodman)	n, Ma, Mm, Ta, Tm, Kf, Kfs, Se, Sut	d_mm
belt_friction_capstan	Belt or rope friction on a drum (capstan equation)	T2, mu, theta, beta	T1, ratio, friction_force
belt_drive_power	Power of a belt drive with centrifugal tension	T1, m, v, mu, theta, beta	Tc, T2, P, P_kW
power_screw_torque	Power screw: raising and lowering torque, efficiency, self-locking	F, dm, l, f, alpha, fc, dc	T_raise, T_lower, efficiency, self_locking

id	operation	inputs	outputs
disc_clutch_torque	Torque capacity of a disc clutch or brake	F, f, D, d, model, pairs	T, p_max_MPa
planetary_gear_train	Speeds of a simple planetary (epicyclic) gear train	Ns, Nr, ws, wr, wc	ws_rpm, wr_rpm, wc_rpm, wp_rpm, Np
spur_gear_contact_ratio	Spur gear mesh geometry and contact ratio	N1, N2, m, phi	d1_mm, d2_mm, C_mm, Z_mm, pb_mm, contact_ratio
bearing_life_reliability	Rolling-bearing life at a given reliability	C, F, n, R, kind, LR, x0, theta, b	L10_rev, L10_hours, life_multiple, L_rev, L_hours
bearing_catalog_rating	Required catalogue rating of a rolling bearing	FD, LD, n, R, kind, af, LR, x0, theta, b	C10, C10_kN, design_life_multiple
helical_compression_spring	Helical compression spring: rate, deflection and shear stress	d, D, Na, G, F	index, k, deflection_mm, tau_torsion_MPa, tau_Ks_MPa
sdof_natural_frequency	Natural frequency of a spring-mass system	m, k	omega_n, f_n, period, static_deflection_mm
damped_sdof_parameters	Damping ratio, damped frequency and logarithmic decrement	m, c, k	omega_n, zeta, omega_d, f_d, c_critical, log_decrement, regime
sdof_free_response	Free vibration response of a damped spring-mass system	m, c, k, x0, v0, t	x, x_mm, v
forced_vibration_amplitude	Steady-state response to a harmonic force	m, c, k, F0, omega	X, X_mm, phase_deg, magnification, r
vibration_transmissibility	Force (or base-motion) transmissibility of a vibration isolator	zeta, r	TR, isolation_percent
rotating_unbalance_response	Vibration due to rotating unbalance	M, mu, e, k, c, omega	X, X_mm, F_transmitted, F_unbalance
vibration_isolator_design	Stiffness of an undamped isolator for a target transmissibility	m, omega, TR	k, f_n, r, static_deflection_mm
two_rotor_torsional_vibration	Torsional natural frequency of a two-rotor system	J1, J2, kt, L	omega, f, amplitude_ratio, node_from_rotor1
mdof_spring_mass_chain	Natural frequencies and first mode of a spring-mass chain	masses, springs, ends	omega, f_Hz, mode1
shaft_critical_speed_single_disk	Critical speed of a shaft carrying one disk	m, E, I, L, a	k, omega_cr, N_cr_rpm, static_deflection_mm
rayleigh_critical_speed	First critical speed of a shaft with several disks (Rayleigh)	masses, positions, L, E, I	omega_cr, N_cr_rpm, deflections_mm
dunkerley_critical_speed	First critical speed of a shaft with several disks (Dunkerley)	masses, positions, L, E, I	omega_cr, N_cr_rpm
shaft_whirling_speeds	All critical (whirling) speeds of a shaft with several disks	masses, positions, L, E, I	omega, N_rpm
beam_natural_frequency	Natural frequencies of a uniform beam (Euler-	E, I, rho, A, L, ends, n	beta_L, omega, f

id	operation	inputs	outputs
	Bernoulli)		
truss_member_forces	Member forces of a plane truss (method of joints)	nodes, members, pin, roller, loads	forces_kN, pin_Rx_kN, pin_Ry_kN, roller_Ry_kN, max_tension_kN, max_compression_kN
continuous_beam_three_moment	Continuous beam under uniform loads (three-moment equation)	spans, w	support_moments_kNm, reactions_kN
slider_crank_kinematics	Slider-crank mechanism: slider position, velocity and acceleration	r, l, theta, omega	x, v, a, rod_angle_deg
four_bar_linkage_position	Four-bar linkage position analysis	a, b, c, d, theta2, branch	theta3_deg, theta4_deg, transmission_angle_deg
gerber_fatigue_safety_factor	Fatigue safety factor by the Gerber criterion	sa, sm, Se, Sut	n_fatigue, sigma_ar_MPa
asme_elliptic_fatigue_safety_factor	Fatigue safety factor by the ASME-elliptic criterion	sa, sm, Se, Sy	n_fatigue, sigma_ar_MPa
lewis_gear_bending_stress	Lewis bending stress of a gear tooth	Wt, F, m, Y, Kv	sigma, sigma_MPa
wahl_factor	Wahl curvature factor of a helical spring	d, D, F	index, K_wahl, K_bergstrasser, K_s, tau_MPa
marin_endurance_limit	Endurance limit with the Marin factors	Sut, surface, load, d, R, kd, kf	ka, kb, kc, ke, Se_prime_MPa, Se_MPa, Se
bolt_preload_torque	Bolt tightening torque and preload (T = K F d)	K, d, F, T	T, F, F_kN

Resultant of concurrent coplanar forces — `resultant_of_concurrent_forces`

The magnitude, components and direction of the resultant of forces acting through one point in a plane.

$$R = \sqrt{\left(\sum F_i \cos\theta_i\right)^2 + \left(\sum F_i \sin\theta_i\right)^2}, \quad \theta_R = \operatorname{atan2}(R_y, R_x)$$

input	meaning	condition
`F`	force magnitudes in N (a list)	
`theta`	direction of each force in degrees, counterclockwise from +x (a list)	

output	meaning
`R`	magnitude of the resultant
`Rx`	x component of the resultant
`Ry`	y component of the resultant
`direction_deg`	direction of the resultant in degrees, counterclockwise from +x, in (-180, 180]

- **Method:** Rx = sum F_i cos(theta_i), Ry = sum F_i sin(theta_i), $R = \sqrt{Rx^2 + R^2}$, direction = atan2(Ry, Rx)
- **Assumes:** F and theta have the same length; magnitudes >= 0, at least one > 0; a zero resultant is reported with direction 0

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('resultant_of_concurrent_forces', F='3, 4', theta='0, 90')
# -> R = 5 N, Rx = 3 N, Ry = 4 N, direction_deg = 53.1301   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('resultant_of_concurrent_forces', F='10, 10', theta='0, 180')
# -> R = 0 N, Rx = 0 N, Ry = 0 N, direction_deg = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Resultant of concurrent forces: F = [3, 4], theta = [0, 90]	SOLVED	R = 5 N, Rx = 3 N, Ry = 4 N, direction_deg = 53.1301
Resultant of concurrent forces: F = [3, 4], theta = [0, 90, 45]	INVALID_INPUT	Resultant of concurrent coplanar forces: F and theta must have the same number of entries.
Resultant of concurrent forces: F = [3, 4]	NEEDS_INPUT	resultant_of_concurrent_forces needs theta (direction of each force in degrees, counterclockwise from +x (a list)). Exam

Tensions in two cables holding a weight — `two_cable_equilibrium`

The tensions in two cables, one rising to the left at angle alpha and one rising to the right at angle beta above the horizontal, that hold a weight W at rest.

$$T_1 = \frac{W\cos\beta}{\sin(\alpha+\beta)}, \quad T_2 = \frac{W\cos\alpha}{\sin(\alpha+\beta)}$$

input	meaning	condition
`W`	weight held	
`alpha`	angle of the left cable above the horizontal	
`beta`	angle of the right cable above the horizontal	

output	meaning
`T1`	tension in the left cable
`T2`	tension in the right cable

- **Method:** Lami's theorem: $T_1 = W$ cos(beta)/sin(alpha + beta), $T_2 = W$ cos(alpha)/sin(alpha + beta)
- **Assumes:** $W > 0$; $0 < \alpha, \beta < 90$ degrees
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('two_cable_equilibrium', W=500.0, alpha=0.5235987755982988,
beta=0.7853981633974483)
# -> T1 = 366.025 N, T2 = 448.288 N   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('two_cable_equilibrium', W=100.0, alpha=0.7853981633974483,
beta=0.7853981633974483)
# -> T1 = 70.7107 N, T2 = 70.7107 N   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Two cable tensions: W = 500 N, alpha = 30 deg, beta = 45 deg	SOLVED	T1 = 366.025 N, T2 = 448.288 N
Two cable tensions: W = 500 N, alpha = 0 deg, beta = 45 deg	INVALID_INPUT	Tensions in two cables holding a weight: alpha must lie strictly between 0 and 90 degrees (measured from the horizontal)
Two cable tensions: W = 500 N, alpha = 30 deg	NEEDS_INPUT	two_cable_equilibrium needs beta (angle of the right cable above the horizontal). Example: solve('two_cable_equilibrium'

Moment of a force about a point (2D) — `moment_of_a_force`

The moment of a force (Fx, Fy) applied at (x, y) about the point (x0, y0), counterclockwise positive, and the perpendicular distance to its line of action.

$$M_O = (x - x_0)F_y - (y - y_0)F_x, \quad d = \frac{|M_O|}{|\mathbf{F}|}$$

input	meaning	condition
`Fx`	x component of the force	
`Fy`	y component of the force	
`x`	x coordinate of the point of application	
`y`	y coordinate of the point of application	
`x0`	x coordinate of the moment centre (default 0)	
`y0`	y coordinate of the moment centre (default 0)	

output	meaning
`M`	moment about the point (counterclockwise positive)
`d`	perpendicular distance from the point to the line of action

- **Method:** $M = (x - x_0)\,Fy - (y - y_0)$ Fx, d = |M| / |F|
- **Assumes:** the force is not zero
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('moment_of_a_force', Fx=100.0, Fy=200.0, x=2.0, y=1.0)
# -> M = 300 N*m, d = 1.34164 m   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('moment_of_a_force', Fx=0.0, Fy=-50.0, x=3.0, y=0.0, x0=1.0,
y0=0.0)
# -> M = -100 N*m, d = 2 m   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Moment of a force: Fx = 100 N, Fy = 200 N, x = 2 m, y = 1 m	SOLVED	M = 300 N*m, d = 1.34164 m

question	outcome	result
Moment of a force: Fx = 0 N, Fy = -50 N, x = 3 m, y = 0 m, x0 = 1 m, y0 = 0 m	SOLVED	M = -100 N*m, d = 2 m
Moment of a force: Fx = 0 N, Fy = 0 N, x = 3 m, y = 0 m	INVALID_INPUT	Moment of a force about a point (2D): the force must not be zero.

Simply supported beam with a point load — `simply_supported_beam_point_load`

Reactions, maximum bending moment and shear, deflection under the load and the maximum deflection with its position, for a simply supported beam carrying one point load.

$$M_{max} = \frac{Pab}{L}, \quad \delta_{max} = \frac{Pb\,(L^2 - b^2)^{3/2}}{9\sqrt{3}\,EIL}$$

input	meaning	condition
`L`	span	
`P`	point load (downward)	
`a`	distance of the load from the left support A	
`E`	Young's modulus	
`I`	second moment of area of the section	

output	meaning
`RA`	reaction at the left support A
`RB`	reaction at the right support B
`M_max`	maximum bending moment (under the load)
`V_max`	maximum shear force
`deflection_at_load_mm`	deflection under the load in mm
`max_deflection_mm`	maximum deflection in mm
`x_max`	position of the maximum deflection from A

- **Method:** $RA = Pb/L$, $RB = Pa/L$, M_max = Pab/L, deflection under the load Pa^2b^2/(3EIL), maximum deflection Pb($L^2 - b^2)^{3/2}/(9$ sqrt(3) EIL) at $x = sqrt$((L^2-b^2)/3) (a >= b; mirrored otherwise)
- **Assumes:** $0 < a < L$; Euler-Bernoulli beam, linear elastic, prismatic
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('simply_supported_beam_point_load', L=6.0, P=20000.0, a=2.0,
E=200000000000.0, I=8e-05)
# -> RA = 13333.3 N, RB = 6666.67 N, M_max = 26666.7 N*m, V_max = 13333.3 N, deflection_at_load_mm =
4.44444, max_deflection_mm = 4.8385, x_max = 2.73401 m   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('simply_supported_beam_point_load', L=4.0, P=1000.0, a=2.0,
E=1.0, I=1.0)
# -> RA = 500 N, RB = 500 N, M_max = 1000 N*m, V_max = 500 N, deflection_at_load_mm = 1.33333e+06,
max_deflection_mm = 1.33333e+06, x_max = 2 m   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Simply supported beam with point load: L = 6 m, P = 20 kN, a = 2 m, E = 200 GPa, I = 8e-5 m^4	SOLVED	RA = 13333.3 N, RB = 6666.67 N, M_max = 26666.7 N*m, V_max = 13333.3 N, deflection_at_load_mm = 4.44444, max_deflection_mm = 4.8385, x_max = 2.73401 m
Simply supported beam with point load: L = 6 m, P = 20 kN, a = 7 m, E = 200 GPa, I = 8e-5 m^4	INVALID_INPUT	Simply supported beam with a point load: the load position a must lie strictly between the supports (0 < a < L).
Simply supported beam with point load: L = 6 m, P = 20 kN, a = 2 m	NEEDS_INPUT	simply_supported_beam_point_load needs E, I (Young's modulus; second moment of area of the section). Example: solve('sim

Simply supported beam with a uniformly distributed load — `simply_supported_beam_udl`

Reactions, maximum bending moment and shear, maximum (midspan) deflection and end slope of a simply supported beam under a uniform load over its whole span.

$$M_{max} = \frac{wL^2}{8}, \quad \delta_{max} = \frac{5wL^4}{384EI}$$

input	meaning	condition
`L`	span	
`w`	uniform load per unit length (downward)	
`E`	Young's modulus	
`I`	second moment of area of the section	

output	meaning
`RA`	reaction at A
`RB`	reaction at B
`M_max`	maximum bending moment (midspan)
`V_max`	maximum shear force (at the supports)
`max_deflection_mm`	maximum deflection (midspan) in mm
`end_slope_deg`	slope at the supports in degrees

- **Method:** $R = wL/2$, M_max = wL^2/8, delta_max = 5wL^4/(384EI), theta_end = wL^3/(24EI)
- **Assumes:** L, w, E, $I > 0$; Euler-Bernoulli beam, small deflections
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('simply_supported_beam_udl', L=5.0, w=10000.0, E=200000000000.0,
I=4e-05)
# -> RA = 25000 N, RB = 25000 N, M_max = 31250 N*m, V_max = 25000 N, max_deflection_mm = 10.1725,
end_slope_deg = 0.373019   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Simply supported beam with uniform load: L = 5 m, w = 10 kN/m, E = 200 GPa, I = 4e-5 m^4	SOLVED	RA = 25000 N, RB = 25000 N, M_max = 31250 N*m, V_max = 25000 N, max_deflection_mm = 10.1725, end_slope_deg = 0.373019
Simply supported beam with UDL: L = 5 m, w = 10 kN/m, E = 200 GPa, I = 4e-5 m^4	SOLVED	RA = 25000 N, RB = 25000 N, M_max = 31250 N*m, V_max = 25000 N, max_deflection_mm = 10.1725, end_slope_deg = 0.373019
Simply supported beam with uniform load: L = -5 m, w = 10 kN/m, E = 200 GPa, I = 4e-5 m^4	INVALID_INPUT	Simply supported beam with a uniformly distributed load: L and w must be a positive number.

Simply supported beam with several point loads — `simply_supported_beam_point_loads`

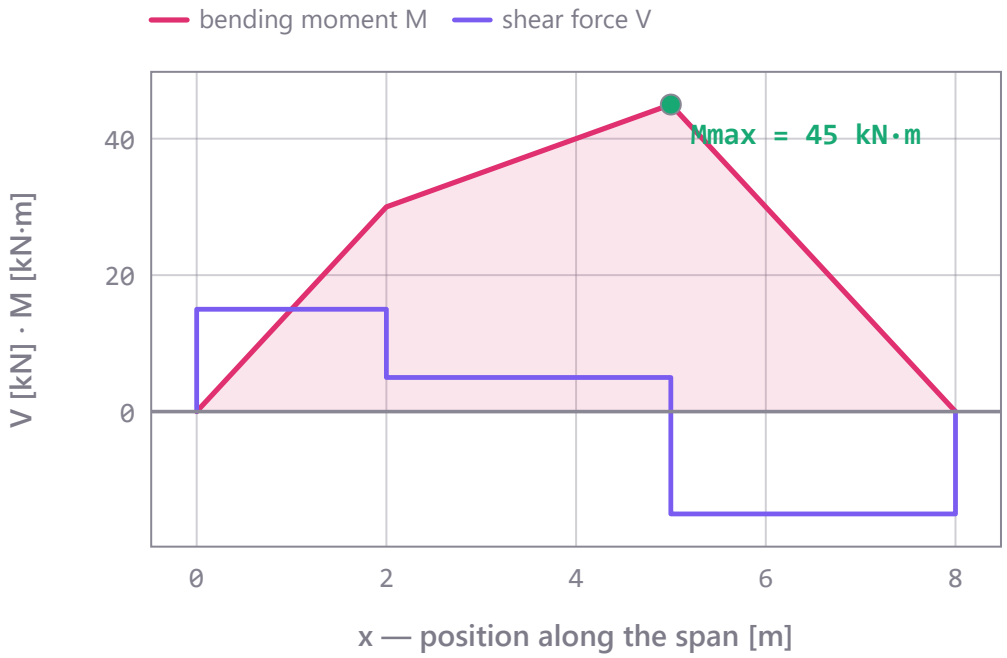
Support reactions, the maximum bending moment and its position, and the maximum shear for a simply supported beam carrying several downward point loads.

$$R_B = \frac{\sum P_i x_i}{L}, \quad R_A = \sum P_i - R_B, \quad M(x) = R_A x - \sum_{x_i < x} P_i (x - x_i)$$

input	meaning	condition
`L`	span	
`x`	load positions from the left support A, in m (a list)	
`P`	point loads in kN, downward (a list, same order as x)	

output	meaning
`RA`	reaction at A in kN
`RB`	reaction at B in kN
`M_max`	maximum bending moment in kNm
`x_Mmax`	position of the maximum moment from A, in m (the first one if several are equal)
`V_max`	maximum shear force in kN

- **Method:** moments about A: RB = sum P_i x_i / L, RA = sum P_i - RB; M(x) = RA x - sum P_i (x - x_i) for $x_i < x$, largest at a load point
- **Assumes:** $0 \le x_i \le L$; $P_i > 0$ (downward); x and P of equal length
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Shear-force and bending-moment diagrams of a simply supported beam of span 8 m with point loads 10, 20 kN at x = 2, 5 m, drawn from the reactions the package returned (RA = 15 kN, RB = 15 kN); marked: the maximum moment it returned, x = 5 m, M = 45 kN·m.

```
mechanical_engineering_ai999x.solve('simply_supported_beam_point_loads', L=8.0, x='2, 5', P='10, 20')
# -> RA = 15, RB = 15, M_max = 45, x_Mmax = 5, V_max = 15   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('simply_supported_beam_point_loads', L=6.0, x='3', P='12')
# -> RA = 6, RB = 6, M_max = 18, x_Mmax = 3, V_max = 6   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Simply supported beam with point loads: L = 8 m, x = [2, 5], P = [10, 20]	SOLVED	RA = 15, RB = 15, M_max = 45, x_Mmax = 5, V_max = 15
Simply supported beam with point loads: L = 8 m, x = [2, 9], P = [10, 20]	INVALID_INPUT	Simply supported beam with several point loads: every load position must lie on the span (0 <= x <= L).
Simply supported beam with point loads: L = 8 m, x = [2, 5]	NEEDS_INPUT	simply_supported_beam_point_loads needs P (point loads in kN, downward (a list, same order as x)). Example: solve('simpl

Cantilever with a point load — `cantilever_point_load`

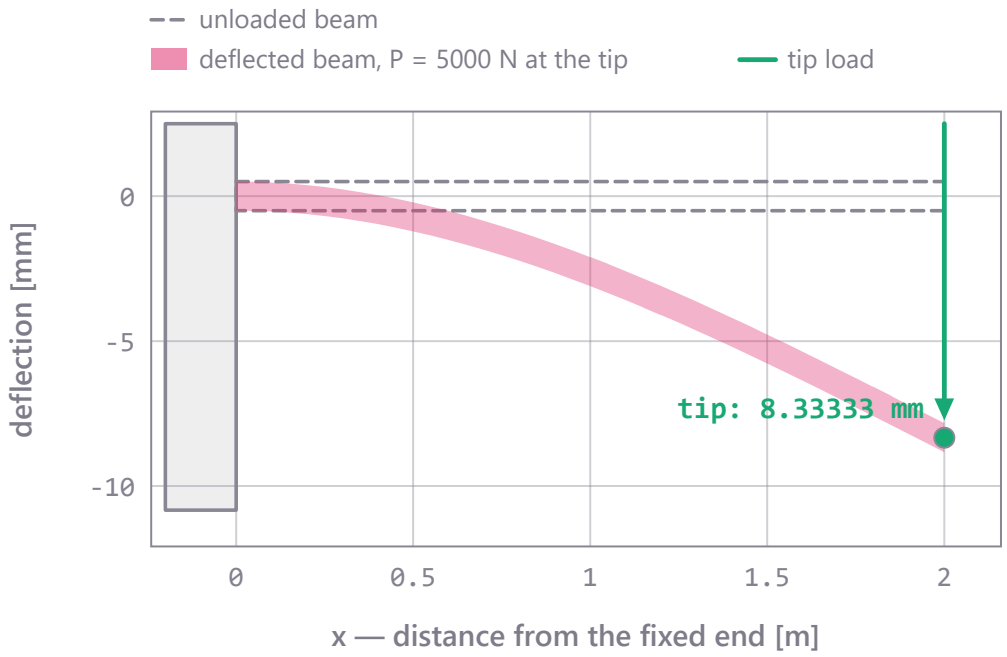
Reaction, fixed-end moment, deflection under the load, tip deflection and tip slope of a cantilever carrying one point load at distance a from the fixed end (default: at the free end).

$$\delta_{tip} = \frac{Pa^2(3L - a)}{6EI}, \quad \theta_{tip} = \frac{Pa^2}{2EI}$$

input	meaning	condition
`L`	length of the cantilever	
`P`	point load (downward)	
`E`	Young's modulus	
`I`	second moment of area	
`a`	distance of the load from the fixed end (default L: at the free end)	

output	meaning
`R`	vertical reaction at the fixed end
`M_fixed`	hogging moment at the fixed end
`deflection_at_load_mm`	deflection under the load in mm
`tip_deflection_mm`	deflection of the free end in mm
`tip_slope_deg`	slope at the free end in degrees

- **Method:** $M = Pa$, deflection under the load $Pa^3/(3EI)$, slope there $Pa^2/(2EI)$, tip deflection Pa^2 (3L - a)/(6EI)
- **Assumes:** $0 < a \leq L$; Euler-Bernoulli beam, small deflections
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Deflected shape of a cantilever (L = 2 m, E = 2.0000e+11 Pa, I = 8.0000e-06 m^4) under a tip load P = 5000 N: the deflection at each of 40 positions x is the tip deflection the package returns for the load placed at x (Maxwell's reciprocal theorem). Marked: x = 2 m, tip deflection = 8.33333 mm (downward); the wall and beam depth are drawn for orientation.

```
mechanical_engineering_ai999x.solve('cantilever_point_load', L=2.0, P=5000.0, E=200000000000.0,
I=8e-06)
# -> R = 5000 N, M_fixed = 10000 N*m, deflection_at_load_mm = 8.33333, tip_deflection_mm = 8.33333,
tip_slope_deg = 0.358099   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('cantilever_point_load', L=3.0, P=4000.0, a=2.0,
E=70000000000.0, I=2e-05)
# -> R = 4000 N, M_fixed = 8000 N*m, deflection_at_load_mm = 7.61905, tip_deflection_mm = 13.3333,
tip_slope_deg = 0.327404   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Cantilever with end load: L = 2 m, P = 5 kN, E = 200 GPa, I = 8e-6 m^4	SOLVED	R = 5000 N, M_fixed = 10000 N*m, deflection_at_load_mm = 8.33333, tip_deflection_mm = 8.33333, tip_slope_deg = 0.358099
Cantilever with point load: L = 3 m, P = 4 kN, a = 2 m, E = 70 GPa, I = 2e-5 m^4	SOLVED	R = 4000 N, M_fixed = 8000 N*m, deflection_at_load_mm = 7.61905, tip_deflection_mm = 13.3333, tip_slope_deg = 0.327404
Cantilever with point load: L = 3 m, P = 4 kN, a = 4 m, E = 70 GPa, I = 2e-5 m^4	INVALID_INPUT	Cantilever with a point load: the load position a (from the fixed end) must satisfy 0 < a <= L.

Cantilever with a uniformly distributed load — `cantilever_udl`

Reaction, fixed-end moment, tip deflection and tip slope of a cantilever under a uniform load over its whole length.

$$\delta_{tip} = \frac{wL^4}{8EI}, \quad \theta_{tip} = \frac{wL^3}{6EI}$$

input	meaning	condition
`L`	length of the cantilever	
`w`	uniform load per unit length (downward)	
`E`	Young's modulus	
`I`	second moment of area	

output	meaning
`R`	vertical reaction at the fixed end
`M_fixed`	hogging moment at the fixed end
`tip_deflection_mm`	deflection of the free end in mm
`tip_slope_deg`	slope at the free end in degrees

- **Method:** $R = wL$, $M = wL^2/2$, delta_tip = wL^4/(8EI), theta_tip = wL^3/(6EI)
- **Assumes:** L, w, E, $I > 0$; Euler-Bernoulli beam, small deflections
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('cantilever_udl', L=3.0, w=2000.0, E=200000000000.0, I=1e-05)
# -> R = 6000 N, M_fixed = 9000 N*m, tip_deflection_mm = 10.125, tip_slope_deg = 0.257831   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Cantilever with uniform load: L = 3 m, w = 2 kN/m, E = 200 GPa, I = 1e-5 m^4	SOLVED	R = 6000 N, M_fixed = 9000 N*m, tip_deflection_mm = 10.125, tip_slope_deg = 0.257831
Cantilever with UDL: L = 3 m, w = 2 kN/m, E = 200 GPa, I = 0 m^4	INVALID_INPUT	Cantilever with a uniformly distributed load: E and I must be a positive number.

Cantilever with a triangular (linearly varying) load — `cantilever_triangular_load`

Reaction, fixed-end moment, tip deflection and tip slope of a cantilever whose load falls linearly from w at the fixed end to zero at the free end.

$$\delta_{tip} = \frac{w_0 L^4}{30EI}, \quad \theta_{tip} = \frac{w_0 L^3}{24EI}$$

input	meaning	condition
`L`	length of the cantilever	
`w`	load intensity at the fixed end (downward)	
`E`	Young's modulus	
`I`	second moment of area	

output	meaning
`R`	vertical reaction at the fixed end
`M_fixed`	hogging moment at the fixed end
`tip_deflection_mm`	deflection of the free end in mm
`tip_slope_deg`	slope at the free end in degrees

- **Method:** $R = wL/2$, $M = wL^2/6$, delta_tip = wL^4/(30EI), theta_tip = wL^3/(24EI)
- **Assumes:** L, w, E, $I > 0$; maximum intensity at the fixed end; Euler-Bernoulli beam
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('cantilever_triangular_load', L=3.0, w=6000.0, E=200000000000.0,
I=2e-05)
# -> R = 9000 N, M_fixed = 9000 N*m, tip_deflection_mm = 4.05, tip_slope_deg = 0.0966866   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Cantilever with triangular load: L = 3 m, w = 6 kN/m, E = 200 GPa, I = 2e-5 m^4	SOLVED	R = 9000 N, M_fixed = 9000 N*m, tip_deflection_mm = 4.05, tip_slope_deg = 0.0966866
Cantilever with triangular load: L = 3 m, w = 6 kN/m	NEEDS_INPUT	cantilever_triangular_load needs E, I (Young's modulus; second moment of area). Example: solve('cantilever_triangular_lo

Fixed-fixed (built-in) beam with a uniformly distributed load — `fixed_fixed_beam_udl`

Reactions, fixed-end (hogging) moments, midspan (sagging) moment and maximum deflection of a beam built in at both ends under a uniform load.

$$M_{end} = \frac{wL^2}{12}, \quad M_{mid} = \frac{wL^2}{24}, \quad \delta_{max} = \frac{wL^4}{384EI}$$

input	meaning	condition
`L`	span	
`w`	uniform load per unit length (downward)	
`E`	Young's modulus	
`I`	second moment of area	

output	meaning
`RA`	reaction at A
`RB`	reaction at B
`M_end`	hogging moment at each fixed end
`M_mid`	sagging moment at midspan
`max_deflection_mm`	maximum (midspan) deflection in mm

- **Method:** $R = wL/2$, M_end = wL^2/12, M_mid = wL^2/24, delta_max = wL^4/(384EI)
- **Assumes:** L, w, E, $I > 0$; both ends fully fixed against rotation and deflection
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('fixed_fixed_beam_udl', L=6.0, w=12000.0, E=200000000000.0,
I=6e-05)
# -> RA = 36000 N, RB = 36000 N, M_end = 36000 N*m, M_mid = 18000 N*m, max_deflection_mm = 3.375
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Fixed-fixed beam with uniform load: L = 6 m, w = 12 kN/m, E = 200 GPa, I = 6e-5 m^4	SOLVED	RA = 36000 N, RB = 36000 N, M_end = 36000 N*m, M_mid = 18000 N*m, max_deflection_mm = 3.375
Built-in beam with UDL: L = 6 m, w = 12 kN/m, E = 200 GPa, I = 6e-5 m^4	SOLVED	RA = 36000 N, RB = 36000 N, M_end = 36000 N*m, M_mid = 18000 N*m, max_deflection_mm = 3.375

Fixed-fixed (built-in) beam with a point load — `fixed_fixed_beam_point_load`

Reactions, fixed-end moments, the moment under the load and the deflection under the load of a beam built in at both ends carrying one point load.

$$M_A = \frac{Pab^2}{L^2}, \quad M_B = \frac{Pa^2b}{L^2}, \quad \delta_P = \frac{Pa^3b^3}{3EIL^3}$$

input	meaning	condition
`L`	span	
`P`	point load (downward)	
`a`	distance of the load from end A	

input	meaning	condition
`E`	Young's modulus	
`I`	second moment of area	

output	meaning
`RA`	reaction at A
`RB`	reaction at B
`MA`	hogging moment at A
`MB`	hogging moment at B
`M_load`	sagging moment under the load
`deflection_at_load_mm`	deflection under the load in mm

- **Method:** MA = Pab^2/L^2, $MB = Pa^{2}b/L^{2}$, $RA = Pb^{2}$(3a + b)/L^3, $RB = Pa^{2}$(a + 3b)/L^3, M_load = 2Pa^2b^2/L^3, delta_load = Pa^3b^3/(3EIL^3)
- **Assumes:** $0 < a < L$; both ends fully fixed
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('fixed_fixed_beam_point_load', L=6.0, P=30000.0, a=2.0,
E=200000000000.0, I=5e-05)
# -> RA = 22222.2 N, RB = 7777.78 N, MA = 26666.7 N*m, MB = 13333.3 N*m, M_load = 17777.8 N*m,
deflection_at_load_mm = 2.37037   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Fixed-fixed beam with point load: L = 6 m, P = 30 kN, a = 2 m, E = 200 GPa, I = 5e-5 m^4	SOLVED	RA = 22222.2 N, RB = 7777.78 N, MA = 26666.7 N*m, MB = 13333.3 N*m, M_load = 17777.8 N*m, deflection_at_load_mm = 2.37037
Fixed-fixed beam with point load: L = 6 m, P = 30 kN, a = 0 m, E = 200 GPa, I = 5e-5 m^4	INVALID_INPUT	Fixed-fixed (built-in) beam with a point load: the load position a must lie strictly between the ends (0 < a < L).

Propped cantilever with a uniformly distributed load — `propped_cantilever_udl`

Reactions, fixed-end moment, maximum span moment and its position, and the maximum deflection with its position, for a beam fixed at A and simply supported at B under a uniform load.

$$R_B = \frac{3wL}{8}, \quad M_A = \frac{wL^2}{8}, \quad \delta_{max} \approx \frac{wL^4}{185EI}$$

input	meaning	condition
`L`	span	
`w`	uniform load per unit length (downward)	
`E`	Young's modulus	
`I`	second moment of area	

output	meaning
`RA`	reaction at the fixed end A
`RB`	reaction at the prop B
`MA`	hogging moment at A
`M_span`	largest sagging moment in the span
`x_M_span`	its position from A
`max_deflection_mm`	maximum deflection in mm
`x_max`	position of the maximum deflection from A

- **Method:** $RB = 3wL/8$, $RA = 5wL/8$, $MA = wL^2/8$, M_span = 9wL^2/128 at $5L/8$, delta_max = (39 + 55 sqrt(33)) wL^4/(65536 EI) at $x = L(15 - sqrt(33))/16$
- **Assumes:** L, w, E, $I > 0$; A fully fixed, B a simple support
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('propped_cantilever_udl', L=8.0, w=10000.0, E=200000000000.0,
I=0.0001)
# -> RA = 50000 N, RB = 30000 N, MA = 80000 N*m, M_span = 45000 N*m, x_M_span = 5 m,
max_deflection_mm = 11.0922, x_max = 4.62772 m   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Propped cantilever with uniform load: L = 8 m, w = 10 kN/m, E = 200 GPa, I = 1e-4 m^4	SOLVED	RA = 50000 N, RB = 30000 N, MA = 80000 N*m, M_span = 45000 N*m, x_M_span = 5 m, max_deflection_mm = 11.0922, x_max = 4.62772 m

Propped cantilever with a point load — `propped_cantilever_point_load`

Reactions, fixed-end moment, moment under the load and deflection under the load of a beam fixed at A and simply supported at B with one point load.

$$R_B = \frac{Pa^2(3L - a)}{2L^3}, \quad M_A = \frac{Pab(L + b)}{2L^2}$$

input	meaning	condition
`L`	span	
`P`	point load (downward)	
`a`	distance of the load from the fixed end A	
`E`	Young's modulus	
`I`	second moment of area	

output	meaning
`RA`	reaction at the fixed end A
`RB`	reaction at the prop B
`MA`	hogging moment at A

output	meaning
`M_load`	sagging moment under the load
`deflection_at_load_mm`	deflection under the load in mm

- **Method:** $RB = Pa^2$(3L - a)/(2L^3), $RA = P - RB$, MA = Pab(L + b)/(2L^2), M_load = RB b, delta_load = Pa^3b^2(3L + b)/(12EIL^3)
- **Assumes:** $0 < a < L$; A fully fixed, B a simple support
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('propped_cantilever_point_load', L=4.0, P=16000.0, a=2.0,
E=200000000000.0, I=2e-05)
# -> RA = 11000 N, RB = 5000 N, MA = 12000 N*m, M_load = 10000 N*m, deflection_at_load_mm = 2.33333
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Propped cantilever with point load: L = 4 m, P = 16 kN, a = 2 m, E = 200 GPa, I = 2e-5 m^4	SOLVED	RA = 11000 N, RB = 5000 N, MA = 12000 N*m, M_load = 10000 N*m, deflection_at_load_mm = 2.33333

Section properties of a solid rectangle — `rectangular_section_properties`

Area, second moments of area, elastic section moduli, radii of gyration and polar moment of a solid rectangle b wide and h deep.

$$I_x = \frac{bh^3}{12}, \quad Z_x = \frac{bh^2}{6}, \quad r_x = \sqrt{I_x / A}$$

input	meaning	condition
`b`	width (horizontal)	
`h`	depth (vertical)	

output	meaning
`A`	cross-sectional area
`Ix`	second moment of area about the horizontal centroidal axis
`Iy`	second moment of area about the vertical centroidal axis
`Zx`	elastic section modulus about x (Ix / distance to the extreme fibre)
`Zy`	elastic section modulus about y
`rx`	radius of gyration about x
`ry`	radius of gyration about y
`J_polar`	polar second moment about the centroid (Ix + Iy; not the torsion constant)

- **Method:** $A = bh$, $Ix = bh^3/12$, $Iy = hb^3/12$, $Zx = bh^2/6$, $Zy = hb^2/6$, $r = \sqrt{\frac{I}{A}}$
- **Assumes:** b, $h > 0$

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('rectangular_section_properties', b=0.1, h=0.2)
# -> A = 0.02 m^2, Ix = 6.66667e-05 m^4, Iy = 1.66667e-05 m^4, Zx = 0.000666667 m^3, Zy =
0.000333333 m^3, rx = 0.057735 m, ry = 0.0288675 m, J_polar = 8.33333e-05 m^4   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Rectangular section: b = 100 mm, h = 200 mm	SOLVED	A = 0.02 m^2, Ix = 6.66667e-05 m^4, Iy = 1.66667e-05 m^4, Zx = 0.000666667 m^3, Zy = 0.000333333 m^3, rx = 0.057735 m, ry = 0.0288675 m, J_polar = 8.33333e-05 m^4
Rectangular section: b = 100 mm, h = 0 mm	INVALID_INPUT	Section properties of a solid rectangle: b and h must be a positive number.

Section properties of a solid circle — `circular_section_properties`

Area, second moment of area, polar moment, elastic and plastic section moduli and radius of gyration of a solid circular section.

$$I = \frac{\pi d^4}{64}, \quad J = \frac{\pi d^4}{32}, \quad Z = \frac{\pi d^3}{32}$$

input	meaning	condition
`d`	diameter	

output	meaning
`A`	area
`I`	second moment of area about a diameter
`J`	polar second moment of area (torsion constant of a circle)
`Z`	elastic section modulus
`Zp`	plastic section modulus
`r`	radius of gyration

- **Method:** $A = pi\ d^2/4$, $I = pi\ d^4/64$, $J = pi\ d^4/32$, $Z = pi\ d^3/32$, $Zp = d^3/6$, $r = d/4$
- **Assumes:** $d > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('circular_section_properties', d=0.05)
# -> A = 0.0019635 m^2, I = 3.06796e-07 m^4, J = 6.13592e-07 m^4, Z = 1.22718e-05 m^3, Zp =
2.08333e-05 m^3, r = 0.0125 m   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Circular section: d = 50 mm	SOLVED	A = 0.0019635 m^2, I = 3.06796e-07 m^4, J = 6.13592e-07 m^4, Z = 1.22718e-05 m^3, Zp = 2.08333e-05 m^3, r = 0.0125 m

Section properties of a hollow circle (tube) — `hollow_circular_section_properties`

Area, second moment of area, polar moment, elastic and plastic section moduli and radius of gyration of a circular tube.

$$I = \frac{\pi(D^4 - d^4)}{64}, \quad J = \frac{\pi(D^4 - d^4)}{32}$$

input	meaning	condition
`D`	outer diameter	
`d`	inner diameter	

output	meaning
`A`	area
`I`	second moment of area about a diameter
`J`	polar second moment of area (torsion constant)
`Z`	elastic section modulus
`Zp`	plastic section modulus
`r`	radius of gyration

- **Method:** $A = pi$(D^2 - d^2)/4, $I = pi$(D^4 - d^4)/64, $J = 2I$, $Z = 2I/D$, $Zp = (D^3 - d^3)/6$, $r = \frac{\sqrt{D^2+d^2}}{4}$
- **Assumes:** $D > d \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('hollow_circular_section_properties', D=0.06, d=0.04)
# -> A = 0.0015708 m^2, I = 5.10509e-07 m^4, J = 1.02102e-06 m^4, Z = 1.7017e-05 m^3, Zp = 2.53333e-05 m^3, r = 0.0180278 m   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Hollow circular section: D = 60 mm, d = 40 mm	SOLVED	A = 0.0015708 m^2, I = 5.10509e-07 m^4, J = 1.02102e-06 m^4, Z = 1.7017e-05 m^3, Zp = 2.53333e-05 m^3, r = 0.0180278 m
Hollow circular section: D = 40 mm, d = 60 mm	INVALID_INPUT	Section properties of a hollow circle (tube): the inner diameter must be smaller than the outer diameter.

Section properties of a hollow rectangle (box) — `hollow_rectangular_section_properties`

Area, second moments, elastic section moduli, radii of gyration and the plastic modulus about x of a rectangular hollow section of uniform wall thickness.

$$I_x = \frac{BH^3 - bh^3}{12}, \quad Z_{p,x} = \frac{BH^2 - bh^2}{4}$$

input	meaning	condition
`B`	outer width	
`H`	outer depth	
`t`	wall thickness	

output	meaning
`A`	cross-sectional area
`Ix`	second moment of area about the horizontal centroidal axis
`Iy`	second moment of area about the vertical centroidal axis
`Zx`	elastic section modulus about x (Ix / distance to the extreme fibre)
`Zy`	elastic section modulus about y
`rx`	radius of gyration about x
`ry`	radius of gyration about y
`Zpx`	plastic section modulus about x

- **Method:** outer rectangle minus inner rectangle: $Ix = (BH^3 - bh^3)/12$ with $b = B - 2t$, $h = H - 2t$; Zpx = (BH^2 - bh^2)/4
- **Assumes:** B, H, $t > 0$; 2t < min(B, H); square corners (no corner radii)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('hollow_rectangular_section_properties', B=0.1, H=0.2, t=0.01)
# -> A = 0.0056 m^2, Ix = 2.77867e-05 m^4, Iy = 8.98667e-06 m^4, Zx = 0.000277867 m^3, Zy =
0.000179733 m^3, rx = 0.0704408 m, ry = 0.0400595 m, Zpx = 0.000352 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Box section: B = 100 mm, H = 200 mm, t = 10 mm	SOLVED	A = 0.0056 m^2, Ix = 2.77867e-05 m^4, Iy = 8.98667e-06 m^4, Zx = 0.000277867 m^3, Zy = 0.000179733 m^3, rx = 0.0704408 m, ry = 0.0400595 m, Zpx = 0.000352 m^3
Box section: B = 100 mm, H = 200 mm, t = 60 mm	INVALID_INPUT	Section properties of a hollow rectangle (box): the wall thickness must be less than half the smaller outer dimension.

Section properties of a symmetric I-section — `i_section_properties`

Area, second moments, elastic and plastic section moduli, radii of gyration and the shape factor of a doubly symmetric I-section (flanges b x tf, web tw, overall depth h, no fillets).

$$I_x = \frac{bh^3 - (b - t_w)(h - 2t_f)^3}{12}, \quad Z_{p,x} = bt_f(h - t_f) + \frac{t_w(h - 2t_f)^2}{4}$$

input	meaning	condition
`b`	flange width	
`h`	overall depth	
`tf`	flange thickness	
`tw`	web thickness	

output	meaning
`A`	cross-sectional area
`Ix`	second moment of area about the horizontal centroidal axis
`Iy`	second moment of area about the vertical centroidal axis
`Zx`	elastic section modulus about x (Ix / distance to the extreme fibre)
`Zy`	elastic section modulus about y
`rx`	radius of gyration about x
`ry`	radius of gyration about y
`Zpx`	plastic section modulus about x
`Zpy`	plastic section modulus about y
`shape_factor_x`	shape factor Zpx / Zx

- **Method:** three rectangles with the parallel-axis theorem: Ix = sum($bh^3/12 + A$ d^2); plastic moduli about the axes of symmetry: Zpx = b tf (h - tf) + tw $(h - 2tf)^2/4$, Zpy = tf $b^2/2 + (h - 2tf)\,tw^2/4$
- **Assumes:** 2 $tf < h$; $tw < b$; no root fillets
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('i_section_properties', b=0.15, h=0.3, tf=0.012, tw=0.008)
# -> A = 0.005808 m^2, Ix = 8.87092e-05 m^4, Iy = 6.76178e-06 m^4, Zx = 0.000591395 m^3, Zy =
9.0157e-05 m^3, rx = 0.123586 m, ry = 0.0341206 m, Zpx = 0.000670752 m^3, Zpy = 0.000139416 m^3,
shape_factor_x = 1.13419   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
I-section: b = 150 mm, h = 300 mm, tf = 12 mm, tw = 8 mm	SOLVED	A = 0.005808 m^2, Ix = 8.87092e-05 m^4, Iy = 6.76178e-06 m^4, Zx = 0.000591395 m^3, Zy = 9.0157e-05 m^3, rx = 0.123586 m, ry = 0.0341206 m, Zpx = 0.000670752 m^3, Zpy = 0.000139416 m^3, shape_factor_x = 1.13419
I-section: b = 150 mm, h = 300 mm, tf = 160 mm, tw = 8 mm	INVALID_INPUT	Section properties of a symmetric I-section: the flanges (2 tf) must be thinner than the overall depth h.

Section properties of a T-section — `t_section_properties`

Area, centroid height, second moments, top and bottom section moduli and radii of gyration of a T-section (flange on top).

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i}, \quad I_x = \sum \left(\frac{b_i h_i^3}{12} + A_i (y_i - \bar{y})^2 \right)$$

input	meaning	condition
`bf`	flange width	
`tf`	flange thickness	
`h`	overall depth	
`tw`	stem (web) thickness	

output	meaning
`A`	area
`y_bar`	height of the centroid above the bottom of the stem
`Ix`	second moment about the horizontal centroidal axis
`Iy`	second moment about the vertical axis of symmetry
`Z_top`	section modulus for the top fibre
`Z_bottom`	section modulus for the bottom fibre
`rx`	radius of gyration about x
`ry`	radius of gyration about y

- **Method:** two rectangles: y_bar = sum(A y)/sum(A); Ix = sum($bh^3/12 + A$ (y - y_bar)^2); Z = Ix / distance to the fibre
- **Assumes:** $tf < h$; $tw \leq bf$; no fillets
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('t_section_properties', bf=0.12, tf=0.02, h=0.15, tw=0.015)
# -> A = 0.00435 m^2, y_bar = 0.106379 m, Ix = 8.87797e-06 m^4, Iy = 2.91656e-06 m^4, Z_top =
0.000203527 m^3, Z_bottom = 8.34558e-05 m^3, rx = 0.0451765 m, ry = 0.0258935 m   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
T-section: bf = 120 mm, tf = 20 mm, h = 150 mm, tw = 15 mm	SOLVED	A = 0.00435 m^2, y_bar = 0.106379 m, Ix = 8.87797e-06 m^4, Iy = 2.91656e-06 m^4, Z_top = 0.000203527 m^3, Z_bottom = 8.34558e-05 m^3, rx = 0.0451765 m, ry = 0.0258935 m
T-section: bf = 120 mm, tf = 200 mm, h = 150 mm, tw = 15 mm	INVALID_INPUT	Section properties of a T-section: the flange thickness must be less than the overall depth.

Section properties of a channel (C) section — `channel_section_properties`

Area, centroid distance from the back of the web, second moments, section moduli and radii of gyration of a channel with square corners.

$$\bar{x} = \frac{\sum A_i x_i}{\sum A_i}, \quad I_y = \sum \left(\frac{h_i b_i^3}{12} + A_i (x_i - \bar{x})^2 \right)$$

input	meaning	condition
`b`	flange width (including the web)	

input	meaning	condition
`h`	overall depth	
`tf`	flange thickness	
`tw`	web thickness	

output	meaning
`A`	area
`x_bar`	distance of the centroid from the back of the web
`Ix`	second moment about the horizontal axis of symmetry
`Iy`	second moment about the vertical centroidal axis
`Zx`	section modulus about x
`Zy_min`	smaller section modulus about y (flange tips)
`rx`	radius of gyration about x
`ry`	radius of gyration about y

- **Method:** three rectangles (two flanges b x tf, web tw x $(h - 2tf)$) with the parallel-axis theorem
- **Assumes:** 2 $tf < h$; $tw < b$; square corners, no taper
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('channel_section_properties', b=0.08, h=0.2, tf=0.01, tw=0.007)
# -> A = 0.00286 m^2, x_bar = 0.0239196 m, Ix = 1.78553e-05 m^4, Iy = 1.79757e-06 m^4, Zx =
0.000178553 m^3, Zy_min = 3.20535e-05 m^3, rx = 0.0790134 m, ry = 0.0250704 m   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Channel section: b = 80 mm, h = 200 mm, tf = 10 mm, tw = 7 mm	SOLVED	A = 0.00286 m^2, x_bar = 0.0239196 m, Ix = 1.78553e-05 m^4, Iy = 1.79757e-06 m^4, Zx = 0.000178553 m^3, Zy_min = 3.20535e-05 m^3, rx = 0.0790134 m, ry = 0.0250704 m

Section properties of an angle (L) section, with principal axes — `angle_section_properties`

Area, centroid, centroidal second moments and product of inertia, principal second moments, the principal angle and the least radius of gyration of an equal or unequal angle.

$$I_{1,2} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}, \quad \tan 2\varphi = \frac{-2I_{xy}}{I_x - I_y}$$

input	meaning	condition
`b`	horizontal leg length	
`h`	vertical leg length	
`t`	thickness (both legs)	

output	meaning
`A`	area

output	meaning
`x_bar`	centroid distance from the back of the vertical leg
`y_bar`	centroid distance from the bottom of the horizontal leg
`Ix`	centroidal second moment about x
`Iy`	centroidal second moment about y
`Ixy`	centroidal product of inertia (integral of x y dA)
`I1`	major principal second moment
`I2`	minor principal second moment
`principal_angle_deg`	angle of the major principal axis from x, degrees, in (-90, 90]
`r_min`	least radius of gyration

- **Method:** two rectangles with the parallel-axis theorem (also for Ixy); I1,$\mathbf{2} = (Ix + Iy)/2$ +- sqrt(((Ix - Iy)/2)^2 + Ixy^2), tan(2 phi) = -2 Ixy/(Ix - Iy)
- **Assumes:** $t < b$ and $t < h$; square corners, no root radius
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('angle_section_properties', b=0.1, h=0.1, t=0.01)
# -> A = 0.0019 m^2, x_bar = 0.0286842 m, y_bar = 0.0286842 m, Ix = 1.80004e-06 m^4, Iy = 1.80004e-06 m^4, Ixy = -1.06579e-06 m^4, I1 = 2.86583e-06 m^4, I2 = 7.34254e-07 m^4, principal_angle_deg = 45, r_min = 0.0196583 m   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Angle section: b = 100 mm, h = 100 mm, t = 10 mm	SOLVED	A = 0.0019 m^2, x_bar = 0.0286842 m, y_bar = 0.0286842 m, Ix = 1.80004e-06 m^4, Iy = 1.80004e-06 m^4, Ixy = -1.06579e-06 m^4, I1 = 2.86583e-06 m^4, I2 = 7.34254e-07 m^4, principal_angle_deg = 45, r_min = 0.0196583 m
Angle section: b = 100 mm, h = 150 mm, t = 200 mm	INVALID_INPUT	Section properties of an angle (L) section, with principal axes: the thickness must be less than both leg lengths.

Composite section of rectangles (parallel-axis theorem) — `composite_section_properties`

Area, centroid, centroidal second moments, product of inertia and principal second moments of a section built from non-overlapping rectangles.

$$I_x = \sum_i \left(\frac{b_i h_i^3}{12} + A_i (y_i - \bar{y})^2 \right), \quad I_{xy} = \sum_i A_i (x_i - \bar{x})(y_i - \bar{y})$$

input	meaning	condition
`rects`	one row per rectangle: width b, height h, centre x, centre y, all in mm	

output	meaning
`A`	area in mm^2
`x_bar`	centroid x in mm

output	meaning
`y_bar`	centroid y in mm
`Ix`	centroidal second moment about x in mm^4
`Iy`	centroidal second moment about y in mm^4
`Ixy`	centroidal product of inertia in mm^4
`I1`	major principal second moment in mm^4
`I2`	minor principal second moment in mm^4

- **Method:** parallel-axis theorem: x_bar = sum A x / sum A, Ix = sum($bh^3/12 + A$ (y - y_bar)^2), Ixy = sum A (x - x_bar)(y - y_bar)
- **Assumes:** rectangles do not overlap (touching is allowed); 1 to 20 rectangles, all dimensions in mm
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('composite_section_properties', rects='200 20 0 10; 20 300 0
170')
# -> A = 10000, x_bar = 0, y_bar = 106, Ix = 1.06573e+08, Iy = 1.35333e+07, Ixy = 0, I1 =
1.06573e+08, I2 = 1.35333e+07   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Composite section: rects = [200 20 0 10; 20 300 0 170]	SOLVED	A = 10000, x_bar = 0, y_bar = 106, Ix = 1.06573e+08, Iy = 1.35333e+07, Ixy = 0, I1 = 1.06573e+08, I2 = 1.35333e+07
Composite section: rects = [200 20 0 10; 20 300 0 15]	INVALID_INPUT	Composite section of rectangles (parallel-axis theorem): rectangles 1 and 2 overlap: the parts of a composite section mu

Section properties of a polygonal section — `polygon_section_properties`

Area, centroid, centroidal second moments, product of inertia and principal second moments of any simple polygon given by its vertices.

$$A = \tfrac{1}{2}\sum c_i, \quad I_x = \tfrac{1}{12}\sum c_i\,(y_i^2 + y_i y_{i+1} + y_{i+1}^2) - A\bar{y}^2$$

input	meaning	condition
`vertices`	vertices in order (either sense), one row 'x y' per vertex, in mm	

output	meaning
`A`	area in mm^2
`x_bar`	centroid x in mm
`y_bar`	centroid y in mm
`Ix`	centroidal second moment about x in mm^4
`Iy`	centroidal second moment about y in mm^4
`Ixy`	centroidal product of inertia in mm^4
`I1`	major principal second moment in mm^4

output	meaning
`I2`	minor principal second moment in mm^4

- **Method:** Green's theorem over the edges: $A = 1/2$ sum c_i, $Ix = 1/12$ sum c_i (y_i^2 + y_i y_i+1 + y_i+1^2) with $c_i = x_i\ y_i + 1 - x_i + 1$ y_i; then shifted to the centroid
- **Assumes:** 3 to 60 distinct vertices; a simple polygon (edges do not cross); no holes
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('polygon_section_properties', vertices='0 0; 100 0; 100 200; 0
200')
# -> A = 20000, x_bar = 50, y_bar = 100, Ix = 6.66667e+07, Iy = 1.66667e+07, Ixy = 0, I1 =
6.66667e+07, I2 = 1.66667e+07   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('polygon_section_properties', vertices='0 0; 0 30; 40 0')
# -> A = 600, x_bar = 13.3333, y_bar = 10, Ix = 30000, Iy = 53333.3, Ixy = -20000, I1 = 64820.7, I2
= 18512.6   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Polygon section: vertices = [0 0; 100 0; 100 200; 0 200]	SOLVED	A = 20000, x_bar = 50, y_bar = 100, Ix = 6.66667e+07, Iy = 1.66667e+07, Ixy = 0, I1 = 6.66667e+07, I2 = 1.66667e+07
Polygon section: vertices = [0 0; 100 100; 100 0; 0 100]	INVALID_INPUT	Section properties of a polygonal section: the polygon must be simple (its edges must not cross).

Torsion of a solid circular shaft — `solid_shaft_torsion`

Maximum shear stress, polar moment and angle of twist of a solid circular shaft under a torque.

$$\tau_{max} = \frac{16T}{\pi d^3}, \quad \varphi = \frac{TL}{GJ}, \quad J = \frac{\pi d^4}{32}$$

input	meaning	condition
`T`	applied torque	
`d`	diameter	
`L`	length	
`G`	shear modulus	

output	meaning
`tau_max_MPa`	maximum shear stress (outer surface) in MPa
`tau_inner_MPa`	shear stress at the inner surface in MPa (0 for a solid shaft)
`twist_rad`	angle of twist
`twist_deg`	angle of twist in degrees
`J`	polar second moment of area

- **Method:** $J = pi\ d^4/32$, tau_max = T (d/2)/J = 16T/(pi d^3), $\phi = TL/(GJ)$

- **Assumes:** T, d, L, $G > 0$; linear elastic, circular section, uniform torque
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('solid_shaft_torsion', T=1200.0, d=0.05, L=1.5, G=80000000000.0)
# -> tau_max_MPa = 48.8924, tau_inner_MPa = 0, twist_rad = 0.0366693 rad, twist_deg = 2.101, J =
6.13592e-07 m^4   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solid shaft torsion: T = 1.2 kN*m, d = 50 mm, L = 1.5 m, G = 80 GPa	SOLVED	tau_max_MPa = 48.8924, tau_inner_MPa = 0, twist_rad = 0.0366693 rad, twist_deg = 2.101, J = 6.13592e-07 m^4
Solid shaft torsion: T = 1.2 kN*m, d = 50 mm, L = 1.5 m	NEEDS_INPUT	solid_shaft_torsion needs G (shear modulus). Example: solve('solid_shaft_torsion', T='1200 N*m', d='0.05 m', L='1.5 m',

Torsion of a hollow circular shaft — `hollow_shaft_torsion`

Shear stresses at the outer and inner surfaces, polar moment and angle of twist of a hollow circular shaft under a torque.

$$\tau_{max} = \frac{16TD}{\pi(D^4 - d^4)}, \quad \varphi = \frac{TL}{GJ}$$

input	meaning	condition
`T`	applied torque	
`D`	outer diameter	
`d`	inner diameter	
`L`	length	
`G`	shear modulus	

output	meaning
`tau_max_MPa`	maximum shear stress (outer surface) in MPa
`tau_inner_MPa`	shear stress at the inner surface in MPa (0 for a solid shaft)
`twist_rad`	angle of twist
`twist_deg`	angle of twist in degrees
`J`	polar second moment of area

- **Method:** $J = pi\,(D^4 - d^4)/32$, tau_max = T $(D/2)/J$, tau_inner = T $(d/2)/J$, $\phi = TL/(GJ)$
- **Assumes:** $D > d > 0$; linear elastic
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('hollow_shaft_torsion', T=2000.0, D=0.06, d=0.04, L=2.0,
G=80000000000.0)
# -> tau_max_MPa = 58.7649, tau_inner_MPa = 39.1766, twist_rad = 0.0489708 rad, twist_deg = 2.80582,
J = 1.02102e-06 m^4   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Hollow shaft torsion: T = 2 kN*m, D = 60 mm, d = 40 mm, L = 2 m, G = 80 GPa	SOLVED	tau_max_MPa = 58.7649, tau_inner_MPa = 39.1766, twist_rad = 0.0489708 rad, twist_deg = 2.80582, J = 1.02102e-06 m^4
Hollow shaft torsion: T = 2 kN*m, D = 40 mm, d = 60 mm, L = 2 m, G = 80 GPa	INVALID_INPUT	Torsion of a hollow circular shaft: the inner diameter must be smaller than the outer diameter.

Torque and minimum diameter of a power-transmission shaft — `shaft_power_transmission`

The torque a solid shaft transmits at a given power and speed, and the smallest diameter that keeps the torsional shear stress within an allowable value.

$$T = \frac{P}{\omega}, \quad d_{min} = \left(\frac{16T}{\pi \tau_{allow}}\right)^{1/3}$$

input	meaning	condition
`P`	transmitted power	
`n`	rotational speed (rpm or rad/s)	
`tau_allow`	allowable shear stress	

output	meaning
`T`	transmitted torque
`d_min_mm`	minimum solid-shaft diameter in mm
`frequency_Hz`	rotational frequency in Hz

- **Method:** $T = P/\omega$, d_min = (16T/(pi tau_allow))^$(1/3)$
- **Assumes:** P, n, tau_allow > 0; pure torsion (no bending), solid circular shaft
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('shaft_power_transmission', P=50000.0, n=157.07963267948966,
tau_allow=40000000.0)
# -> T = 318.31 N*m, d_min_mm = 34.3495, frequency_Hz = 25   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Shaft power transmission: P = 50 kW, n = 1500 rpm, tau_allow = 40 MPa	SOLVED	T = 318.31 N*m, d_min_mm = 34.3495, frequency_Hz = 25
Shaft power transmission: P = 50 kW, n = 1500 rpm, tau_allow = 0 MPa	INVALID_INPUT	Torque and minimum diameter of a power-transmission shaft: P, n and the allowable shear stress must be a positive number

Angle of twist of a stepped shaft — `stepped_shaft_twist`

Total angle of twist and the largest shear stress of a shaft made of solid circular segments in series carrying one torque.

$$\varphi = \sum_i \frac{T L_i}{G J_i}, \quad J_i = \frac{\pi d_i^4}{32}$$

input	meaning	condition
`T`	torque carried by every segment	
`lengths`	segment lengths in mm (a list)	
`diameters`	segment diameters in mm (a list, same order)	
`G`	shear modulus	

output	meaning
`twist_rad`	total angle of twist
`twist_deg`	total angle of twist in degrees
`tau_max_MPa`	largest shear stress (smallest segment) in MPa

- **Method:** phi = sum T L_i/(G J_i), $J_i = pi\ d_i^4/32$, tau_max = max 16T/(pi d_i^3)
- **Assumes:** equal-length lists; the same torque in every segment; stress concentrations at the steps not included
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('stepped_shaft_twist', T=500.0, lengths='400, 600',
diameters='40, 30', G=80000000000.0)
# -> twist_rad = 0.0571042 rad, twist_deg = 3.27183, tau_max_MPa = 94.314   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Stepped shaft: T = 500 N*m, lengths = [400, 600], diameters = [40, 30], G = 80 GPa	SOLVED	twist_rad = 0.0571042 rad, twist_deg = 3.27183, tau_max_MPa = 94.314
Stepped shaft: T = 500 N*m, lengths = [400, 600], diameters = [40], G = 80 GPa	INVALID_INPUT	Angle of twist of a stepped shaft: lengths and diameters must have the same number of entries.

Shaft fixed at both ends with a torque between them — `statically_indeterminate_shaft`

The reaction torques at both fixed ends, the shear stress in each part and the rotation at the loaded section of a shaft built in at both ends.

$$T_A = T_0 \frac{b/J_2}{a/J_1 + b/J_2}, \quad T_B = T_0 - T_A$$

input	meaning	condition
`T0`	torque applied between the ends	
`a`	length of part 1 (from end A to the load)	
`b`	length of part 2 (from the load to end B)	
`d1`	diameter of part 1	

input	meaning	condition
`d2`	diameter of part 2	
`G`	shear modulus	

output	meaning
`TA`	reaction torque at A
`TB`	reaction torque at B
`tau1_MPa`	maximum shear stress in part 1, MPa
`tau2_MPa`	maximum shear stress in part 2, MPa
`twist_at_load_deg`	rotation of the loaded section in degrees

- **Method:** compatibility: TA a/(G J1) = TB b/(G J2), equilibrium $TA + TB = T_0$
- **Assumes:** both ends fully fixed; solid circular parts of one material
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('statically_indeterminate_shaft', T0=1000.0, a=0.5, b=1.5,
d1=0.05, d2=0.05, G=80000000000.0)
# -> TA = 750 N*m, TB = 250 N*m, tau1_MPa = 30.5577, tau2_MPa = 10.1859, twist_at_load_deg =
0.437708   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Statically indeterminate shaft: T0 = 1 kN*m, a = 0.5 m, b = 1.5 m, d1 = 50 mm, d2 = 50 mm, G = 80 GPa	SOLVED	TA = 750 N*m, TB = 250 N*m, tau1_MPa = 30.5577, tau2_MPa = 10.1859, twist_at_load_deg = 0.437708

Stepped bar under axial loads at its joints — `axial_bar_segments`

Internal axial force, normal stress of every segment and the total elongation of a bar fixed at its left end, made of segments in series, with axial loads at the joints.

$$\delta = \sum_i \frac{N_i L_i}{A_i E}, \quad \sigma_i = \frac{N_i}{A_i}$$

input	meaning	condition
`lengths`	segment lengths in mm, from the fixed end (a list)	
`areas`	segment cross-sectional areas in mm^2 (a list)	
`loads`	axial load at the far end of each segment in kN, positive pulling away from the support (a list)	
`E`	Young's modulus (all segments)	

output	meaning
`elongation_mm`	total elongation of the bar in mm (negative = shortening)
`N_kN`	internal axial force of each segment in kN (tension positive)
`stress_MPa`	normal stress of each segment in MPa
`max_abs_stress_MPa`	largest stress magnitude in MPa

- **Method:** N_i = sum of the loads beyond segment i; $sigma_i = N_i / A_i$; delta = sum N_i L_i/(A_i E)
- **Assumes:** equal-length lists; left end fixed, right end free; linear elastic, no buckling
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('axial_bar_segments', lengths='300, 500', areas='400, 200',
loads='-20, 30', E=200000000000.0)
# -> elongation_mm = 0.4125, N_kN = [10.0, 30.0], stress_MPa = [25.0, 150.0], max_abs_stress_MPa =
150   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Stepped bar: lengths = [300, 500], areas = [400, 200], loads = [-20, 30], E = 200 GPa	SOLVED	elongation_mm = 0.4125, N_kN = [10.0, 30.0], stress_MPa = [25.0, 150.0], max_abs_stress_MPa = 150
Stepped bar: lengths = [300, 500], areas = [400], loads = [-20, 30], E = 200 GPa	INVALID_INPUT	Stepped bar under axial loads at its joints: lengths, areas and loads must have the same number of entries (one per segm

Bar fixed at both ends with an axial load at a step — `bar_fixed_at_both_ends`

Reactions, stresses in both parts and the displacement of the loaded joint of a two-part bar held between two rigid supports, loaded axially at the joint.

$$R_A = P\frac{k_1}{k_1+k_2}, \quad k_i = \frac{A_i E_i}{L_i}, \quad \delta = \frac{P}{k_1+k_2}$$

input	meaning	condition
`P`	axial load at the joint (towards end B)	
`L1`	length of part 1 (end A to the joint)	
`A1`	area of part 1	
`E1`	modulus of part 1	
`L2`	length of part 2	
`A2`	area of part 2	
`E2`	modulus of part 2	

output	meaning
`RA`	reaction at A
`RB`	reaction at B
`stress1_MPa`	stress in part 1 in MPa (tension positive)
`stress2_MPa`	stress in part 2 in MPa (compression negative)
`displacement_mm`	displacement of the joint in mm

- **Method:** compatibility: the joint moves the same amount in both parts; $RA = P\,k_1/(k_1+k_2)$, $k = AE/L$
- **Assumes:** rigid supports, no gap; linear elastic, no buckling of the compressed part

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('bar_fixed_at_both_ends', P=60000.0, L1=0.4, A1=0.0005,
E1=200000000000.0, L2=0.4, A2=0.0005, E2=200000000000.0)
# -> RA = 30000 N, RB = 30000 N, stress1_MPa = 60, stress2_MPa = -60, displacement_mm = 0.12
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Bar fixed at both ends: P = 60 kN, L1 = 0.4 m, A1 = 500 mm^2, E1 = 200 GPa, L2 = 0.4 m, A2 = 500 mm^2, E2 = 200 GPa	SOLVED	RA = 30000 N, RB = 30000 N, stress1_MPa = 60, stress2_MPa = -60, displacement_mm = 0.12

Plane stress transformation to rotated axes — `plane_stress_transformation`

The normal and shear stresses on axes rotated counterclockwise by theta from x, for a plane state of stress.

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

input	meaning	condition
`sx`	normal stress sigma_x	
`sy`	normal stress sigma_y	
`txy`	shear stress tau_xy	
`theta`	rotation of the axes, counterclockwise from x	

output	meaning
`sx_t_MPa`	sigma_x' in MPa
`sy_t_MPa`	sigma_y' in MPa
`txy_t_MPa`	tau_x'y' in MPa

- **Method:** sigma_x' = $(sx + sy)/2 + (sx - sy)/2$ cos 2theta + txy sin 2theta; tau_x'y' = -$(sx - sy)/2$ sin 2theta + txy cos 2theta
- **Assumes:** plane stress; tension positive, theta counterclockwise
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('plane_stress_transformation', sx=80000000.0, sy=-40000000.0,
txy=25000000.0, theta=0.5235987755982988)
# -> sx_t_MPa = 71.6506, sy_t_MPa = -31.6506, txy_t_MPa = -39.4615   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Plane stress transformation: sx = 80 MPa, sy = -40 MPa, txy = 25 MPa, theta = 30 deg	SOLVED	sx_t_MPa = 71.6506, sy_t_MPa = -31.6506, txy_t_MPa = -39.4615

question	outcome	result
Plane stress transformation: sx = 80 MPa, sy = -40 MPa, txy = 25 MPa	NEEDS_INPUT	plane_stress_transformation needs theta (rotation of the axes, counterclockwise from x). Example: solve('plane_stress_tr

Principal stresses and maximum shear (Mohr's circle, plane stress) — `principal_stresses_2d`

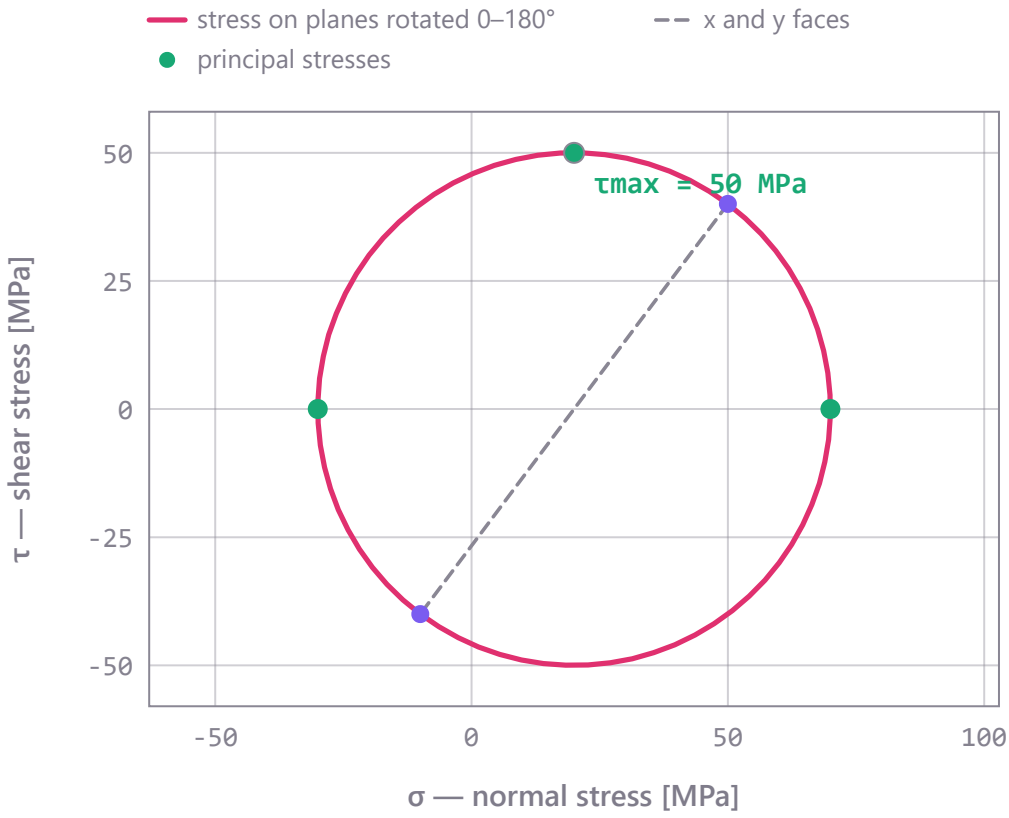
Principal stresses, their direction, the maximum in-plane and absolute shear stresses and the von Mises stress of a plane state of stress.

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}, \quad \tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

input	meaning	condition
`sx`	normal stress sigma_x	
`sy`	normal stress sigma_y	
`txy`	shear stress tau_xy	

output	meaning
`s1_MPa`	larger in-plane principal stress, MPa
`s2_MPa`	smaller in-plane principal stress, MPa
`tau_max_inplane_MPa`	maximum in-plane shear stress (radius of Mohr's circle), MPa
`theta_p_deg`	angle from x to the sigma_1 direction, degrees in (-90, 90] (0 when the circle is a point)
`tau_abs_max_MPa`	absolute maximum shear stress including $sigma_z = 0$, MPa
`von_mises_MPa`	von Mises equivalent stress, MPa

- **Method:** sigma_1,2 $= (sx + sy)/2$ +- R, $R = \sqrt{\left(\frac{sx-sy}{2}\right)^2 + txy^2}$; tan $2theta_p = 2$ txy/(sx - sy); tau_abs = max($s_1 - s_2$, |s1|, |s2|)/2
- **Assumes:** plane stress $(sigma_z = 0)$; tension positive
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Mohr's circle for σx = 50 MPa, σy = -10 MPa, τxy = 40 MPa: the normal and shear stress on planes rotated by 0–180° (73 states computed by the package's stress transformation) trace the circle; the principal stresses it returned are σ1 = 70 MPa and σ2 = -30 MPa at θp = 26.5651°; marked: centre σ = 20 MPa, τmax = 50 MPa.

```
mechanical_engineering_ai999x.solve('principal_stresses_2d', sx=50000000.0, sy=-10000000.0,
txy=40000000.0)
# -> s1_MPa = 70, s2_MPa = -30, tau_max_inplane_MPa = 50, theta_p_deg = 26.5651, tau_abs_max_MPa =
50, von_mises_MPa = 88.8819   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('principal_stresses_2d', sx=100000000.0, sy=0.0, txy=0.0)
# -> s1_MPa = 100, s2_MPa = 0, tau_max_inplane_MPa = 50, theta_p_deg = 0, tau_abs_max_MPa = 50,
von_mises_MPa = 100   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Principal stresses: sx = 50 MPa, sy = -10 MPa, txy = 40 MPa	SOLVED	s1_MPa = 70, s2_MPa = -30, tau_max_inplane_MPa = 50, theta_p_deg = 26.5651, tau_abs_max_MPa = 50, von_mises_MPa = 88.8819
Mohr's circle: sx = 50 MPa, sy = -10 MPa, txy = 40 MPa	SOLVED	s1_MPa = 70, s2_MPa = -30, tau_max_inplane_MPa = 50, theta_p_deg = 26.5651, tau_abs_max_MPa = 50, von_mises_MPa = 88.8819

Principal stresses of a three-dimensional stress state — `principal_stresses_3d`

The three principal stresses, the absolute maximum shear, von Mises, hydrostatic and octahedral shear stresses of a general 3D state of stress.

$$\det(\boldsymbol{\sigma} - \sigma\mathbf{I}) = 0, \quad \tau_{max} = \frac{\sigma_1 - \sigma_3}{2}, \quad \tau_{oct} = \frac{\sqrt{2}}{3}\sigma_{vm}$$

input	meaning	condition
`sx`	sigma_x	
`sy`	sigma_y	
`sz`	sigma_z	
`txy`	tau_xy	
`tyz`	tau_yz	
`tzx`	tau_zx	

output	meaning
`s1_MPa`	largest principal stress, MPa
`s2_MPa`	intermediate principal stress, MPa
`s3_MPa`	smallest principal stress, MPa
`tau_max_MPa`	absolute maximum shear stress $(s_1 - s_3)/2$, MPa
`von_mises_MPa`	von Mises stress, MPa
`hydrostatic_MPa`	mean (hydrostatic) stress, MPa
`tau_oct_MPa`	octahedral shear stress, MPa

- **Method:** eigenvalues of the stress tensor by Jacobi rotations; $sigma_vm = sqrt$(((sx-sy)^2 + (sy-sz)^2 + (sz-sx)^2)/2 + 3(txy^2 + tyz^2 + tzx^2)); tau_oct = sqrt(2)/3 sigma_vm
- **Assumes:** symmetric stress tensor; tension positive
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('principal_stresses_3d', sx=100000000.0, sy=0.0, sz=0.0,
txy=0.0, tyz=0.0, tzx=0.0)
# -> s1_MPa = 100, s2_MPa = 0, s3_MPa = 0, tau_max_MPa = 50, von_mises_MPa = 100, hydrostatic_MPa =
33.3333, tau_oct_MPa = 47.1405   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('principal_stresses_3d', sx=0.0, sy=0.0, sz=0.0, txy=50000000.0,
tyz=0.0, tzx=0.0)
# -> s1_MPa = 50, s2_MPa = 0, s3_MPa = -50, tau_max_MPa = 50, von_mises_MPa = 86.6025,
hydrostatic_MPa = 0, tau_oct_MPa = 40.8248   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Principal stresses 3D: sx = 100 MPa, sy = 0 MPa, sz = 0 MPa, txy = 0 MPa, tyz = 0 MPa, tzx = 0 MPa	SOLVED	s1_MPa = 100, s2_MPa = 0, s3_MPa = 0, tau_max_MPa = 50, von_mises_MPa = 100, hydrostatic_MPa = 33.3333, tau_oct_MPa = 47.1405
Principal stresses 3D: sx = 100 MPa, sy = 0 MPa, sz = 0 MPa	NEEDS_INPUT	principal_stresses_3d needs txy, tyz, tzx (tau_xy; tau_yz; tau_zx). Example: solve('principal_stresses_3d', sx='1e+08 Pa

Von Mises and Tresca equivalent stresses and safety factors — `yield_criteria_safety_factor`

Equivalent stresses by the distortion-energy (von Mises) and maximum-shear (Tresca) criteria and the corresponding factors of safety against yielding.

$$\sigma_{vm} = \sqrt{\tfrac{1}{2}\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right]}, \quad n = \frac{S_y}{\sigma_{eq}}$$

input	meaning	condition
`Sy`	yield strength	
`sx`	sigma_x	
`sy`	sigma_y (default 0)	
`sz`	sigma_z (default 0)	
`txy`	tau_xy (default 0)	
`tyz`	tau_yz (default 0)	
`tzx`	tau_zx (default 0)	

output	meaning
`von_mises_MPa`	von Mises stress, MPa
`tresca_MPa`	Tresca equivalent stress $(s_1 - s_3)$, MPa
`n_von_mises`	safety factor by distortion energy
`n_tresca`	safety factor by maximum shear

- **Method:** $sigma_vm = sqrt$(((sx-sy)^2 + (sy-sz)^2 + (sz-sx)^2)/2 + 3(txy^2+tyz^2+tzx^2)), $sigma_T = s_1 - s_3$, $n = Sy/sigma_eq$
- **Assumes:** $Sy > 0$; not a purely hydrostatic stress; ductile, isotropic material
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('yield_criteria_safety_factor', Sy=350000000.0, sx=120000000.0,
sy=-40000000.0, txy=60000000.0)
# -> von_mises_MPa = 177.764, tresca_MPa = 200, n_von_mises = 1.9689, n_tresca = 1.75   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Yield safety factor: Sy = 350 MPa, sx = 120 MPa, sy = -40 MPa, txy = 60 MPa	SOLVED	von_mises_MPa = 177.764, tresca_MPa = 200, n_von_mises = 1.9689, n_tresca = 1.75
Yield safety factor: Sy = 350 MPa, sx = 100 MPa, sy = 100 MPa, sz = 100 MPa	INVALID_INPUT	Von Mises and Tresca equivalent stresses and safety factors: the stress state is purely hydrostatic: neither criterion p

Safety factors of a brittle material (maximum normal stress, Coulomb-Mohr, modified Mohr) — `brittle_failure_mohr`

Factors of safety of a brittle material under plane stress by the maximum-normal-stress, brittle Coulomb-Mohr and modified-Mohr criteria (proportional loading).

$$\frac{\sigma_1}{S_{ut}} - \frac{\sigma_3}{S_{uc}} = \frac{1}{n}, \quad \frac{(S_{uc}-S_{ut})\sigma_A}{S_{uc}S_{ut}} - \frac{\sigma_B}{S_{uc}} = \frac{1}{n}$$

input	meaning	condition
`sA`	first in-plane principal stress	
`sB`	second in-plane principal stress	
`Sut`	ultimate tensile strength	
`Suc`	ultimate compressive strength (positive number)	

output	meaning
`n_max_normal`	safety factor, maximum normal stress theory
`n_coulomb_mohr`	safety factor, brittle Coulomb-Mohr
`n_modified_mohr`	safety factor, modified Mohr

- **Method:** BCM: 1/n = s1/Sut - s3/Suc (with s1 = max(sA, sB, 0), s3 = min(sA, sB, 0)); MM: 1/n = (Suc - Sut) sA/(Suc Sut) - sB/Suc when $sA > 0 > sB$ and |sB| > sA; MNS: n = min(Sut/s1, Suc/|s3|)
- **Assumes:** plane stress (third principal stress zero); Sut, Suc > 0 (Suc given as a magnitude); proportional loading
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('brittle_failure_mohr', sA=90000000.0, sB=-50000000.0,
Sut=210000000.0, Suc=750000000.0)
# -> n_max_normal = 2.33333, n_coulomb_mohr = 2.01923, n_modified_mohr = 2.33333   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('brittle_failure_mohr', sA=50000000.0, sB=-300000000.0,
Sut=210000000.0, Suc=750000000.0)
# -> n_max_normal = 2.5, n_coulomb_mohr = 1.56716, n_modified_mohr = 1.75   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Brittle failure: sA = 90 MPa, sB = -50 MPa, Sut = 210 MPa, Suc = 750 MPa	SOLVED	n_max_normal = 2.33333, n_coulomb_mohr = 2.01923, n_modified_mohr = 2.33333
Brittle failure: sA = 90 MPa, sB = -50 MPa, Sut = 210 MPa, Suc = -750 MPa	INVALID_INPUT	Safety factors of a brittle material (maximum normal stress, Coulomb-Mohr, modified Mohr): Sut and Suc must be a positiv

Strain rosette: strains, principal strains and principal stresses — `strain_rosette`

From the readings of a three-gauge rosette (rectangular $0/45/90$ or delta $0/60/120$ degrees): the strain components, the principal strains and their direction, and the principal stresses (plane stress).

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}, \quad \sigma_1 = \frac{E(\varepsilon_1 + \nu\varepsilon_2)}{1-\nu^2}$$

input	meaning	condition
`ea`	strain of gauge a (at 0 degrees), microstrain	
`eb`	strain of gauge b (45 or 60 degrees), microstrain	
`ec`	strain of gauge c (90 or 120 degrees), microstrain	

input	meaning	condition
`E`	Young's modulus	
`nu`	Poisson's ratio	
`layout`	rosette layout: rectangular $(0/45/90)$ or delta $(0/60/120)$	

output	meaning
`ex`	strain eps_x, microstrain
`ey`	strain eps_y, microstrain
`gxy`	shear strain gamma_xy, microstrain
`e1`	larger principal strain, microstrain
`e2`	smaller principal strain, microstrain
`theta_p_deg`	angle from gauge a to the eps_1 direction, degrees in (-90, 90]
`s1_MPa`	larger principal stress, MPa
`s2_MPa`	smaller principal stress, MPa

- **Method:** rectangular: $ex = ea$, $ey = ec$, gxy = 2eb - ea - ec; delta: $ex = ea$, $ey = (2eb + 2ec - ea)/3$, gxy = 2(eb - ec)/sqrt(3); e1,$2 = (ex + ey)/2$ +- sqrt(((ex-ey)/2)^2 + (gxy/2)^2); $s_1 = E$(e1 + ν e2)/(1 - nu^2)
- **Assumes:** gauge angles measured counterclockwise from gauge a; plane stress, linear elastic isotropic; $0 \le \nu < 0.5$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('strain_rosette', ea=400.0, eb=300.0, ec=-100.0,
E=200000000000.0, nu=0.3, layout='rectangular')
# -> ex = 400, ey = -100, gxy = 300, e1 = 441.548, e2 = -141.548, theta_p_deg = 15.4819, s1_MPa =
87.7106, s2_MPa = -1.99633   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Strain rosette: ea = 400, eb = 300, ec = -100, E = 200 GPa, nu = 0.3, layout = rectangular	SOLVED	ex = 400, ey = -100, gxy = 300, e1 = 441.548, e2 = -141.548, theta_p_deg = 15.4819, s1_MPa = 87.7106, s2_MPa = -1.99633
Strain rosette: ea = 400, eb = 300, ec = -100, E = 200 GPa, nu = 0.3, layout = star	INVALID_INPUT	layout: one of rectangular, delta; got 'star'.

Generalized Hooke's law: strains from a 3D stress state — `generalized_hookes_law`

The six strain components and the volumetric strain of an isotropic linear-elastic material under a general three-dimensional stress state.

$$\varepsilon_x = \frac{1}{E}\left[\sigma_x - \nu(\sigma_y + \sigma_z)\right], \quad \gamma_{xy} = \frac{\tau_{xy}}{G}, \quad G = \frac{E}{2(1+\nu)}$$

input	meaning	condition
`sx`	sigma_x	

input	meaning	condition
`sy`	sigma_y	
`sz`	sigma_z	
`txy`	tau_xy	
`tyz`	tau_yz	
`tzx`	tau_zx	
`E`	Young's modulus	
`nu`	Poisson's ratio	

output	meaning
`ex`	eps_x, microstrain
`ey`	eps_y, microstrain
`ez`	eps_z, microstrain
`gxy`	gamma_xy, microstrain
`gyz`	gamma_yz, microstrain
`gzx`	gamma_zx, microstrain
`volumetric`	volumetric strain eps_x + eps_y + eps_z, microstrain

- **Method:** eps_x = (sx - nu(sy + sz))/E (and cyclic), $\gamma = \tau / G$ with G = E/(2(1 + nu))
- **Assumes:** isotropic, linear elastic; $-1 < \nu < 0.5$; small strains
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('generalized_hookes_law', sx=100000000.0, sy=50000000.0, sz=0.0,
txy=30000000.0, tyz=0.0, tzx=0.0, E=200000000000.0, nu=0.3)
# -> ex = 425, ey = 100, ez = -225, gxy = 390, gyz = 0, gzx = 0, volumetric = 300   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Generalized Hooke's law: sx = 100 MPa, sy = 50 MPa, sz = 0 MPa, txy = 30 MPa, tyz = 0 MPa, tzx = 0 MPa, E = 200 GPa, nu = 0.3	SOLVED	ex = 425, ey = 100, ez = -225, gxy = 390, gyz = 0, gzx = 0, volumetric = 300
Generalized Hooke's law: sx = 100 MPa, sy = 50 MPa, sz = 0 MPa, txy = 30 MPa, tyz = 0 MPa, tzx = 0 MPa, E = 200 GPa, nu = 0.6	INVALID_INPUT	Generalized Hooke's law: strains from a 3D stress state: Poisson's ratio must lie between -1 and 0.5.

Thin-walled pressure vessel stresses (cylinder or sphere) — `thin_walled_pressure_vessel`

Hoop and longitudinal stresses and the in-plane and absolute maximum shear stresses in the wall of a thin cylindrical or spherical pressure vessel.

$$\sigma_h = \frac{pr}{t}, \quad \sigma_l = \frac{pr}{2t}$$

input	meaning	condition
`p`	internal gauge pressure	
`r`	inner radius	
`t`	wall thickness	
`shape`	vessel shape	

output	meaning
`hoop_MPa`	hoop (circumferential) stress, MPa
`longitudinal_MPa`	longitudinal (axial) stress, MPa
`tau_inplane_MPa`	maximum in-plane shear stress, MPa
`tau_abs_MPa`	absolute maximum shear stress at the outer surface, MPa

- **Method:** cylinder: $sigma_h = pr/t$, $sigma_l = pr/(2t)$; sphere: $\sigma = pr/(2t)$; $tau_abs = sigma_h/2$ (radial stress 0 at the outer surface)
- **Assumes:** $r/t \geq 10$ (thin wall); closed ends, internal pressure only; r is the inner radius
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('thin_walled_pressure_vessel', p=2000000.0, r=0.5, t=0.01,
shape='cylinder')
# -> hoop_MPa = 100, longitudinal_MPa = 50, tau_inplane_MPa = 25, tau_abs_MPa = 50   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('thin_walled_pressure_vessel', p=2000000.0, r=0.5, t=0.01,
shape='sphere')
# -> hoop_MPa = 50, longitudinal_MPa = 50, tau_inplane_MPa = 0, tau_abs_MPa = 25   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Thin-walled pressure vessel: p = 2 MPa, r = 500 mm, t = 10 mm, shape = cylinder	SOLVED	hoop_MPa = 100, longitudinal_MPa = 50, tau_inplane_MPa = 25, tau_abs_MPa = 50
Thin-walled pressure vessel: p = 2 MPa, r = 50 mm, t = 10 mm, shape = cylinder	INVALID_INPUT	Thin-walled pressure vessel stresses (cylinder or sphere): r/t is below 10: the wall is not thin (use the thick-walled c

Thick-walled cylinder under pressure (Lame equations) — `thick_walled_cylinder`

Radial and hoop stresses at a radius, the hoop stresses at both surfaces, the maximum shear at the bore and the closed-end axial stress of a thick cylinder under internal and external pressure.

$$\sigma_{r,\theta} = \frac{p_i a^2 - p_o b^2}{b^2 - a^2} \mp \frac{(p_i - p_o) a^2 b^2}{(b^2 - a^2) r^2}$$

input	meaning	condition
`pi`	internal pressure	
`po`	external pressure	
`a`	inner radius	

input	meaning	condition
`b`	outer radius	
`r`	radius at which to evaluate (default: the bore, r = a)	

output	meaning
`sr_MPa`	radial stress at r, MPa
`st_MPa`	hoop stress at r, MPa
`st_inner_MPa`	hoop stress at the bore, MPa
`st_outer_MPa`	hoop stress at the outer surface, MPa
`tau_max_inner_MPa`	maximum shear stress at the bore, MPa
`sz_closed_MPa`	axial stress with closed ends, MPa

- **Method:** $sigma_r = A - B/r^2$, $sigma_t = A + B/r^2$ with A = (pi $a^2 - po$ b^2)/(b^2 - a^2), $B = (pi - po)\ a^2\ b^2/(b^2 - a^2)$; closed ends $sigma_z = A$
- **Assumes:** $a < b$; pressures >= 0, not both zero; $a \le r \le b$; linear elastic
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('thick_walled_cylinder', pi=100000000.0, po=0.0, a=0.05, b=0.1)
# -> sr_MPa = -100, st_MPa = 166.667, st_inner_MPa = 166.667, st_outer_MPa = 66.6667,
tau_max_inner_MPa = 133.333, sz_closed_MPa = 33.3333   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Thick-walled cylinder: pi = 100 MPa, po = 0 MPa, a = 50 mm, b = 100 mm	SOLVED	sr_MPa = -100, st_MPa = 166.667, st_inner_MPa = 166.667, st_outer_MPa = 66.6667, tau_max_inner_MPa = 133.333, sz_closed_MPa = 33.3333
Thick-walled cylinder: pi = 100 MPa, po = 0 MPa, a = 100 mm, b = 50 mm	INVALID_INPUT	Thick-walled cylinder under pressure (Lame equations): the inner radius a must be smaller than the outer radius b.

Shrink (press) fit: interface pressure and hoop stresses — `shrink_fit_pressure`

The contact pressure of an interference fit between a hub and a solid or hollow shaft of the same material, and the hoop stresses it causes at the interface.

$$p = \frac{E\delta}{2R^3}\,\frac{(r_o^2 - R^2)(R^2 - r_i^2)}{r_o^2 - r_i^2}$$

input	meaning	condition
`delta`	radial interference (shaft radius minus hub bore radius)	
`R`	nominal interface radius	
`ro`	outer radius of the hub	
`E`	Young's modulus (both parts)	
`ri`	bore radius of a hollow shaft (default 0: solid)	

output	meaning
`p_MPa`	interface pressure, MPa
`hub_hoop_MPa`	hoop stress in the hub at the interface, MPa
`shaft_hoop_MPa`	hoop stress in the shaft at the interface, MPa

- **Method:** $p = E\,\delta/(2R^3)\;(ro^2 - R^2)(R^2 - ri^2)/(ro^2 - ri^2)$; hub $sigma_t = p\;(ro^2 + R^2)/(ro^2 - R^2)$; shaft $sigma_t = -p\;(R^2 + ri^2)/(R^2 - ri^2)$
- **Assumes:** $ri < R < ro$; both parts of the same material; plane stress, elastic, long hub
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('shrink_fit_pressure', delta=2e-05, R=0.025, ro=0.05,
E=207000000000.0)
# -> p_MPa = 62.1, hub_hoop_MPa = 103.5, shaft_hoop_MPa = -62.1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Shrink fit: delta = 0.02 mm, R = 25 mm, ro = 50 mm, E = 207 GPa	SOLVED	p_MPa = 62.1, hub_hoop_MPa = 103.5, shaft_hoop_MPa = -62.1
Shrink fit: delta = 0.02 mm, R = 60 mm, ro = 50 mm, E = 207 GPa	INVALID_INPUT	Shrink (press) fit: interface pressure and hoop stresses: need ri < R < ro (shaft bore < interface < hub outer radius).

Stresses in a rotating disk (solid or with a central hole) — `rotating_disk_stresses`

The largest hoop and radial stresses, the radius of the largest radial stress and the hoop stress at the rim of a thin disk of uniform thickness spinning at constant speed.

$$\sigma_\theta(r_i) = \frac{3+\nu}{4}\rho\omega^2\left(r_o^2 + \frac{1-\nu}{3+\nu}r_i^2\right), \quad \sigma_{r,max} = \frac{3+\nu}{8}\rho\omega^2(r_o - r_i)^2$$

input	meaning	condition
`rho`	density	
`omega`	angular speed (rpm or rad/s)	
`nu`	Poisson's ratio	
`ro`	outer radius	
`ri`	radius of the central hole (default 0: solid)	

output	meaning
`hoop_max_MPa`	largest hoop stress (at the bore, or the centre of a solid disk), MPa
`radial_max_MPa`	largest radial stress, MPa
`r_radial_max`	radius of the largest radial stress (sqrt(ri ro); 0 for a solid disk)
`hoop_rim_MPa`	hoop stress at the outer rim, MPa

- **Method:** solid: $sigma_r = sigma_t = (3+\nu)/8$ rho w^2 ro^2 at the centre; annular: sigma_t(ri) = (3 + nu)/4 rho w^2 (ro^2 + (1 - nu)/(3 + nu) ri^2), sigma_r,max = (3 + nu)/8 rho w^2 $(ro - ri)^2$ at sqrt(ri ro)
- **Assumes:** thin disk of uniform thickness, plane stress; free inner and outer edges; $0 \le \nu < 0.5$

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('rotating_disk_stresses', rho=7800.0, omega=314.1592653589793,
nu=0.3, ro=0.25, ri=0.05)
# -> hoop_max_MPa = 40.0311, radial_max_MPa = 12.7022, r_radial_max = 0.111803 m, hoop_rim_MPa =
10.0078   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('rotating_disk_stresses', rho=7800.0, omega=314.1592653589793,
nu=0.3, ro=0.25)
# -> hoop_max_MPa = 19.8472, radial_max_MPa = 19.8472, r_radial_max = 0 m, hoop_rim_MPa = 8.42001
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Rotating disk: rho = 7800 kg/m^3, omega = 3000 rpm, nu = 0.3, ro = 250 mm, ri = 50 mm	SOLVED	hoop_max_MPa = 40.0311, radial_max_MPa = 12.7022, r_radial_max = 0.111803 m, hoop_rim_MPa = 10.0078
Rotating disk: rho = 7800 kg/m^3, omega = 3000 rpm, nu = 0.3, ro = 250 mm, ri = 300 mm	INVALID_INPUT	Stresses in a rotating disk (solid or with a central hole): the bore radius must be smaller than the outer radius.

Euler buckling load of a column with given end conditions — `euler_buckling`

The critical (Euler) buckling load of an ideal elastic column, its effective length factor and length, and with the area also the critical stress and slenderness ratio.

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

input	meaning	condition
`E`	Young's modulus	
`I`	least second moment of area	
`L`	column length	
`ends`	end conditions: pinned-pinned, fixed-free, fixed-fixed or fixed-pinned	
`A`	cross-sectional area (optional: gives the stress and slenderness)	

output	meaning
`P_cr`	critical buckling load
`P_cr_kN`	critical load in kN
`K`	effective length factor (theoretical)
`Le`	effective length K L
`sigma_cr_MPa`	critical stress P_cr/A, MPa (with A)
`slenderness`	slenderness ratio K L / r (with A)

- **Method:** $P_cr = pi^2$ E I/(K L)^2
- **Assumes:** ideal straight elastic column, centric load; theoretical K: 1, 2, 0.5, 0.6992 (fixed-pinned, from tan x = x); valid only below the proportional limit

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('euler_buckling', E=200000000000.0, I=8e-06, L=4.0,
ends='pinned-pinned')
# -> P_cr = 986960 N, P_cr_kN = 986.96, K = 1, Le = 4 m   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('euler_buckling', E=200000000000.0, I=8e-06, L=4.0, ends='fixed-
pinned', A=0.005)
# -> P_cr = 2.01907e+06 N, P_cr_kN = 2019.07, K = 0.699156, Le = 2.79662 m, sigma_cr_MPa = 403.815,
slenderness = 69.9156   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Euler buckling: E = 200 GPa, I = 8e-6 m^4, L = 4 m, ends = pinned-pinned	SOLVED	P_cr = 986960 N, P_cr_kN = 986.96, K = 1, Le = 4 m
Euler buckling: E = 200 GPa, I = 8e-6 m^4, L = 4 m, ends = fixed-fixed, A = 5000 mm^2	SOLVED	P_cr = 3.94784e+06 N, P_cr_kN = 3947.84, K = 0.5, Le = 2 m, sigma_cr_MPa = 789.568, slenderness = 50
Euler buckling: E = 200 GPa, I = 8e-6 m^4, L = 4 m, ends = hinged	INVALID_INPUT	ends: one of pinned-pinned, fixed-free, fixed-fixed, fixed-pinned; got 'hinged'.

Critical stress of an intermediate column (Johnson parabola / Euler) — `johnson_column`

The critical stress of a column by the Johnson parabola below the transition slenderness and by Euler above it, with the transition slenderness and (with the area) the critical load.

$$\sigma_{cr} = S_y - \frac{1}{E}\left(\frac{S_y}{2\pi}\frac{KL}{r}\right)^2 \quad (KL/r < C_c), \quad C_c = \sqrt{\frac{2\pi^2 E}{S_y}}$$

input	meaning	condition
`Sy`	yield strength	
`E`	Young's modulus	
`L`	column length	
`r`	least radius of gyration	
`ends`	end conditions (theoretical K)	
`A`	cross-sectional area (optional: gives the load)	

output	meaning
`slenderness`	slenderness ratio K L / r
`transition_slenderness`	transition slenderness $C_c = sqrt$(2 pi^2 E / Sy)
`regime`	Johnson or Euler
`sigma_cr_MPa`	critical stress, MPa
`P_cr_kN`	critical load in kN (with A)

- **Method:** $C_c = sqrt(2\ pi^2$ E/Sy); $KL/r < C_c$: sigma_cr = Sy - (Sy KL/(2 pi r))^$2/E$; otherwise $sigma_cr = pi^2\ E/(KL/r)^2$
- **Assumes:** centric load; theoretical effective length factors
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('johnson_column', Sy=250000000.0, E=200000000000.0, L=1.5,
r=0.02, ends='pinned-pinned')
# -> slenderness = 75, transition_slenderness = 125.664, regime = Johnson, sigma_cr_MPa = 205.474
(SOLVED)
```

```
mechanical_engineering_ai999x.solve('johnson_column', Sy=250000000.0, E=200000000000.0, L=4.0,
r=0.02, ends='pinned-pinned')
# -> slenderness = 200, transition_slenderness = 125.664, regime = Euler, sigma_cr_MPa = 49.348
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Johnson column: Sy = 250 MPa, E = 200 GPa, L = 1.5 m, r = 20 mm, ends = pinned-pinned	SOLVED	slenderness = 75, transition_slenderness = 125.664, regime = Johnson, sigma_cr_MPa = 205.474
Johnson column: Sy = 250 MPa, E = 200 GPa, L = 1.5 m, ends = pinned-pinned	NEEDS_INPUT	johnson_column needs r (least radius of gyration). Example: solve('johnson_column', Sy='2.5e+08 Pa', E='2e+11 Pa', L='1.

Eccentrically loaded column (secant formula) — `secant_column`

The maximum compressive stress, midspan deflection and largest bending moment of a pinned column under a load with equal end eccentricities.

$$\sigma_{max} = \frac{P}{A}\left[1 + \frac{ec}{r^2}\sec\left(\frac{L}{2}\sqrt{\frac{P}{EI}}\right)\right]$$

input	meaning	condition
`P`	compressive load	
`e`	eccentricity of the load	
`L`	length (pinned ends)	
`A`	area	
`I`	second moment of area in the bending plane	
`c`	distance from the centroid to the extreme fibre	
`E`	Young's modulus	

output	meaning
`sigma_max_MPa`	maximum compressive stress (magnitude), MPa
`deflection_mm`	midspan lateral deflection, mm
`M_max`	largest bending moment P (e + deflection)
`P_over_Pcr`	P / P_Euler

- **Method:** sigma_max = P/A [1 + (e c/r^2) sec($L/2$ sqrt(P/(EI)))], deflection = e [sec(kL/2) - 1]
- **Assumes:** pinned ends, equal eccentricities on the same side; P below the Euler load; elastic
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('secant_column', P=200000.0, e=0.01, L=3.0, A=0.004, I=1.2e-05,
c=0.075, E=200000000000.0)
# -> sigma_max_MPa = 63.771, deflection_mm = 1.01679, M_max = 2203.36 N*m, P_over_Pcr = 0.0759909
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Secant column: P = 200 kN, e = 10 mm, L = 3 m, A = 4000 mm^2, I = 1.2e-5 m^4, c = 75 mm, E = 200 GPa	SOLVED	sigma_max_MPa = 63.771, deflection_mm = 1.01679, M_max = 2203.36 N*m, P_over_Pcr = 0.0759909
Secant column: P = 5000 kN, e = 10 mm, L = 3 m, A = 4000 mm^2, I = 1.2e-5 m^4, c = 75 mm, E = 200 GPa	INVALID_INPUT	Eccentrically loaded column (secant formula): the load reaches the Euler load of the pinned column: no equilibrium.

Combined axial force and biaxial bending of a rectangular section —

`combined_axial_and_bending_stress`

Normal stresses at the four corners of a rectangular section carrying an axial force and bending moments about both axes, with the extreme values.

$$\sigma = \frac{N}{A} + \frac{M_x y}{I_x} + \frac{M_y x}{I_y}$$

input	meaning	condition
`b`	width (along x)	
`h`	depth (along y)	
`N`	axial force (tension positive)	
`Mx`	moment about x (positive: tension at +y)	
`My`	moment about y (positive: tension at +x)	

output	meaning
`sigma_corners_MPa`	stresses at the corners (+x,+y), (-x,+y), (-x,-y), (+x,-y), MPa
`sigma_max_MPa`	largest (most tensile) stress, MPa
`sigma_min_MPa`	smallest (most compressive) stress, MPa
`sigma_axial_MPa`	uniform axial part N/A, MPa

- **Method:** sigma(x, y) = N/A + Mx y/Ix + My x/Iy at the corners (x = +−$b/2$, y = +-h/2)
- **Assumes:** linear elastic, plane sections; moments about the centroidal axes
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('combined_axial_and_bending_stress', b=0.1, h=0.2, N=-50000.0,
Mx=8000.0, My=2000.0)
# -> sigma_corners_MPa = [15.5, 3.5, -20.5, -8.5], sigma_max_MPa = 15.5, sigma_min_MPa = -20.5,
sigma_axial_MPa = -2.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Combined axial and bending stress: b = 100 mm, h = 200 mm, N = -50 kN, Mx = 8 kN*m, My = 2 kN*m	SOLVED	sigma_corners_MPa = [15.5, 3.5, -20.5, -8.5], sigma_max_MPa = 15.5, sigma_min_MPa = -20.5, sigma_axial_MPa = -2.5
Combined axial and bending stress: b = 100 mm, h = 200 mm, N = -50 kN, Mx = 8 kN*m	NEEDS_INPUT	combined_axial_and_bending_stress needs My (moment about y (positive: tension at +x)). Example: solve('combined_axial_an

Eccentrically loaded bolt group (elastic method) — `eccentric_bolt_group`

The direct shear per bolt, the largest resultant bolt force and which bolt carries it, for a group of identical bolts under a vertical load applied with a horizontal eccentricity.

$$\mathbf{F}_i = -\frac{P}{n}\hat{\mathbf{y}} + \frac{Pe}{\sum r_j^2}\left(y_i - \bar{y},\ -(x_i - \bar{x})\right)$$

input	meaning	condition
`bolts`	bolt positions, one row 'x y' per bolt, in mm	
`P`	vertical load (downward)	
`e`	horizontal distance of the load line from the bolt-group centroid (positive to the right)	

output	meaning
`direct_kN`	direct shear per bolt, kN
`max_bolt_force_kN`	largest resultant bolt force, kN
`critical_bolt`	number of the most loaded bolt (1-based, the first if several are equal)
`polar_sum_mm2`	sum of r^2 about the centroid, mm^2

- **Method:** direct F/n plus torsional M r_i / sum r^2 perpendicular to r_i, with $M = P$ e about the bolt centroid; vector sum per bolt
- **Assumes:** identical bolts, rigid plate, elastic method; 2 to 40 bolts, not all at one point
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('eccentric_bolt_group', bolts='0 0; 0 100; 100 0; 100 100',
P=20000.0, e=0.25)
# -> direct_kN = 5, max_bolt_force_kN = 21.5058, critical_bolt = 3, polar_sum_mm2 = 20000   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Bolt group: bolts = [0 0; 0 100; 100 0; 100 100], P = 20 kN, e = 250 mm	SOLVED	direct_kN = 5, max_bolt_force_kN = 21.5058, critical_bolt = 3, polar_sum_mm2 = 20000
Bolt group: bolts = [0 0; 0 0], P = 20 kN, e = 250 mm	INVALID_INPUT	Eccentrically loaded bolt group (elastic method): the bolts must not all coincide.

Fatigue safety factor by the modified Goodman criterion — `goodman_fatigue_safety_factor`

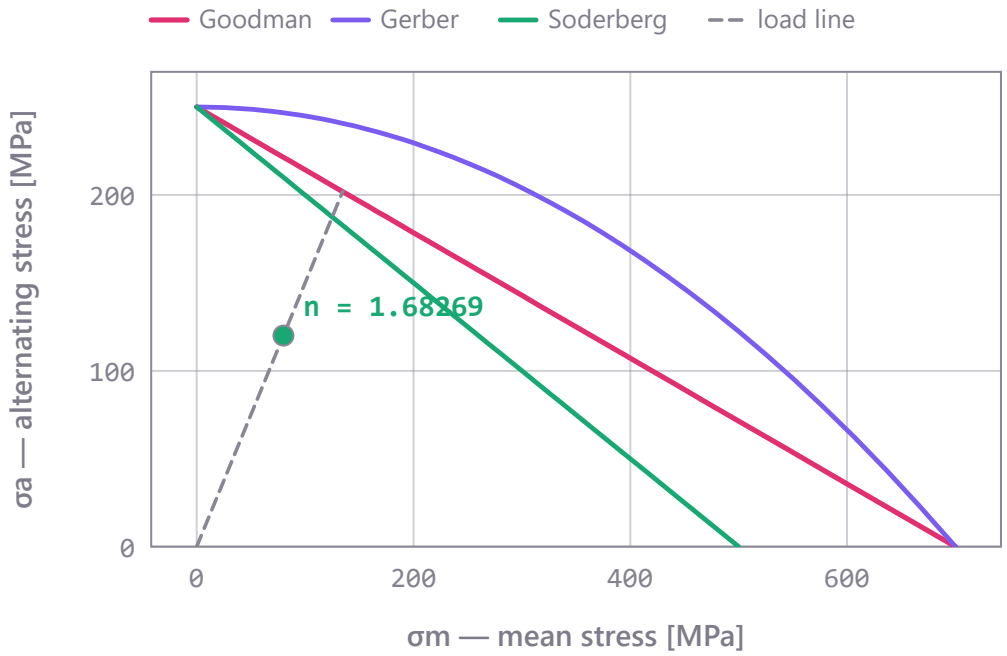
The fatigue factor of safety along a proportional load line by the modified Goodman line, the equivalent fully reversed amplitude, and (with Sy) Langer's first-cycle yield factor.

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{n}$$

input	meaning	condition
`sa`	alternating stress amplitude	
`sm`	mean stress (>= 0)	
`Se`	fully corrected endurance limit	
`Sut`	ultimate tensile strength	
`Sy`	yield strength (optional: first-cycle yield check)	

output	meaning
`n_fatigue`	fatigue safety factor (Goodman, proportional loading)
`sigma_ar_MPa`	equivalent fully reversed amplitude sa/(1 - sm/Sut), MPa
`n_yield`	Langer first-cycle yield factor Sy/(sa + sm) (with Sy)

- **Method:** 1/n = sa/Se + sm/Sut; sigma_ar = sa/(1 - sm/Sut); $n_y = Sy/(sa + sm)$
- **Assumes:** $sm \geq 0$ (compressive means not covered); sm < Sut; proportional loading (the load line through the origin)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Fatigue failure envelopes for Se = 250 MPa, Sut = 700 MPa, Sy = 500 MPa: each point is the stress state at which the package's safety factor along a load line equals 1 (47 load lines per criterion). Marked: the operating point σm = 80 MPa, σa = 120 MPa, Goodman safety factor n = 1.68269.

```
mechanical_engineering_ai999x.solve('goodman_fatigue_safety_factor', sa=120000000.0, sm=80000000.0,
Se=250000000.0, Sut=700000000.0, Sy=500000000.0)
# -> n_fatigue = 1.68269, sigma_ar_MPa = 135.484, n_yield = 2.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Goodman safety factor: sa = 120 MPa, sm = 80 MPa, Se = 250 MPa, Sut = 700 MPa	SOLVED	n_fatigue = 1.68269, sigma_ar_MPa = 135.484
Goodman safety factor: sa = 120 MPa, sm = -80 MPa, Se = 250 MPa, Sut = 700 MPa	INVALID_INPUT	Fatigue safety factor by the modified Goodman criterion: the mean stress (compressive means are not covered) must be zer

Fatigue safety factor by the Soderberg criterion — `soderberg_fatigue_safety_factor`

The conservative Soderberg fatigue factor of safety (the line from Se to the yield strength) along a proportional load line, and the equivalent amplitude.

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_y} = \frac{1}{n}$$

input	meaning	condition
`sa`	alternating stress amplitude	
`sm`	mean stress (>= 0)	
`Se`	fully corrected endurance limit	
`Sy`	yield strength	

output	meaning
`n_fatigue`	fatigue safety factor (Soderberg)

output	meaning
`sigma_ar_MPa`	equivalent fully reversed amplitude $sa/(1 - sm/Sy)$, MPa

- **Method:** $1/n = sa/Se + sm/Sy$
- **Assumes:** $sm \geq 0$; $sm < Sy$; proportional loading
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('soderberg_fatigue_safety_factor', sa=100000000.0,
sm=60000000.0, Se=220000000.0, Sy=450000000.0)
# -> n_fatigue = 1.70103, sigma_ar_MPa = 115.385   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Soderberg safety factor: sa = 100 MPa, sm = 60 MPa, Se = 220 MPa, Sy = 450 MPa	SOLVED	n_fatigue = 1.70103, sigma_ar_MPa = 115.385
Soderberg safety factor: sa = 100 MPa, sm = 500 MPa, Se = 220 MPa, Sy = 450 MPa	INVALID_INPUT	Fatigue safety factor by the Soderberg criterion: the mean stress must be below the strength.

Fatigue life with a mean-stress correction (Morrow, Goodman, SWT, Walker) and Basquin's law — `fatigue_life_mean_stress_correction`

The equivalent fully reversed stress amplitude of a cycle with a mean stress, by a chosen correction, and the cycles to failure from Basquin's law $sigma_ar = sf' (2N)^b$.

$$\sigma_{ar} = \frac{\sigma_a}{1 - \sigma_m/\sigma'_f}, \quad \sigma_{ar} = \sqrt{\sigma_{max}\sigma_a}, \quad N_f = \tfrac{1}{2}\left(\frac{\sigma_{ar}}{\sigma'_f}\right)^{1/b}$$

input	meaning	condition
`sa`	stress amplitude	
`sm`	mean stress	
`sf`	fatigue strength coefficient sf' (Morrow and Goodman use it as the strength)	
`b`	fatigue strength exponent (negative)	
`method`	mean-stress correction	
`gamma`	Walker exponent (walker only)	

output	meaning
`sigma_ar_MPa`	equivalent fully reversed amplitude, MPa
`Nf`	cycles to failure
`log10_Nf`	log10 of the cycles to failure

- **Method:** Morrow: sigma_ar = sa/(1 - sm/sf'); SWT: $sigma_ar = sqrt$(sigma_max sa); Walker: sigma_max^(1-gamma) sa^γ; Nf = (sigma_ar/sf')^$(1/b)/2$
- **Assumes:** $-0.2 < b < 0$; Morrow/Goodman: $sm < sf'$; SWT/Walker: sigma_max > 0; goodman here uses sf' as the strength (Morrow form)

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('fatigue_life_mean_stress_correction', sa=300000000.0,
sm=100000000.0, sf=1200000000.0, b=-0.09, method='swt')
# -> sigma_ar_MPa = 346.41, Nf = 494791, log10_Nf = 5.69442   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('fatigue_life_mean_stress_correction', sa=300000000.0,
sm=100000000.0, sf=1200000000.0, b=-0.09, method='morrow')
# -> sigma_ar_MPa = 327.273, Nf = 930363, log10_Nf = 5.96865   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Fatigue life with mean stress: sa = 300 MPa, sm = 100 MPa, sf = 1200 MPa, b = -0.09, method = swt	SOLVED	sigma_ar_MPa = 346.41, Nf = 494791, log10_Nf = 5.69442
Fatigue life with mean stress: sa = 300 MPa, sm = 100 MPa, sf = 1200 MPa, b = -0.09, method = walker	INVALID_INPUT	Fatigue life with a mean-stress correction (Morrow, Goodman, SWT, Walker) and Basquin's law: the Walker exponent gamma m

Fatigue life from the S-N line (10^3 to 10^6 cycles) — `sn_fatigue_life`

Cycles to failure of a fully reversed amplitude on the straight log-log S-N line through f*Sut at 10^3 cycles and Se at 10^6 cycles, with the line's constants a and b.

$$a = \frac{(fS_{ut})^2}{S_e}, \quad b = -\frac{1}{3}\log_{10}\frac{fS_{ut}}{S_e}, \quad N = \left(\frac{\sigma_a}{a}\right)^{1/b}$$

input	meaning	condition
`Sut`	ultimate tensile strength	
`Se`	endurance limit	
`f`	fatigue strength fraction at 10^3 cycles (e.g. 0.9)	
`sa`	fully reversed stress amplitude	

output	meaning
`a_MPa`	S-N constant a, MPa
`b`	S-N exponent b
`N`	cycles to failure
`log10_N`	log10 of the cycles to failure

- **Method:** a = (f Sut)^2/Se, $b = -log10$(f Sut/Se)/3, $N = (sa/a)^{1/b}$
- **Assumes:** $Se < sa < f$ Sut; $0 < f < 1$; fully reversed loading
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('sn_fatigue_life', Sut=700000000.0, Se=300000000.0, f=0.85,
sa=400000000.0)
# -> a_MPa = 1180.08, b = -0.0991319, N = 54912.6, log10_N = 4.73967   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
S-N fatigue life: Sut = 700 MPa, Se = 300 MPa, f = 0.85, sa = 400 MPa	SOLVED	a_MPa = 1180.08, b = -0.0991319, N = 54912.6, log10_N = 4.73967
S-N fatigue life: Sut = 700 MPa, Se = 300 MPa, f = 0.85, sa = 250 MPa	INVALID_INPUT	Fatigue life from the S-N line (10^3 to 10^6 cycles): the amplitude must lie between Se (infinite life) and f*Sut (low-c

Fatigue strength at a given number of cycles (S-N line) — `sn_fatigue_strength`

The fully reversed fatigue strength at N cycles (10^3 to 10^6) on the straight log-log S-N line through f*Sut at 10^3 and Se at 10^6 cycles.

$$S_f = aN^b$$

input	meaning	condition
`Sut`	ultimate tensile strength	
`Se`	endurance limit	
`f`	fatigue strength fraction at 10^3 cycles	
`N`	number of cycles (10^3 to 10^6)	

output	meaning
`Sf_MPa`	fatigue strength at N cycles, MPa

- **Method:** $Sf = a\,N^b$ with a = (f Sut)^2/Se, $b = -log10$(f Sut/Se)/3
- **Assumes:** $10^3 \le N \le 10^6$; $0 < f < 1$, f Sut > Se
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('sn_fatigue_strength', Sut=700000000.0, Se=300000000.0, f=0.85,
N=1000.0)
# -> Sf_MPa = 595   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('sn_fatigue_strength', Sut=700000000.0, Se=300000000.0, f=0.85,
N=1000000.0)
# -> Sf_MPa = 300   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
S-N fatigue strength: Sut = 700 MPa, Se = 300 MPa, f = 0.85, N = 1000000	SOLVED	Sf_MPa = 300
S-N fatigue strength: Sut = 700 MPa, Se = 300 MPa, f = 0.85, N = 100	INVALID_INPUT	Fatigue strength at a given number of cycles (S-N line): N must lie between 10^3 and 10^6 cycles.

Miner's rule damage with a single-slope S-N curve — `miner_damage_sn_curve`

The Palmgren-Miner damage sum of stress-range blocks on an S-N curve $N = Nc\,(Sc/S)^m$, and how many repeats of the spectrum the part survives.

$$D = \sum_i \frac{n_i}{N_i}, \quad N_i = N_c\Big(\frac{S_c}{S_i}\Big)^m$$

input	meaning	condition
`ranges`	stress ranges in MPa (a list)	
`counts`	number of cycles of each range (a list)	
`Sc`	reference stress range of the curve at Nc cycles, MPa	
`m`	S-N slope (e.g. 3 for welded details)	
`Nc`	cycles at Sc (default 2e6)	

output	meaning
`damage`	Miner damage sum D
`repeats_to_failure`	repeats of the spectrum until $D = 1$

- **Method:** D = sum n_i / N_i, $N_i = Nc\,(Sc/S_i)^m$
- **Assumes:** linear damage accumulation (Palmgren-Miner); a single-slope curve without endurance limit
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('miner_damage_sn_curve', ranges='90', counts='1000000', Sc=90.0, m=3.0)
# -> damage = 0.5, repeats_to_failure = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Miner's rule: ranges = [90], counts = [1000000], Sc = 90, m = 3	SOLVED	damage = 0.5, repeats_to_failure = 2
Miner's rule: ranges = [120, 80], counts = [20000], Sc = 90, m = 3	INVALID_INPUT	Miner's rule damage with a single-slope S-N curve: ranges and counts must have the same number of entries.

Fatigue damage on the trilinear (Eurocode 3) S-N curve — `eurocode_fatigue_damage`

The Miner damage of stress-range blocks on the trilinear S-N curve of a detail category (slope 3 to 5e6 cycles, slope 5 to 1e8, cut-off below), with the constant-amplitude and cut-off limits.

$$N = 2\cdot 10^6\Big(\frac{\Delta\sigma_C}{\Delta\sigma}\Big)^3 \ (\Delta\sigma \geq \Delta\sigma_D), \quad N = 5\cdot 10^6\Big(\frac{\Delta\sigma_D}{\Delta\sigma}\Big)^5$$

input	meaning	condition
`ranges`	stress ranges in MPa (a list)	
`counts`	cycles of each range (a list)	

input	meaning	condition
`dsc`	detail category: stress range at 2e6 cycles, MPa	

output	meaning
`damage`	Miner damage sum
`delta_sigma_D`	constant-amplitude fatigue limit (5e6 cycles), MPa
`delta_sigma_L`	cut-off limit (1e8 cycles), MPa
`repeats_to_failure`	repeats of the spectrum until $D = 1$

- **Method:** $N = 2e6$ (dsc/S)^3 for $S \geq sD = 0.737$ dsc; $N = 5e6\ (sD/S)^5$ for $sL \leq S < sD$ with $sL = 0.549$ sD; no damage below sL
- **Assumes:** partial safety factors not applied $(gamma_Mf = gamma_Ff = 1)$; at least one range above the cut-off
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('eurocode_fatigue_damage', ranges='90', counts='1000000',
dsc=90.0)
# -> damage = 0.5, delta_sigma_D = 66.3126, delta_sigma_L = 36.4242, repeats_to_failure = 2
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Eurocode fatigue damage: ranges = [90], counts = [1000000], dsc = 90	SOLVED	damage = 0.5, delta_sigma_D = 66.3126, delta_sigma_L = 36.4242, repeats_to_failure = 2
Eurocode fatigue damage: ranges = [10], counts = [1000000], dsc = 90	INVALID_INPUT	Fatigue damage on the trilinear (Eurocode 3) S-N curve: every range is below the cut-off limit: no damage.

Rainflow cycle counting of a load history (ASTM E1049) — `rainflow_counting`

The cycles in a load or stress history by rainflow counting (three-point method, half cycles for the residue), as distinct ranges with their counts.

$$|x_3 - x_2| \geq |x_2 - x_1| \Rightarrow \text{count } |x_2 - x_1|$$

input	meaning	condition
`history`	the load or stress history, point by point (a list; any one unit)	

output	meaning
`ranges`	distinct ranges, largest first (in the history's unit)
`counts`	cycles of each range (half cycles count 0.5)
`total_cycles`	total number of cycles
`max_range`	largest range

- **Method:** reversals (peaks and valleys), then the three-point rule: when a range Y is not larger than the next range X, count Y (a half cycle if it contains the start) and remove its points

- **Assumes:** at least 3 points, not constant; ASTM $E1049-85$: the first and last points are reversals, the residue counts as half cycles; ranges equal to 12 significant digits are merged
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('rainflow_counting', history='0, 5, -3, 4, -2, 6, -4, 1, 0')
# -> ranges = [10.0, 9.0, 8.0, 6.0, 5.0, 1.0], counts = [0.5, 0.5, 0.5, 1.0, 1.0, 0.5], total_cycles
= 4, max_range = 10   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Rainflow counting: history = [0, 5, -3, 4, -2, 6, -4, 1, 0]	SOLVED	ranges = [10.0, 9.0, 8.0, 6.0, 5.0, 1.0], counts = [0.5, 0.5, 0.5, 1.0, 1.0, 0.5], total_cycles = 4, max_range = 10
Rainflow counting: history = [2, 2, 2]	INVALID_INPUT	Rainflow cycle counting of a load history (ASTM E1049): the history is constant: there are no cycles.

Fatigue damage of a stress history (rainflow counting + Miner's rule) — `rainflow_fatigue_damage`

The Miner damage of one pass of a stress history: rainflow cycles (ASTM, half cycles) on a single-slope S-N curve $N = Nc\,(Sc/S)^m$, and the passes to failure.

$$D = \sum_i \frac{n_i}{N_c}\left(\frac{S_i}{S_c}\right)^m$$

input	meaning	condition
`history`	the stress history in MPa, point by point (a list)	
`Sc`	reference stress range at Nc cycles, MPa	
`m`	S-N slope	
`Nc`	cycles at Sc (default 2e6)	

output	meaning
`damage`	damage of one pass of the history
`repeats_to_failure`	passes until $D = 1$
`cycles`	cycles counted per pass

- **Method:** rainflow ranges S_i with counts n_i; D = sum n_i S_i^m/(Nc Sc^m)
- **Assumes:** ASTM $E1049-85$ rainflow counting, half cycles weighted 0.5; no mean-stress correction; single-slope S-N curve, no endurance limit
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('rainflow_fatigue_damage', history='0, 90, 0', Sc=90.0, m=3.0)
# -> damage = 5e-07, repeats_to_failure = 2e+06, cycles = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Rainflow fatigue damage: history = [0, 90, 0], Sc = 90, m = 3	SOLVED	damage = 5e-07, repeats_to_failure = 2e+06, cycles = 1
Rainflow fatigue damage: history = [0, 90], Sc = 90, m = 3	INVALID_INPUT	history: needs at least 3 entries, got 2.

Shaft fatigue safety factor (DE-Goodman) — `shaft_fatigue_safety_factor`

The fatigue factor of safety of a rotating solid shaft under alternating and mean bending and torsion by the distortion-energy Goodman criterion, with the von Mises stresses and (with Sy) the first-cycle yield factor.

$$\frac{1}{n} = \frac{16}{\pi d^3}\left\{\frac{\sqrt{4(K_f M_a)^2 + 3(K_{fs} T_a)^2}}{S_e} + \frac{\sqrt{4(K_f M_m)^2 + 3(K_{fs} T_m)^2}}{S_{ut}}\right\}$$

input	meaning	condition
`d`	shaft diameter	
`Ma`	alternating bending moment	
`Mm`	mean bending moment	
`Ta`	alternating torque	
`Tm`	mean torque	
`Kf`	fatigue stress-concentration factor, bending	
`Kfs`	fatigue stress-concentration factor, torsion	
`Se`	fully corrected endurance limit	
`Sut`	ultimate tensile strength	
`Sy`	yield strength (optional)	

output	meaning
`n_fatigue`	fatigue safety factor (DE-Goodman)
`sigma_a_vm_MPa`	alternating von Mises stress, MPa
`sigma_m_vm_MPa`	mean von Mises stress, MPa
`n_yield`	first-cycle yield factor (with Sy)

- **Method:** sigma_a' = sqrt((32 Kf Ma/(pi d^3))^$\mathbf{2} + \mathbf{3}$(16 Kfs Ta/(pi d^3))^2), likewise sigma_m'; $1/n = \boldsymbol{sigma_a}$'/Se + sigma_m'/Sut
- **Assumes:** solid round shaft, no axial load; Kf, Kfs >= 1; an alternating moment or torque present
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('shaft_fatigue_safety_factor', d=0.04, Ma=150.0, Mm=0.0, Ta=0.0,
Tm=200.0, Kf=1.6, Kfs=1.4, Se=200000000.0, Sut=700000000.0)
# -> n_fatigue = 4.06308, sigma_a_vm_MPa = 38.1972, sigma_m_vm_MPa = 38.593   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Shaft fatigue safety factor: d = 40 mm, Ma = 150 N*m, Mm = 0 N*m, Ta = 0 N*m, Tm = 200 N*m, Kf = 1.6, Kfs = 1.4, Se = 200 MPa, Sut = 700 MPa	SOLVED	n_fatigue = 4.06308, sigma_a_vm_MPa = 38.1972, sigma_m_vm_MPa = 38.593
Shaft fatigue safety factor: d = 40 mm, Ma = 150 N*m, Mm = 0 N*m, Ta = 0 N*m, Tm = 200 N*m, Kf = 0.6, Kfs = 1.4, Se = 200 MPa, Sut = 700 MPa	INVALID_INPUT	Shaft fatigue safety factor (DE-Goodman): fatigue stress-concentration factors are >= 1.

Shaft diameter for a design factor (DE-Goodman) — `shaft_diameter_de_goodman`

The smallest solid-shaft diameter giving a required fatigue design factor under alternating and mean bending and torsion (distortion-energy Goodman).

$$d = \left(\frac{16n}{\pi}\left\{\frac{\sqrt{4(K_f M_a)^2 + 3(K_{fs} T_a)^2}}{S_e} + \frac{\sqrt{4(K_f M_m)^2 + 3(K_{fs} T_m)^2}}{S_{ut}}\right\}\right)^{1/3}$$

input	meaning	condition
`n`	required fatigue design factor (>= 1)	
`Ma`	alternating bending moment	
`Mm`	mean bending moment	
`Ta`	alternating torque	
`Tm`	mean torque	
`Kf`	fatigue stress-concentration factor, bending	
`Kfs`	fatigue stress-concentration factor, torsion	
`Se`	fully corrected endurance limit	
`Sut`	ultimate tensile strength	

output	meaning
`d_mm`	required diameter, mm

- **Method:** d = (16n/pi [sqrt(4(Kf Ma)^2 + 3(Kfs Ta)^2)/Se + sqrt(4(Kf Mm)^2 + 3(Kfs Tm)^2)/Sut])^$(1/3)$
- **Assumes:** solid round shaft; the stress-concentration factors for the final size are assumed known; $n \geq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('shaft_diameter_de_goodman', n=1.5, Ma=150.0, Mm=0.0, Ta=0.0,
Tm=200.0, Kf=1.6, Kfs=1.4, Se=200000000.0, Sut=700000000.0)
# -> d_mm = 28.6949   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Shaft diameter DE-Goodman: n = 1.5, Ma = 150 N*m, Mm = 0 N*m, Ta = 0 N*m, Tm = 200 N*m, Kf = 1.6, Kfs = 1.4, Se = 200 MPa, Sut = 700 MPa	SOLVED	d_mm = 28.6949

question	outcome	result
Shaft diameter DE-Goodman: n = 0.5, Ma = 150 N*m, Mm = 0 N*m, Ta = 0 N*m, Tm = 200 N*m, Kf = 1.6, Kfs = 1.4, Se = 200 MPa, Sut = 700 MPa	INVALID_INPUT	Shaft diameter for a design factor (DE-Goodman): the design factor n must be at least 1.

Belt or rope friction on a drum (capstan equation) — `belt_friction_capstan`

The largest tight-side tension a flat belt, rope or V-belt can hold for a given slack-side tension, friction coefficient and wrap angle.

$$\frac{T_1}{T_2} = e^{\mu\theta} \quad \left(e^{\mu\theta/\sin(\beta/2)} \text{ for a V-belt}\right)$$

input	meaning	condition
`T2`	slack-side tension	
`mu`	coefficient of friction	
`theta`	angle of wrap	
`beta`	included groove angle of a V-belt (optional; flat belt when absent)	

output	meaning
`T1`	tight-side tension at impending slip
`ratio`	tension ratio T_1/T_2
`friction_force`	net friction force $T_1 - T_2$

- **Method:** $T_1/T_2 = exp$(mu theta), V-belt exp(mu theta / sin(beta/2))
- **Assumes:** impending slip, no centrifugal effect; $mu > 0$, wrap > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('belt_friction_capstan', T2=200.0, mu=0.3,
theta=3.141592653589793)
# -> T1 = 513.266 N, ratio = 2.56633, friction_force = 313.266 N   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('belt_friction_capstan', T2=200.0, mu=0.3,
theta=3.141592653589793, beta=0.6283185307179586)
# -> T1 = 4222.74 N, ratio = 21.1137, friction_force = 4022.74 N   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Capstan equation: T2 = 200 N, mu = 0.3, theta = 180 deg	SOLVED	T1 = 513.266 N, ratio = 2.56633, friction_force = 313.266 N
Capstan equation: T2 = 200 N, mu = 0, theta = 180 deg	INVALID_INPUT	Belt or rope friction on a drum (capstan equation): mu and the wrap angle must be a positive number.

Power of a belt drive with centrifugal tension — `belt_drive_power`

The centrifugal tension, slack-side tension and transmitted power of a flat or V-belt running at the allowable tight-side tension and impending slip.

$$\frac{T_1 - mv^2}{T_2 - mv^2} = e^{\mu\theta}, \quad P = (T_1 - T_2)v$$

input	meaning	condition
`T1`	allowable tight-side tension	
`m`	belt mass per unit length	
`v`	belt speed	
`mu`	coefficient of friction	
`theta`	wrap angle on the smaller pulley	
`beta`	included groove angle for a V-belt (optional)	

output	meaning
`Tc`	centrifugal tension m v^2
`T2`	slack-side tension
`P`	transmitted power
`P_kW`	transmitted power in kW

- **Method:** $(T_1 - Tc)/(T_2 - Tc) = exp$(mu theta) (mu/sin(beta/2) for a V-belt), $Tc = m\,v^2$, $P = (T_1 - T_2)$ v
- **Assumes:** impending slip on the smaller pulley; m $v^2 < T_1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('belt_drive_power', T1=1200.0, m=0.4, v=20.0, mu=0.3,
theta=2.792526803190927)
# -> Tc = 160 N, T2 = 609.987 N, P = 11800.3 W, P_kW = 11.8003   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Belt drive power: T1 = 1.2 kN, m = 0.4 kg/m, v = 20 m/s, mu = 0.3, theta = 160 deg	SOLVED	Tc = 160 N, T2 = 609.987 N, P = 11800.3 W, P_kW = 11.8003
Belt drive power: T1 = 100 N, m = 0.4 kg/m, v = 20 m/s, mu = 0.3, theta = 160 deg	INVALID_INPUT	Power of a belt drive with centrifugal tension: the centrifugal tension m v^2 reaches the allowable tension: the belt tr

Power screw: raising and lowering torque, efficiency, self-locking — `power_screw_torque`

The torque to raise and to lower a load with a square or Acme power screw (thread and collar friction), the raising efficiency and whether the screw is self-locking.

$$T_R = \frac{F d_m}{2}\,\frac{l + \pi f d_m \sec\alpha}{\pi d_m - f l \sec\alpha} + \frac{F f_c d_c}{2}$$

input	meaning	condition
`F`	axial load	
`dm`	mean thread diameter	
`l`	lead	
`f`	thread friction coefficient	
`alpha`	thread half-angle (0 for square, 14.5 deg for Acme; default 0)	
`fc`	collar friction coefficient (default 0)	
`dc`	collar mean diameter (needed with fc)	

output	meaning
`T_raise`	torque to raise the load
`T_lower`	torque to lower the load (negative: the load lowers itself)
`efficiency`	efficiency when raising
`self_locking`	yes if the thread alone holds the load

- **Method:** $T_R = F\,dm/2$ (l + pi f dm sec a)/(pi dm - f l sec a) + F fc $dc/2$; $T_L = F\,dm/2$ (pi f dm sec a - l)/(pi dm + f l sec a) + F fc $dc/2$; $e = F$ l/(2 pi T_R)
- **Assumes:** $\alpha < 45$ degrees; thread friction taken as f/cos(alpha); single thrust collar
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('power_screw_torque', F=6400.0, dm=0.03, l=0.012, f=0.08,
fc=0.08, dc=0.04)
# -> T_raise = 30.3479 N*m, T_lower = 5.74271 N*m, efficiency = 0.402766, self_locking = no
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Power screw: F = 6.4 kN, dm = 30 mm, l = 12 mm, f = 0.08, fc = 0.08, dc = 40 mm	SOLVED	T_raise = 30.3479 N*m, T_lower = 5.74271 N*m, efficiency = 0.402766, self_locking = no
Power screw: F = 6.4 kN, dm = 30 mm, l = 12 mm, f = 0.08, fc = 0.08	INVALID_INPUT	Power screw: raising and lowering torque, efficiency, self-locking: a collar friction coefficient needs the collar mean

Torque capacity of a disc clutch or brake — `disc_clutch_torque`

The friction torque and the largest contact pressure of a plate (disc) clutch or brake with an annular friction face, by the uniform-wear or uniform-pressure assumption.

$$T_{wear} = \frac{F f (D + d)}{4}, \quad T_{press} = \frac{F f}{3}\,\frac{D^3 - d^3}{D^2 - d^2}$$

input	meaning	condition
`F`	axial actuating force	
`f`	coefficient of friction	
`D`	outer diameter of the friction face	

input	meaning	condition
`d`	inner diameter of the friction face	
`model`	uniform wear (worn-in) or uniform pressure (new)	
`pairs`	number of friction-surface pairs (default 1)	

output	meaning
`T`	torque capacity
`p_max_MPa`	largest contact pressure, MPa

- **Method:** uniform wear: $T = F$ f $(D + d)/4$, p_max = 2F/(pi d (D - d)); uniform pressure: $T = F$ f (D^3 - d^3)/(3(D^2 - d^2)), p = 4F/(pi $(D^2 - d^2)$); times the number of pairs
- **Assumes:** $d < D$; the same force on every pair
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('disc_clutch_torque', F=4000.0, f=0.3, D=0.2, d=0.1,
model='uniform wear')
# -> T = 90 N*m, p_max_MPa = 0.254648   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('disc_clutch_torque', F=4000.0, f=0.3, D=0.2, d=0.1,
model='uniform pressure', pairs=2)
# -> T = 186.667 N*m, p_max_MPa = 0.169765   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Disc clutch: F = 4 kN, f = 0.3, D = 200 mm, d = 100 mm, model = uniform wear	SOLVED	T = 90 N*m, p_max_MPa = 0.254648
Disc clutch: F = 4 kN, f = 0.3, D = 200 mm, d = 100 mm, model = uniform pressure, pairs = 2	SOLVED	T = 186.667 N*m, p_max_MPa = 0.169765
Disc clutch: F = 4 kN, f = 0.3, D = 100 mm, d = 200 mm, model = uniform wear	INVALID_INPUT	Torque capacity of a disc clutch or brake: the inner diameter must be smaller than the outer diameter.

Speeds of a simple planetary (epicyclic) gear train — `planetary_gear_train`

The third speed of a sun-planet-ring-carrier train from the other two (any member may be held: speed 0), with the planets' absolute speed and tooth number.

$$\frac{\omega_s - \omega_c}{\omega_r - \omega_c} = -\frac{N_r}{N_s}$$

input	meaning	condition
`Ns`	teeth on the sun	
`Nr`	teeth on the ring (internal gear)	
`ws`	sun speed (signed; rpm or rad/s)	
`wr`	ring speed (signed)	
`wc`	carrier (arm) speed (signed)	

output	meaning
`ws_rpm`	sun speed, rpm
`wr_rpm`	ring speed, rpm
`wc_rpm`	carrier speed, rpm
`wp_rpm`	absolute speed of each planet, rpm
`Np`	teeth on each planet

- **Method:** Willis: $(ws - wc)/(wr - wc) = -Nr/Ns$; planet: $wp = wc - (ws - wc)$ Ns/Np, $Np = (Nr - Ns)/2$
- **Assumes:** exactly two of the three speeds given; $Nr > Ns$, Nr - Ns even; same sign convention for every speed
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('planetary_gear_train', Ns=20, Nr=80, ws=125.66370614359171,
wr=0.0)
# -> ws_rpm = 1200, wr_rpm = 0, wc_rpm = 240, wp_rpm = -400, Np = 30   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('planetary_gear_train', Ns=20, Nr=80, wc=10.471975511965976,
wr=0.0)
# -> ws_rpm = 500, wr_rpm = 0, wc_rpm = 100, wp_rpm = -166.667, Np = 30   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Planetary gear train: Ns = 20, Nr = 80, ws = 1200 rpm, wr = 0 rpm	SOLVED	ws_rpm = 1200, wr_rpm = 0, wc_rpm = 240, wp_rpm = -400, Np = 30
Planetary gear train: Ns = 20, Nr = 81, ws = 1200 rpm, wr = 0 rpm	INVALID_INPUT	Speeds of a simple planetary (epicyclic) gear train: need Nr > Ns > 0 with Nr - Ns even (the planets have (Nr - Ns)/2 te
Planetary gear train: Ns = 20, Nr = 80, ws = 1200 rpm, wr = 0 rpm, wc = 10 rpm	INVALID_INPUT	Speeds of a simple planetary (epicyclic) gear train: give exactly two of the three speeds ws, wr, wc.

Spur gear mesh geometry and contact ratio — `spur_gear_contact_ratio`

Pitch diameters, centre distance, length of the path of contact, base pitch and contact ratio of a pair of standard full-depth spur gears.

$$Z = \sqrt{r_{a1}^2 - r_{b1}^2} + \sqrt{r_{a2}^2 - r_{b2}^2} - C\sin\phi, \quad m_c = \frac{Z}{\pi m \cos\phi}$$

input	meaning	condition
`N1`	teeth on the pinion	
`N2`	teeth on the gear	
`m`	module (e.g. 4 mm)	
`phi`	pressure angle	

output	meaning
`d1_mm`	pinion pitch diameter, mm
`d2_mm`	gear pitch diameter, mm
`C_mm`	centre distance, mm
`Z_mm`	length of the path of contact, mm
`pb_mm`	base pitch, mm
`contact_ratio`	contact ratio Z / p_b

- **Method:** $Z = sqrt$(ra1^2 - rb1^2) + sqrt(ra2^2 - rb2^2) - C sin(phi), $rb = r$ cos(phi), $ra = r + m$, $pb = pi$ m cos(phi), $m_c = Z/pb$
- **Assumes:** standard full-depth teeth: addendum = 1 module; standard centre distance; no interference (refused otherwise); 10 <= pressure angle <= 30 degrees
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('spur_gear_contact_ratio', N1=20, N2=50, m=0.004,
phi=0.3490658503988659)
# -> d1_mm = 80, d2_mm = 200, C_mm = 140, Z_mm = 19.552, pb_mm = 11.8085, contact_ratio = 1.65576
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Spur gear contact ratio: N1 = 20, N2 = 50, m = 4 mm, phi = 20 deg	SOLVED	d1_mm = 80, d2_mm = 200, C_mm = 140, Z_mm = 19.552, pb_mm = 11.8085, contact_ratio = 1.65576
Spur gear contact ratio: N1 = 8, N2 = 50, m = 4 mm, phi = 20 deg	INVALID_INPUT	Spur gear mesh geometry and contact ratio: involute interference: an addendum passes the other gear's base-circle tangen

Rolling-bearing life at a given reliability — `bearing_life_reliability`

The rating (L10) life of a ball or roller bearing from its catalogue rating and load, and the life at another reliability from the three-parameter Weibull distribution, in revolutions and hours.

$$L_{10} = L_R\left(\frac{C}{F}\right)^{a}, \quad x = x_0 + (\theta - x_0)\left(\ln\frac{1}{R}\right)^{1/b}$$

input	meaning	condition
`C`	catalogue (basic dynamic) load rating	
`F`	equivalent radial load	
`n`	speed (rpm)	
`R`	required reliability (e.g. 0.95 or 95%)	
`kind`	ball $(a = 3)$ or roller $(a = 10/3)$ bearing	
`LR`	rating life of the catalogue, revolutions (default 1e6)	
`x0`	Weibull guaranteed life multiple (default 0.02)	
`theta`	Weibull characteristic life multiple (default 4.459)	

input	meaning	condition
`b`	Weibull shape (default 1.483)	

output	meaning
`L10_rev`	rating life (90 % reliability), revolutions
`L10_hours`	rating life in hours
`life_multiple`	life at R as a multiple of the rating life
`L_rev`	life at reliability R, revolutions
`L_hours`	life at reliability R, hours

- **Method:** $L10 = LR\,(C/F)^a$; $x = x_0 + (\theta - x_0)$ (ln 1/R)^(1/b); $L = x$ L10; hours = revolutions / (60 n_rpm)
- **Assumes:** $0 < R < 1$; Weibull parameters of the catalogue (defaults: $x_0 = 0.02$, $\theta = 4.459$, b = 1.483); constant load and speed
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('bearing_life_reliability', C=30000.0, F=4000.0,
n=157.07963267948963, R=0.9, kind='ball')
# -> L10_rev = 4.21875e+08, L10_hours = 4687.5, life_multiple = 0.993348, L_rev = 4.19069e+08,
L_hours = 4656.32   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Bearing life: C = 30 kN, F = 4 kN, n = 1500 rpm, R = 0.95, kind = ball	SOLVED	L10_rev = 4.21875e+08, L10_hours = 4687.5, life_multiple = 0.619056, L_rev = 2.61164e+08, L_hours = 2901.83
Bearing life: C = 30 kN, F = 4 kN, n = 1500 rpm, R = 95%, kind = ball	SOLVED	L10_rev = 4.21875e+08, L10_hours = 4687.5, life_multiple = 0.619056, L_rev = 2.61164e+08, L_hours = 2901.83
Bearing life: C = 30 kN, F = 4 kN, n = 1500 rpm, R = 1.2, kind = ball	INVALID_INPUT	Rolling-bearing life at a given reliability: the reliability must lie strictly between 0 and 1.

Required catalogue rating of a rolling bearing — `bearing_catalog_rating`

The basic dynamic load rating C10 a ball or roller bearing needs for a design load, life and speed at a required reliability (three-parameter Weibull, application factor).

$$C_{10} = a_f F_D \left[\frac{x_D}{x_0 + (\theta - x_0)(\ln 1/R_D)^{1/b}}\right]^{1/a}$$

input	meaning	condition
`FD`	design (equivalent radial) load	
`LD`	design life (hours)	
`n`	speed (rpm)	
`R`	required reliability	
`kind`	ball $(a = 3)$ or roller $(a = 10/3)$	

input	meaning	condition
`af`	application (load) factor (default 1)	
`LR`	rating life of the catalogue, revolutions (default 1e6)	
`x0`	Weibull guaranteed life multiple (default 0.02)	
`theta`	Weibull characteristic life multiple (default 4.459)	
`b`	Weibull shape (default 1.483)	

output	meaning
`C10`	required catalogue rating
`C10_kN`	required rating in kN
`design_life_multiple`	design life as a multiple of LR

- **Method:** $C10 = af$ FD [xD/(x0 + (theta - x0)(ln 1/R)^(1/b))]^$(1/a)$, $xD = LD$ n $60/LR$
- **Assumes:** $0 < R < 1$; Weibull defaults $x_0 = 0.02$, $\theta = 4.459$, $b = 1.483$ and $LR = 1e6$ revolutions unless given
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('bearing_catalog_rating', FD=1800.0, LD=108000000.0,
n=52.35987755982988, R=0.99, kind='ball', af=1.2)
# -> C10 = 34567.2 N, C10_kN = 34.5672, design_life_multiple = 900   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Bearing catalogue rating: FD = 1.8 kN, LD = 30000 h, n = 500 rpm, R = 0.99, kind = ball, af = 1.2	SOLVED	C10 = 34567.2 N, C10_kN = 34.5672, design_life_multiple = 900

Helical compression spring: rate, deflection and shear stress — `helical_compression_spring`

The spring index, rate, deflection under a load and the wire shear stress (torsion alone, and with the direct-shear factor Ks = 1 + 0.5/C) of a round-wire helical spring.

$$k = \frac{Gd^4}{8D^3N_a}, \quad \tau = K_s\frac{8FD}{\pi d^3}, \quad K_s = 1 + \frac{0.5}{C}$$

input	meaning	condition
`d`	wire diameter	
`D`	mean coil diameter	
`Na`	number of active coils	
`G`	shear modulus of the wire	
`F`	axial load	

output	meaning
`index`	spring index $C = D/d$
`k`	spring rate

output	meaning
`deflection_mm`	deflection under F, mm
`tau_torsion_MPa`	torsional shear stress 8FD/(pi d^3), MPa
`tau_Ks_MPa`	shear stress with the direct-shear factor Ks, MPa

- **Method:** $k = G$ d^4/(8 D^3 Na), $\delta = F/k$, $\tau = Ks$ 8FD/(pi d^3) with $Ks = 1 + 0.5/C$
- **Assumes:** $C > 2$; close-coiled (small helix angle); curvature (Wahl) correction not included
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('helical_compression_spring', d=0.003, D=0.024, Na=10.0,
G=79000000000.0, F=100.0)
# -> index = 8, k = 5786.13 N/m, deflection_mm = 17.2827, tau_torsion_MPa = 226.354, tau_Ks_MPa =
240.501   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Helical spring: d = 3 mm, D = 24 mm, Na = 10, G = 79 GPa, F = 100 N	SOLVED	index = 8, k = 5786.13 N/m, deflection_mm = 17.2827, tau_torsion_MPa = 226.354, tau_Ks_MPa = 240.501
Helical spring: d = 3 mm, D = 5 mm, Na = 10, G = 79 GPa, F = 100 N	INVALID_INPUT	Helical compression spring: rate, deflection and shear stress: the spring index D/d must exceed 2.

Natural frequency of a spring-mass system — `sdof_natural_frequency`

The natural circular frequency, frequency, period and static deflection (under gravity) of an undamped single-degree-of-freedom spring-mass system.

$$\omega_n = \sqrt{\frac{k}{m}}, \quad f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{g}{\delta_{st}}}$$

input	meaning	condition
`m`	mass	
`k`	spring stiffness	

output	meaning
`omega_n`	natural circular frequency
`f_n`	natural frequency
`period`	period
`static_deflection_mm`	static deflection m g/k in mm (g = 9.80665 m/s^2)

- **Method:** $\omega_n = \sqrt{\frac{k}{m}}$, f_n = omega_n/(2 pi), $T = 1/f_n$, $delta_st = m$ g/k
- **Assumes:** m, $k > 0$; linear spring, no damping
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('sdof_natural_frequency', m=20.0, k=8000.0)
# -> omega_n = 20 rad/s, f_n = 3.1831 Hz, period = 0.314159 s, static_deflection_mm = 24.5166
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
SDOF natural frequency: m = 20 kg, k = 8 kN/m	SOLVED	omega_n = 20 rad/s, f_n = 3.1831 Hz, period = 0.314159 s, static_deflection_mm = 24.5166
Natural frequency of a spring-mass system: m = 20 kg, k = 8000 N/m	SOLVED	omega_n = 20 rad/s, f_n = 3.1831 Hz, period = 0.314159 s, static_deflection_mm = 24.5166
SDOF natural frequency: m = 20 kg	NEEDS_INPUT	sdof_natural_frequency needs k (spring stiffness). Example: solve('sdof_natural_frequency', m='20 kg', k='8000 N/m').

Damping ratio, damped frequency and logarithmic decrement — `damped_sdof_parameters`

The natural frequency, damping ratio, critical damping, damped natural frequency and logarithmic decrement of a viscously damped spring-mass system, and whether it is under- or overdamped.

$$\zeta = \frac{c}{2\sqrt{km}}, \quad \omega_d = \omega_n\sqrt{1-\zeta^2}, \quad \delta = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$$

input	meaning	condition
`m`	mass	
`c`	viscous damping coefficient (Ns/m)	
`k`	stiffness	

output	meaning
`omega_n`	undamped natural frequency
`zeta`	damping ratio c/c_c
`omega_d`	damped natural frequency (0 if not underdamped)
`f_d`	damped natural frequency in Hz
`c_critical`	critical damping 2 sqrt(k m)
`log_decrement`	logarithmic decrement (0 if not underdamped)
`regime`	underdamped, critically damped or overdamped

- **Method:** zeta = c/(2 sqrt(k m)), $omega_d = omega_n$ sqrt(1 - zeta^2), $\delta = 2$ pi zeta/sqrt(1 - zeta^2)
- **Assumes:** m, $k > 0$, $c \geq 0$; linear viscous damping
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('damped_sdof_parameters', m=10.0, c=40.0, k=4000.0)
# -> omega_n = 20 rad/s, zeta = 0.1, omega_d = 19.8997 rad/s, f_d = 3.16714 Hz, c_critical = 400
kg/s, log_decrement = 0.631484, regime = underdamped   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Damped SDOF: m = 10 kg, c = 40 N*s/m, k = 4000 N/m	SOLVED	omega_n = 20 rad/s, zeta = 0.1, omega_d = 19.8997 rad/s, f_d = 3.16714 Hz, c_critical = 400 kg/s, log_decrement = 0.631484, regime = underdamped
Damping ratio: m = 10 kg, c = 800 N*s/m, k = 4000 N/m	SOLVED	omega_n = 20 rad/s, zeta = 2, omega_d = 0 rad/s, f_d = 0 Hz, c_critical = 400 kg/s, log_decrement = 0, regime = overdamped

Free vibration response of a damped spring-mass system — `sdof_free_response`

The displacement and velocity at time t of a damped single-degree-of-freedom system released from an initial displacement and velocity (under-, critically or overdamped).

$$x(t) = e^{-\zeta\omega_n t}\left(x_0\cos\omega_d t + \frac{v_0 + \zeta\omega_n x_0}{\omega_d}\sin\omega_d t\right)$$

input	meaning	condition
`m`	mass	
`c`	damping coefficient	
`k`	stiffness	
`x0`	initial displacement	
`v0`	initial velocity	
`t`	time	

output	meaning
`x`	displacement at t
`x_mm`	displacement at t in mm
`v`	velocity at t

- **Method:** underdamped x = e^(-zeta wn t)(x0 cos wd t + (v0 + zeta wn x0)/wd sin wd t); critical x = (x0 + (v0 + wn x0) t) e^(-wn t); overdamped sum of two exponentials
- **Assumes:** linear viscous damping; no external force; zeta within $1e-9$ of 1 is treated as critical
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('sdof_free_response', m=1.0, c=2.0, k=100.0, x0=0.01, v0=0.0,
t=0.5)
# -> x = 0.000985507 m, x_mm = 0.985507, v = 0.0588697 m/s   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('sdof_free_response', m=1.0, c=20.0, k=100.0, x0=0.01, v0=0.0,
t=0.1)
# -> x = 0.00735759 m, x_mm = 7.35759, v = -0.0367879 m/s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
SDOF free response: m = 1 kg, c = 20 N*s/m, k = 100 N/m, x0 = 10 mm, v0 = 0 m/s, t = 0.1 s	SOLVED	x = 0.00735759 m, x_mm = 7.35759, v = -0.0367879 m/s

question	outcome	result
SDOF free response: m = 1 kg, c = 20 N*s/m, k = 100 N/m, x0 = 10 mm, v0 = 0 m/s	NEEDS_INPUT	sdof_free_response needs t (time). Example: solve('sdof_free_response', m='1 kg', c='2 kg/s', k='100 N/m', x0='0.01 m',

Steady-state response to a harmonic force — `forced_vibration_amplitude`

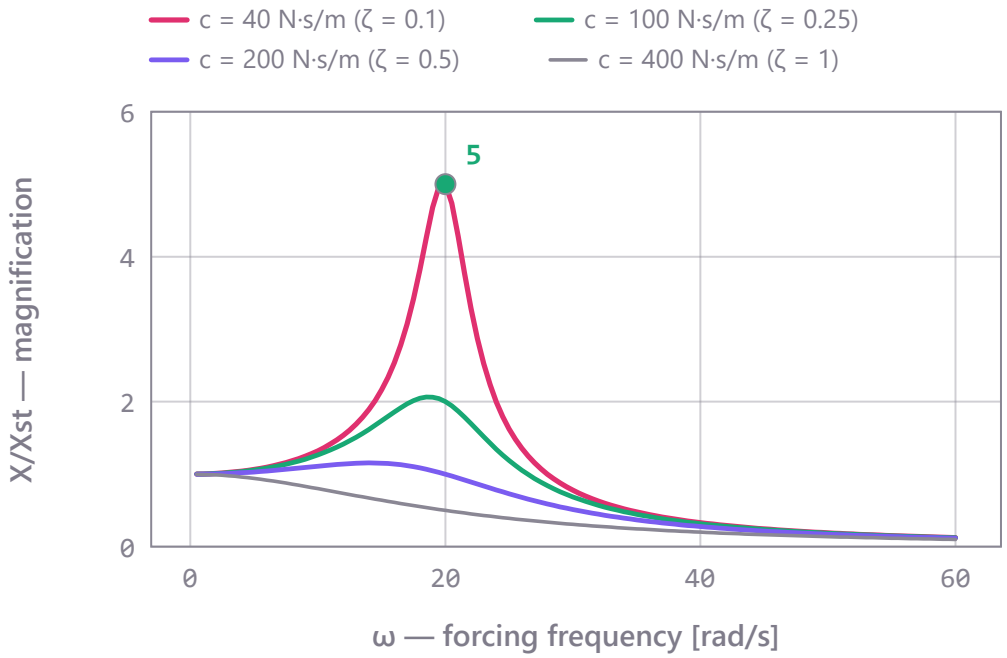
The steady-state amplitude, phase lag and dynamic magnification factor of a damped spring-mass system driven by a harmonic force F0 cos(omega t).

$$X = \frac{F_0/k}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \quad \tan\phi = \frac{2\zeta r}{1-r^2}$$

input	meaning	condition
`m`	mass	
`c`	damping coefficient	
`k`	stiffness	
`F0`	force amplitude	
`omega`	forcing frequency (rad/s or rpm)	

output	meaning
`X`	steady-state amplitude
`X_mm`	amplitude in mm
`phase_deg`	phase lag of the displacement behind the force, degrees (0 to 180)
`magnification`	dynamic magnification X k/F_0
`r`	frequency ratio omega/omega_n

- **Method:** $X = F_0/sqrt$((k - m $w^2)^2 + (c$ w)^2), phi = atan2(c w, k - m w^2)
- **Assumes:** steady state (transient decayed); not undamped resonance
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Steady-state magnification of a forced spring–mass–damper (m = 10 kg, k = 4000 N/m, natural frequency 20 rad/s) for four damping coefficients, 120 forcing frequencies each computed by the package; marked: ω = 20 rad/s, c = 40 N·s/m, magnification = 5.

```
mechanical_engineering_ai999x.solve('forced_vibration_amplitude', m=10.0, c=40.0, k=4000.0,
F0=100.0, omega=20.0)
# -> X = 0.125 m, X_mm = 125, phase_deg = 90, magnification = 5, r = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Forced vibration: m = 10 kg, c = 40 N*s/m, k = 4000 N/m, F0 = 100 N, omega = 20 rad/s	SOLVED	X = 0.125 m, X_mm = 125, phase_deg = 90, magnification = 5, r = 1
Forced vibration: m = 10 kg, c = 0 N*s/m, k = 4000 N/m, F0 = 100 N, omega = 20 rad/s	INVALID_INPUT	Steady-state response to a harmonic force: undamped resonance: the amplitude grows without bound.

Force (or base-motion) transmissibility of a vibration isolator — `vibration_transmissibility`

The ratio of transmitted to applied force amplitude (equally, of mass to base displacement amplitude) for a damped isolator at a frequency ratio, and the isolation achieved.

$$TR = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}}$$

input	meaning	condition
`zeta`	damping ratio	
`r`	frequency ratio omega/omega_n	

output	meaning
`TR`	transmissibility
`isolation_percent`	isolation $(1 - TR)$ in percent (0 when TR >= 1)

- **Method:** $TR = sqrt$((1 + (2 zeta r)^2)/((1 - r^2)^2 + (2 zeta r)^2))
- **Assumes:** $\zeta \geq 0$, $r > 0$; not undamped resonance (r = 1 with zeta = 0)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('vibration_transmissibility', zeta=0.0, r=2.0)
# -> TR = 0.333333, isolation_percent = 66.6667   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('vibration_transmissibility', zeta=0.1, r=1.4142135623730951)
# -> TR = 1, isolation_percent = 4.44089e-14   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Transmissibility: zeta = 0, r = 2	SOLVED	TR = 0.333333, isolation_percent = 66.6667
Transmissibility: zeta = 0, r = 1	INVALID_INPUT	Force (or base-motion) transmissibility of a vibration isolator: undamped resonance: the transmissibility is unbounded.

Vibration due to rotating unbalance — `rotating_unbalance_response`

The steady-state amplitude of a machine on springs and dampers excited by a rotating unbalance, the unbalance force and the force transmitted to the foundation.

$$X = \frac{m_u e \omega^2}{\sqrt{(k - M\omega^2)^2 + (c\omega)^2}}, \quad F_T = X\sqrt{k^2 + (c\omega)^2}$$

input	meaning	condition
`M`	total mass of the machine	
`mu`	unbalanced mass	
`e`	eccentricity of the unbalanced mass	
`k`	total stiffness of the mounts	
`c`	damping coefficient	
`omega`	rotational speed	

output	meaning
`X`	amplitude of the machine
`X_mm`	amplitude in mm
`F_transmitted`	force amplitude transmitted to the foundation
`F_unbalance`	rotating unbalance force mu e ω^2

- **Method:** $X = mu$ e $w^2/sqrt$((k - M $w^2)^2 + (c$ w)^2), $F_T = X$ sqrt(k^2 + (c w)^2)
- **Assumes:** vertical motion only, steady state; $mu < M$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('rotating_unbalance_response', M=100.0, mu=0.5, e=0.05,
k=400000.0, c=0.0, omega=100.0)
# -> X = 0.000416667 m, X_mm = 0.416667, F_transmitted = 166.667 N, F_unbalance = 250 N   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Rotating unbalance: M = 100 kg, mu = 0.5 kg, e = 50 mm, k = 400 kN/m, c = 0 N*s/m, omega = 100 rad/s	SOLVED	X = 0.000416667 m, X_mm = 0.416667, F_transmitted = 166.667 N, F_unbalance = 250 N
Rotating unbalance: M = 100 kg, mu = 200 kg, e = 50 mm, k = 400 kN/m, c = 0 N*s/m, omega = 100 rad/s	INVALID_INPUT	Vibration due to rotating unbalance: the unbalanced mass must be part of the total mass M.

Stiffness of an undamped isolator for a target transmissibility — `vibration_isolator_design`

The mount stiffness, natural frequency, frequency ratio and static deflection an undamped isolator needs so that a machine running at omega transmits only a fraction TR of its force.

$$TR = \frac{1}{r^2 - 1}, \quad k = m\left(\frac{\omega}{r}\right)^2$$

input	meaning	condition
`m`	isolated mass	
`omega`	forcing frequency (rad/s or rpm)	
`TR`	target transmissibility (0 < TR < 1, e.g. 0.1 or 10%)	

output	meaning
`k`	required total stiffness
`f_n`	natural frequency of the mounted system
`r`	frequency ratio omega/omega_n
`static_deflection_mm`	static deflection of the mounts, mm

- **Method:** $TR = 1/(r^2 - 1)$ -> $r = \sqrt{1 + \frac{1}{TR}}$, $k = m\,(\omega/r)^2$
- **Assumes:** undamped isolator; $0 < TR < 1$ (isolation region $r > sqrt(2)$)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('vibration_isolator_design', m=200.0, omega=157.07963267948966,
TR=0.1)
# -> k = 448618 N/m, f_n = 7.53778 Hz, r = 3.31662, static_deflection_mm = 4.37193   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Vibration isolator design: m = 200 kg, omega = 1500 rpm, TR = 0.1	SOLVED	k = 448618 N/m, f_n = 7.53778 Hz, r = 3.31662, static_deflection_mm = 4.37193

question	outcome	result
Vibration isolator design: m = 200 kg, omega = 1500 rpm, TR = 10%	SOLVED	k = 448618 N/m, f_n = 7.53778 Hz, r = 3.31662, static_deflection_mm = 4.37193
Vibration isolator design: m = 200 kg, omega = 1500 rpm, TR = 1.5	INVALID_INPUT	Stiffness of an undamped isolator for a target transmissibility: the target transmissibility must lie strictly between 0

Torsional natural frequency of a two-rotor system — `two_rotor_torsional_vibration`

The torsional natural frequency of two rotors joined by an elastic shaft, the ratio of their vibration amplitudes and (with the shaft length) where the node lies.

$$\omega = \sqrt{\frac{k_t(J_1 + J_2)}{J_1 J_2}}, \quad l_1 = \frac{J_2 L}{J_1 + J_2}$$

input	meaning	condition
`J1`	mass moment of inertia of rotor 1	
`J2`	mass moment of inertia of rotor 2	
`kt`	torsional stiffness of the shaft (N*m/rad)	
`L`	shaft length (optional: gives the node position)	

output	meaning
`omega`	torsional natural frequency
`f`	natural frequency in Hz
`amplitude_ratio`	amplitude of rotor 2 / rotor 1 (negative: opposite phase)
`node_from_rotor1`	distance of the node from rotor 1 (with L)

- **Method:** $\omega^2 = kt$ (J1 + J2)/(J1 J2); amplitude ratio -J1/J2; node at L $J_2/(J_1 + J_2)$ from rotor 1
- **Assumes:** shaft inertia neglected; free-free system (no fixed end)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('two_rotor_torsional_vibration', J1=2.0, J2=6.0, kt=30000.0,
L=1.2)
# -> omega = 141.421 rad/s, f = 22.5079 Hz, amplitude_ratio = -0.333333, node_from_rotor1 = 0.9 m
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Two-rotor torsional vibration: J1 = 2 kg*m^2, J2 = 6 kg*m^2, kt = 30 kN*m/rad, L = 1.2 m	SOLVED	omega = 141.421 rad/s, f = 22.5079 Hz, amplitude_ratio = -0.333333, node_from_rotor1 = 0.9 m
Two-rotor torsional vibration: J1 = 2 kg*m^2, J2 = 6 kg*m^2	NEEDS_INPUT	two_rotor_torsional_vibration needs kt (torsional stiffness of the shaft (N*m/rad)). Example: solve('two_rotor_torsional

Natural frequencies and first mode of a spring-mass chain — `mdof_spring_mass_chain`

All natural frequencies and the first mode shape of a chain of masses joined by springs, fixed to a wall at the left end (and optionally at the right end).

$$(\mathbf{K} - \omega^2 \mathbf{M})\boldsymbol{\phi} = \mathbf{0}$$

input	meaning	condition
`masses`	masses in kg, left to right (a list)	
`springs`	spring stiffnesses in N/m, the wall spring first (a list)	
`ends`	fixed-free (n springs) or fixed-fixed (n + 1 springs)	

output	meaning
`omega`	natural circular frequencies in rad/s, ascending
`f_Hz`	natural frequencies in Hz
`mode1`	first mode shape, largest entry +1

- **Method:** K $\phi = w^2$ M phi with tridiagonal K; solved as the symmetric $M^{-1/2}$ K $M^{-1/2}$ by Jacobi rotations
- **Assumes:** 1 to 12 masses; linear springs, no damping; springs: n (fixed-free) or $n + 1$ (fixed-fixed)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('mdof_spring_mass_chain', masses='2, 1', springs='2000, 1000',
ends='fixed-free')
# -> omega = [22.360679775, 44.72135955], f_Hz = [3.55881271709, 7.11762543417], mode1 = [0.5, 1.0]
(SOLVED)
```

```
mechanical_engineering_ai999x.solve('mdof_spring_mass_chain', masses='1, 1', springs='100, 100,
100', ends='fixed-fixed')
# -> omega = [10.0, 17.3205080757], f_Hz = [1.59154943092, 2.75664447711], mode1 = [1.0, 1.0]
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Spring-mass chain: masses = [2, 1], springs = [2000, 1000], ends = fixed-free	SOLVED	omega = [22.360679775, 44.72135955], f_Hz = [3.55881271709, 7.11762543417], mode1 = [0.5, 1.0]
Spring-mass chain: masses = [2, 1], springs = [2000], ends = fixed-free	INVALID_INPUT	Natural frequencies and first mode of a spring-mass chain: a fixed-free chain of 2 masses needs 2 springs (wall spring f

Critical speed of a shaft carrying one disk — `shaft_critical_speed_single_disk`

The lateral stiffness at the disk, the first critical (whirling) speed and the static deflection of a simply supported massless shaft carrying one disk.

$$k = \frac{3EIL}{a^2 b^2}, \quad \omega_{cr} = \sqrt{\frac{k}{m}}$$

input	meaning	condition
`m`	disk mass	
`E`	Young's modulus of the shaft	
`I`	second moment of area of the shaft	
`L`	bearing span	
`a`	distance of the disk from the left bearing	

output	meaning
`k`	lateral stiffness of the shaft at the disk
`omega_cr`	critical speed
`N_cr_rpm`	critical speed in rpm
`static_deflection_mm`	static deflection under the disk's weight, mm

- **Method:** $k = 3$ E I L/(a^2 b^2), $\omega_{cr} = \sqrt{\frac{k}{m}}$
- **Assumes:** simple (short) bearings, uniform massless shaft; gyroscopic effects neglected; $0 < a < L$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('shaft_critical_speed_single_disk', m=50.0, E=200000000000.0,
I=2e-07, L=1.0, a=0.5)
# -> k = 1.92e+06 N/m, omega_cr = 195.959 rad/s, N_cr_rpm = 1871.27, static_deflection_mm = 0.255382
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Shaft critical speed: m = 50 kg, E = 200 GPa, I = 2e-7 m^4, L = 1 m, a = 0.5 m	SOLVED	k = 1.92e+06 N/m, omega_cr = 195.959 rad/s, N_cr_rpm = 1871.27, static_deflection_mm = 0.255382
Shaft critical speed: m = 50 kg, E = 200 GPa, I = 2e-7 m^4, L = 1 m, a = 1.5 m	INVALID_INPUT	Critical speed of a shaft carrying one disk: the disk must sit between the bearings (0 < a < L).

First critical speed of a shaft with several disks (Rayleigh) — `rayleigh_critical_speed`

Rayleigh's estimate of the first critical speed of a simply supported massless shaft carrying several disks, from the static deflections under the disks' weights.

$$\omega_{cr}^2 = \frac{g \sum m_i y_i}{\sum m_i y_i^2}$$

input	meaning	condition
`masses`	disk masses in kg (a list)	
`positions`	disk positions from the left bearing, in m (a list)	
`L`	bearing span	
`E`	Young's modulus	
`I`	second moment of area of the shaft	

output	meaning
`omega_cr`	first critical speed (Rayleigh, an upper bound)
`N_cr_rpm`	critical speed in rpm
`deflections_mm`	static deflection at each disk, mm

- **Method:** y_i = sum_j a_ij m_j g (influence coefficients of the simply supported beam); $\omega^2 = g$ sum m_i y_i / sum m_i y_i^2
- **Assumes:** simply supported uniform massless shaft; disks at distinct positions inside the span; $g = 9.80665\ m/s^2$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('rayleigh_critical_speed', masses='50', positions='0.5', L=1.0,
E=200000000000.0, I=2e-07)
# -> omega_cr = 195.959 rad/s, N_cr_rpm = 1871.27, deflections_mm = [0.255381510417]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Rayleigh critical speed: masses = [50], positions = [0.5], L = 1 m, E = 200 GPa, I = 2e-7 m^4	SOLVED	omega_cr = 195.959 rad/s, N_cr_rpm = 1871.27, deflections_mm = [0.255381510417]
Rayleigh critical speed: masses = [50, 20], positions = [0.5], L = 1 m, E = 200 GPa, I = 2e-7 m^4	INVALID_INPUT	First critical speed of a shaft with several disks (Rayleigh): masses and positions must be lists of the same length (1

First critical speed of a shaft with several disks (Dunkerley) — `dunkerley_critical_speed`

Dunkerley's lower-bound estimate of the first critical speed of a simply supported massless shaft carrying several disks.

$$\frac{1}{\omega_{cr}^2} = \sum_i m_i a_{ii}, \quad a_{ii} = \frac{a_i^2 b_i^2}{3EIL}$$

input	meaning	condition
`masses`	disk masses in kg (a list)	
`positions`	disk positions from the left bearing, in m (a list)	
`L`	bearing span	
`E`	Young's modulus	
`I`	second moment of area of the shaft	

output	meaning
`omega_cr`	first critical speed (Dunkerley, a lower bound)
`N_cr_rpm`	critical speed in rpm

- **Method:** 1/omega^2 = sum m_i a_ii with $a_i i = b_i^2$ a_i^2/(3 E I L)
- **Assumes:** simply supported uniform massless shaft; disks at distinct positions inside the span

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('dunkerley_critical_speed', masses='50', positions='0.5', L=1.0,
E=200000000000.0, I=2e-07)
# -> omega_cr = 195.959 rad/s, N_cr_rpm = 1871.27   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Dunkerley critical speed: masses = [50], positions = [0.5], L = 1 m, E = 200 GPa, I = 2e-7 m^4	SOLVED	omega_cr = 195.959 rad/s, N_cr_rpm = 1871.27

All critical (whirling) speeds of a shaft with several disks — `shaft_whirling_speeds`

The exact critical speeds of a simply supported massless shaft carrying point-mass disks, from the eigenvalues of its flexibility matrix.

$$\det\left(\mathbf{A}\mathbf{M} - \frac{1}{\omega^2}\mathbf{I}\right) = 0$$

input	meaning	condition
`masses`	disk masses in kg (a list)	
`positions`	disk positions from the left bearing, in m (a list)	
`L`	bearing span	
`E`	Young's modulus	
`I`	second moment of area of the shaft	

output	meaning
`omega`	critical speeds in rad/s, ascending
`N_rpm`	critical speeds in rpm

- **Method:** det(A $M - (1/\omega^2)$ I) = 0 with the influence coefficients a_ij; eigenvalues of the symmetric $M^{1/2}$ A $M^{1/2}$ by Jacobi rotations
- **Assumes:** simply supported uniform massless shaft, point-mass disks (no rotary inertia or gyroscopics); distinct positions
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('shaft_whirling_speeds', masses='50', positions='0.5', L=1.0,
E=200000000000.0, I=2e-07)
# -> omega = [195.959179423], N_rpm = [1871.27232296]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Shaft whirling speeds: masses = [50], positions = [0.5], L = 1 m, E = 200 GPa, I = 2e-7 m^4	SOLVED	omega = [195.959179423], N_rpm = [1871.27232296]

Natural frequencies of a uniform beam (Euler-Bernoulli) — `beam_natural_frequency`

The n-th bending natural frequency of a uniform Euler-Bernoulli beam with simply supported, cantilever, fixed-fixed, fixed-pinned or free-free ends.

$$\omega_n = (\beta_n L)^2 \sqrt{\frac{EI}{\rho A L^4}}, \quad \cos\beta L \cosh\beta L = -1 \text{ (cantilever)}$$

input	meaning	condition
`E`	Young's modulus	
`I`	second moment of area	
`rho`	density	
`A`	cross-sectional area	
`L`	length	
`ends`	end conditions	
`n`	mode number (default 1; free-free counts the elastic modes)	

output	meaning
`beta_L`	eigenvalue beta L
`omega`	natural circular frequency
`f`	natural frequency

- **Method:** omega_n = (beta_n L)^2 sqrt(E I/(rho A L^4)); beta L from $sin = 0$ (SS), cos cosh = -1 (cantilever), cos cosh = 1 (fixed-fixed, free-free), tan = tanh (fixed-pinned)
- **Assumes:** slender uniform beam, no shear deformation or rotary inertia; mode 1 to 6
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('beam_natural_frequency', E=200000000000.0, I=1e-08, rho=7850.0,
A=0.001, L=2.0, ends='simply-supported')
# -> beta_L = 3.14159, omega = 39.384 rad/s, f = 6.26816 Hz   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('beam_natural_frequency', E=200000000000.0, I=1e-08, rho=7850.0,
A=0.001, L=2.0, ends='cantilever')
# -> beta_L = 1.8751, omega = 14.0304 rad/s, f = 2.23301 Hz   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Beam natural frequency: E = 200 GPa, I = 1e-8 m^4, rho = 7850 kg/m^3, A = 1000 mm^2, L = 2 m, ends = simply-supported	SOLVED	beta_L = 3.14159, omega = 39.384 rad/s, f = 6.26816 Hz
Beam natural frequency: E = 200 GPa, I = 1e-8 m^4, rho = 7850 kg/m^3, A = 1000 mm^2, L = 2 m, ends = cantilever, n = 9	INVALID_INPUT	n: must be at most 6, got 9.

Member forces of a plane truss (method of joints) — `truss_member_forces`

The axial force in every member and the support reactions of a statically determinate plane pin-jointed truss on a pin and a roller, under joint loads.

$$\sum_{\text{members at } j} F_m\,\hat{\mathbf{e}}_m + \mathbf{R}_j + \mathbf{P}_j = \mathbf{0}$$

input	meaning	condition
`nodes`	node coordinates, one row 'x y' per node in m (node 1 is the first row)	
`members`	members as pairs of node numbers, e.g. $\mathbf{1-2}$, $\mathbf{2-3}$	
`pin`	node number of the pin support	
`roller`	node number of the roller support (vertical reaction)	
`loads`	joint loads, one row 'node Fx Fy' in kN (up and right positive)	

output	meaning
`forces_kN`	member forces in kN in the order given (tension positive)
`pin_Rx_kN`	horizontal reaction at the pin, kN
`pin_Ry_kN`	vertical reaction at the pin, kN
`roller_Ry_kN`	vertical reaction at the roller, kN
`max_tension_kN`	largest tension, kN
`max_compression_kN`	largest compression (magnitude), kN

- **Method:** equilibrium of every joint (sum $\boldsymbol{Fx = 0}$, sum Fy = 0) as one linear system in the member forces and three reactions, solved by Gaussian elimination
- **Assumes:** pin-jointed, loads at the joints; statically determinate: members + 3 = 2 x nodes, geometrically stable; roller on a horizontal surface
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('truss_member_forces', nodes='0 0; 4 0; 2 3', members='1-2, 2-3,
3-1', pin=1, roller=2, loads='3 0 -10')
# -> forces_kN = [3.33333333333, -6.00925212577, -6.00925212577], pin_Rx_kN = 0, pin_Ry_kN = 5,
roller_Ry_kN = 5, max_tension_kN = 3.33333, max_compression_kN = 6.00925   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Truss member forces: nodes = [0 0; 4 0; 2 3], members = [1-2, 2-3, 3-1], pin = 1, roller = 2, loads = [3 0 -10]	SOLVED	forces_kN = [3.33333333333, -6.00925212577, -6.00925212577], pin_Rx_kN = 0, pin_Ry_kN = 5, roller_Ry_kN = 5, max_tension_kN = 3.33333, max_compression_kN = 6.00925
Truss member forces: nodes = [0 0; 4 0; 2 3], members = [1-2, 2-3], pin = 1, roller = 2, loads = [3 0 -10]	INVALID_INPUT	Member forces of a plane truss (method of joints): not statically determinate: 2 members + 3 reactions must equal 2 x 3

Continuous beam under uniform loads (three-moment equation) — `continuous_beam_three_moment`

The bending moments over the supports and the support reactions of a continuous beam of equal stiffness over 2 to 6 simply supported spans, each with its own uniform load.

$$M_{i-1}L_i + 2M_i(L_i + L_{i+1}) + M_{i+1}L_{i+1} = -\frac{w_i L_i^3 + w_{i+1}L_{i+1}^3}{4}$$

input	meaning	condition
`spans`	span lengths in m, left to right (a list)	
`w`	uniform load on each span in kN/m (a list)	

output	meaning
`support_moments_kNm`	bending moment at each support in kNm, end supports first and last (hogging negative)
`reactions_kN`	reaction at each support in kN

- **Method:** Clapeyron: M_(i-1) $L_i + 2$ M_i (L_i + L_(i+1)) + M_(i+1) L_(i+1) = -(w_i Missing superscript or subscript argument (i+1) L_(i+1)^3)/4; reactions from the span shears
- **Assumes:** constant EI along the beam; simple supports, no settlement; end supports carry no moment
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('continuous_beam_three_moment', spans='6, 6', w='10, 10')
# -> support_moments_kNm = [0.0, -45.0, 0.0], reactions_kN = [22.5, 75.0, 22.5]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Continuous beam: spans = [6, 6], w = [10, 10]	SOLVED	support_moments_kNm = [0.0, -45.0, 0.0], reactions_kN = [22.5, 75.0, 22.5]
Continuous beam: spans = [6, 6], w = [10]	INVALID_INPUT	w: needs at least 2 entries, got 1.

Slider-crank mechanism: slider position, velocity and acceleration — `slider_crank_kinematics`

The slider position, velocity and acceleration and the connecting-rod angle of an in-line slider-crank mechanism at a crank angle, the crank turning at constant speed.

$$x = r\cos\theta + \sqrt{l^2 - r^2\sin^2\theta}, \quad v = \omega\frac{dx}{d\theta}, \quad a = \omega^2\frac{d^2x}{d\theta^2}$$

input	meaning	condition
`r`	crank radius	
`l`	connecting rod length	
`theta`	crank angle from the line of stroke (top dead centre at 0)	
`omega`	constant crank speed (rad/s or rpm)	

output	meaning
`x`	distance of the slider pin from the crank centre
`v`	slider velocity (positive away from the crank)
`a`	slider acceleration
`rod_angle_deg`	connecting-rod angle to the line of stroke, degrees

- **Method:** $x = r$ cos(theta) + sqrt($l^2 - r^2\ sin^2$ theta); $v = dx/dt$ and $a = d^{2x}/dt^2$ with dtheta/dt = omega
- **Assumes:** in-line (zero offset); $l > r$; constant crank speed
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('slider_crank_kinematics', r=0.05, l=0.2, theta=0.0,
omega=100.0)
# -> x = 0.25 m, v = 0 m/s, a = -625 m/s^2, rod_angle_deg = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Slider-crank: r = 50 mm, l = 200 mm, theta = 0 deg, omega = 100 rad/s	SOLVED	x = 0.25 m, v = 0 m/s, a = -625 m/s^2, rod_angle_deg = 0
Slider-crank: r = 50 mm, l = 40 mm, theta = 60 deg, omega = 100 rad/s	INVALID_INPUT	Slider-crank mechanism: slider position, velocity and acceleration: the connecting rod must be longer than the crank.

Four-bar linkage position analysis — `four_bar_linkage_position`

The coupler and output-rocker angles and the transmission angle of a four-bar linkage at a given crank angle, on the open or the crossed branch.

$$K_1 \cos\theta_4 - K_2 \cos\theta_2 + K_3 = \cos(\theta_2 - \theta_4), \quad \theta_4 = 2\arctan\frac{-B \mp \sqrt{B^2 - 4AC}}{2A}$$

input	meaning	condition
`a`	input crank length	
`b`	coupler length	
`c`	output rocker length	
`d`	ground link length	
`theta2`	crank angle from the ground link	
`branch`	assembly branch	

output	meaning
`theta3_deg`	coupler angle, degrees in [-180, 180)
`theta4_deg`	output rocker angle, degrees in [-180, 180)
`transmission_angle_deg`	transmission angle (acute), degrees

- **Method:** Freudenstein's equation: K1 cos $th4 - K_2$ cos $th2 + K_3 = cos$(th2 - th4) solved by the half-angle substitution, $th4 = 2$ atan2(-B -+ sqrt(B^2 - 4AC), 2A); likewise th3
- **Assumes:** ground pivots at (0, 0) and (d, 0); open branch: the rocker pivot lies to the right of the directed coupler; refused when the linkage cannot close
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('four_bar_linkage_position', a=2.0, b=7.0, c=9.0, d=6.0,
theta2=0.5235987755982988, branch='open')
# -> theta3_deg = 88.8372, theta4_deg = 117.286, transmission_angle_deg = 28.4488   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('four_bar_linkage_position', a=2.0, b=7.0, c=9.0, d=6.0,
theta2=0.5235987755982988, branch='crossed')
# -> theta3_deg = -115.211, theta4_deg = -143.66, transmission_angle_deg = 28.4488   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Four-bar linkage: a = 2 m, b = 7 m, c = 9 m, d = 6 m, theta2 = 30 deg, branch = open	SOLVED	theta3_deg = 88.8372, theta4_deg = 117.286, transmission_angle_deg = 28.4488
Four-bar linkage: a = 1 m, b = 1 m, c = 1 m, d = 5 m, theta2 = 90 deg, branch = open	INVALID_INPUT	Four-bar linkage position analysis: the linkage cannot be assembled at this input angle.

Fatigue safety factor by the Gerber criterion — `gerber_fatigue_safety_factor`

The fatigue factor of safety along a proportional load line by the Gerber parabola sa/Se + (sm/Sut)^2 = 1, and the equivalent fully reversed amplitude.

$$\frac{n\sigma_a}{S_e} + \left(\frac{n\sigma_m}{S_{ut}}\right)^2 = 1$$

input	meaning	condition
`sa`	alternating stress amplitude	
`sm`	mean stress (>= 0)	
`Se`	fully corrected endurance limit	
`Sut`	ultimate tensile strength	

output	meaning
`n_fatigue`	fatigue safety factor (Gerber, proportional loading)
`sigma_ar_MPa`	equivalent fully reversed amplitude sa/(1 - (sm/Sut)^2), MPa

- **Method:** n sa/Se + (n sm/Sut)^2 = 1: $n = (1/2)$(Sut/sm)^2 (sa/Se) [$-1 + sqrt$(1 + (2 sm Se/(Sut sa))^2)]
- **Assumes:** $sm \geq 0$ (compressive means not covered); sm < Sut; proportional loading
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('gerber_fatigue_safety_factor', sa=120000000.0, sm=80000000.0,
Se=250000000.0, Sut=700000000.0)
# -> n_fatigue = 1.97698, sigma_ar_MPa = 121.588   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('gerber_fatigue_safety_factor', sa=100000000.0, sm=0.0,
Se=250000000.0, Sut=700000000.0)
# -> n_fatigue = 2.5, sigma_ar_MPa = 100   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Gerber safety factor: sa = 120 MPa, sm = 80 MPa, Se = 250 MPa, Sut = 700 MPa	SOLVED	n_fatigue = 1.97698, sigma_ar_MPa = 121.588
Gerber safety factor: sa = 120 MPa, sm = 800 MPa, Se = 250 MPa, Sut = 700 MPa	INVALID_INPUT	Fatigue safety factor by the Gerber criterion: the mean stress must be below the strength.
Gerber safety factor: sa = 120 MPa, sm = 80 MPa, Se = 250 MPa	NEEDS_INPUT	gerber_fatigue_safety_factor needs Sut (ultimate tensile strength). Example: solve('gerber_fatigue_safety_factor', sa='1

Fatigue safety factor by the ASME-elliptic criterion — `asme_elliptic_fatigue_safety_factor`

The fatigue factor of safety along a proportional load line by the ASME ellipse $(sa/Se)^2 + (sm/Sy)^2 = 1$, and the equivalent fully reversed amplitude.

$$\left(\frac{n\sigma_a}{S_e}\right)^2 + \left(\frac{n\sigma_m}{S_y}\right)^2 = 1$$

input	meaning	condition
`sa`	alternating stress amplitude	
`sm`	mean stress (>= 0)	
`Se`	fully corrected endurance limit	
`Sy`	yield strength	

output	meaning
`n_fatigue`	fatigue safety factor (ASME-elliptic, proportional loading)
`sigma_ar_MPa`	equivalent fully reversed amplitude sa/sqrt(1 - (sm/Sy)^2), MPa

- **Method:** (n $sa/Se)^2 + (n$ sm/Sy)^2 = 1: $n = \frac{1}{\sqrt{\left(\frac{sa}{Se}\right)^2 + \left(\frac{sm}{Sy}\right)^2}}$
- **Assumes:** $sm \geq 0$; $sm < Sy$; proportional loading
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('asme_elliptic_fatigue_safety_factor', sa=100000000.0,
sm=60000000.0, Se=220000000.0, Sy=450000000.0)
# -> n_fatigue = 2.11105, sigma_ar_MPa = 100.901   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('asme_elliptic_fatigue_safety_factor', sa=30000000.0,
sm=40000000.0, Se=100000000.0, Sy=100000000.0)
# -> n_fatigue = 2, sigma_ar_MPa = 32.7327   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
ASME-elliptic safety factor: sa = 100 MPa, sm = 60 MPa, Se = 220 MPa, Sy = 450 MPa	SOLVED	n_fatigue = 2.11105, sigma_ar_MPa = 100.901
ASME-elliptic safety factor: sa = 100 MPa, sm = 500 MPa, Se = 220 MPa, Sy = 450 MPa	INVALID_INPUT	Fatigue safety factor by the ASME-elliptic criterion: the mean stress must be below the strength.
ASME-elliptic safety factor: sa = 100 MPa, sm = 60 MPa, Se = 220 MPa	NEEDS_INPUT	asme_elliptic_fatigue_safety_factor needs Sy (yield strength). Example: solve('asme_elliptic_fatigue_safety_factor', sa=

Lewis bending stress of a gear tooth — `lewis_gear_bending_stress`

The bending stress at the root of a spur-gear tooth by the Lewis equation $\sigma = Kv$ Wt/(F m Y) (SI, module m, form factor Y based on the module).

$$\sigma = \frac{K_v W_t}{F m Y}$$

input	meaning	condition
`Wt`	transmitted (tangential) load	
`F`	face width	
`m`	module	
`Y`	Lewis form factor (from the tooth count and pressure angle, based on the module)	
`Kv`	velocity (dynamic) factor, default 1	

output	meaning
`sigma`	tooth bending stress
`sigma_MPa`	tooth bending stress, MPa

- **Method:** $\sigma = Kv$ Wt/(F m Y)
- **Assumes:** Wt, F, $m > 0$; $0 < Y \leq 1$; $Kv \geq 1$; load at the tip, one tooth in contact
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('lewis_gear_bending_stress', Wt=2500.0, F=0.04, m=0.004,
Y=0.322)
# -> sigma = 4.85248e+07 Pa, sigma_MPa = 48.5248   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('lewis_gear_bending_stress', Wt=1000.0, F=0.05, m=0.002, Y=0.25,
Kv=1.5)
# -> sigma = 6e+07 Pa, sigma_MPa = 60   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Lewis bending stress: Wt = 2.5 kN, F = 40 mm, m = 4 mm, Y = 0.322	SOLVED	sigma = 4.85248e+07 Pa, sigma_MPa = 48.5248
Lewis bending stress: Wt = 2.5 kN, F = 40 mm, m = 4 mm, Y = 0.322, Kv = 0.5	INVALID_INPUT	Lewis bending stress of a gear tooth: the velocity factor Kv must be at least 1.
Lewis bending stress: Wt = 2.5 kN, F = 40 mm, m = 4 mm	NEEDS_INPUT	lewis_gear_bending_stress needs Y (Lewis form factor (from the tooth count and pressure angle, based on the module)). Ex

Wahl curvature factor of a helical spring — `wahl_factor`

The Wahl stress-correction factor $K_W = (4C-1)/(4C-4) + 0.615/C$ of a round-wire helical spring (with the Bergstrasser and direct-shear factors), and with a load the corrected maximum shear stress.

$$K_W = \frac{4C-1}{4C-4} + \frac{0.615}{C}, \quad \tau = K_W \frac{8FD}{\pi d^3}$$

input	meaning	condition
`d`	wire diameter	
`D`	mean coil diameter	
`F`	axial load (optional)	

output	meaning
`index`	spring index $C = D/d$
`K_wahl`	Wahl factor
`K_bergstrasser`	Bergstrasser factor $(4C+2)/(4C-3)$
`K_s`	direct-shear factor $1 + 0.5/C$
`tau_MPa`	maximum shear stress K_W 8FD/(pi d^3), MPa (with F)

- **Method:** $K_W = (4C-1)/(4C-4) + 0.615/C$; $\tau = K_W$ 8 F D/(pi d^3)
- **Assumes:** $C = D/d > 2$; round wire, close-coiled
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('wahl_factor', d=0.003, D=0.024)
# -> index = 8, K_wahl = 1.18402, K_bergstrasser = 1.17241, K_s = 1.0625   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('wahl_factor', d=0.002, D=0.012, F=50.0)
# -> index = 6, K_wahl = 1.2525, K_bergstrasser = 1.2381, K_s = 1.08333, tau_MPa = 239.21   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Wahl factor: d = 3 mm, D = 24 mm	SOLVED	index = 8, K_wahl = 1.18402, K_bergstrasser = 1.17241, K_s = 1.0625

question	outcome	result
Wahl factor: d = 3 mm, D = 5 mm	INVALID_INPUT	Wahl curvature factor of a helical spring: the spring index D/d must exceed 2.
Wahl factor: d = 3 mm	NEEDS_INPUT	wahl_factor needs D (mean coil diameter). Example: solve('wahl_factor', d='0.003 m', D='0.024 m').

Endurance limit with the Marin factors — `marin_endurance_limit`

The corrected endurance limit $Se = ka$ kb kc kd ke kf Se' of a steel part from its tensile strength, surface finish, size, loading and reliability (Se' = 0.5 Sut up to 1400 MPa, else 700 MPa).

$$S_e = k_a k_b k_c k_d k_e k_f \, S_e', \quad k_a = a S_{ut}^{b}, \quad k_e = 1 - 0.08\, z_a$$

input	meaning	condition
`Sut`	ultimate tensile strength	
`surface`	surface finish (ground, machined, cold-drawn, hot-rolled, as-forged)	
`load`	loading (bending, axial, torsion)	
`d`	diameter (needed for bending and torsion, $2.79 - 254$ mm)	
`R`	reliability (default 0.5; written as 0.99 or 99%)	
`kd`	temperature factor (default 1)	
`kf`	miscellaneous-effects factor (default 1)	

output	meaning
`ka`	surface factor a Sut^b
`kb`	size factor
`kc`	load factor
`ke`	reliability factor $1 - 0.08$ z_a
`Se_prime_MPa`	uncorrected endurance limit, MPa
`Se_MPa`	corrected endurance limit, MPa
`Se`	corrected endurance limit

- **Method:** $ka = a$ Sut^b (a, b by surface, Sut in MPa); $kb = 1.24\, d^{-0.107}$ (d <= 51 mm) or 1.51 $d^{-0.157}$; $kc = 1$, 0.85, 0.59; $ke = 1 - 0.08$ z_a
- **Assumes:** steels (Se' = 0.5 Sut, at most 700 MPa); size factor for $2.79 \le d \le 254$ mm (rotating round sections); $0.5 \le R < 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('marin_endurance_limit', Sut=690000000.0, surface='machined',
load='bending', d=0.032, R=0.99)
# -> ka = 0.797777, kb = 0.855797, kc = 1, ke = 0.813892, Se_prime_MPa = 345, Se_MPa = 191.707, Se =
1.91707e+08 Pa   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('marin_endurance_limit', Sut=1600000000.0, surface='ground',
load='axial')
# -> ka = 0.843931, kb = 1, kc = 0.85, ke = 1, Se_prime_MPa = 700, Se_MPa = 502.139, Se =
5.02139e+08 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Marin factors: Sut = 690 MPa, surface = machined, load = bending, d = 32 mm, R = 99%	SOLVED	ka = 0.797777, kb = 0.855797, kc = 1, ke = 0.813892, Se_prime_MPa = 345, Se_MPa = 191.707, Se = 1.91707e+08 Pa
Marin factors: Sut = 1600 MPa, surface = ground, load = axial	SOLVED	ka = 0.843931, kb = 1, kc = 0.85, ke = 1, Se_prime_MPa = 700, Se_MPa = 502.139, Se = 5.02139e+08 Pa
Marin factors: Sut = 690 MPa, surface = machined, load = bending	INVALID_INPUT	Endurance limit with the Marin factors: the diameter d is needed for bending and torsion (size factor).
Marin factors: Sut = 690 MPa, surface = machined	NEEDS_INPUT	marin_endurance_limit needs load (loading (bending, axial, torsion)). Example: solve('marin_endurance_limit', Sut='6.9e+

Bolt tightening torque and preload (T = K F d) — `bolt_preload_torque`

The tightening torque $T = K$ F d that produces a bolt preload F with nut factor K and nominal diameter d, or the preload produced by a torque.

$$T = K F_i d$$

input	meaning	condition
`K`	nut (torque) factor (about 0.2 for unlubricated steel)	
`d`	nominal bolt diameter	
`F`	preload (give F or T)	
`T`	tightening torque (give F or T)	

output	meaning
`T`	tightening torque
`F`	bolt preload
`F_kN`	bolt preload, kN

- **Method:** $T = K$ F d; F = T/(K d)
- **Assumes:** $0.05 \le K \le 0.5$; exactly one of F and T
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
mechanical_engineering_ai999x.solve('bolt_preload_torque', K=0.2, d=0.012, F=25000.0)
# -> T = 60 N*m, F = 25000 N, F_kN = 25   (SOLVED)
```

```
mechanical_engineering_ai999x.solve('bolt_preload_torque', K=0.2, d=0.012, T=60.0)
# -> T = 60 N*m, F = 25000 N, F_kN = 25   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Bolt torque: K = 0.2, d = 12 mm, F = 25 kN	SOLVED	T = 60 N*m, F = 25000 N, F_kN = 25
Bolt torque: K = 0.2, d = 12 mm, T = 60 N*m	SOLVED	T = 60 N*m, F = 25000 N, F_kN = 25
Bolt torque: K = 0.2, d = 12 mm, F = 25 kN, T = 60 N*m	INVALID_INPUT	Bolt tightening torque and preload (T = K F d): give exactly one of the preload F and the torque T.
Bolt torque: K = 0.2, F = 25 kN	NEEDS_INPUT	bolt_preload_torque needs d (nominal bolt diameter). Example: solve('bolt_preload_torque', K=0.2, d='0.012 m', F='25000